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ANSI

ANSI/NETA MTS-2023

2023 MTS

STANDARD FOR

MAINTENANCE TESTING SPECIFICATIONS

FOR ELECTRICAL POWER EQUIPMENT & SYSTEMS

NETA®

ANSI/NETA MTS-2023

AMERICAN NATIONAL STANDARD

STANDARD FOR MAINTENANCE TESTING SPECIFICATIONS

for Electrical Power Equipment and Systems

Secretariat
NETA (InterNational Electrical Testing Association)

Approved by
American National Standards Institute

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FOREWORD

(This Foreword is not part of American National Standard ANSI/NETA MTS-2023)

The InterNational Electrical Testing Association (NETA) was formed in 1972 to establish uniform testing procedures for electrical equipment and apparatus. NETA has been an Accredited Standards Developer for the American National Standards Institute since 1996. NETA's scope of standards activity is different from that of IEEE, NECA, NEMA, and UL. In matters of testing electrical equipment and systems NETA continues to reference other standards developers' documents where applicable. NETA's review and updating of presently published standards takes into account both national and international standards. NETA's standards may be used internationally as well as in the United States. NETA firmly endorses a global standardization. IEC standards as well as American consensus standards are taken into consideration by NETA's ballot pools and reviewing committees.

The first NETA *Maintenance Testing Specifications for Electrical Power Equipment and Systems* was published in 1975. Since 1989, revised editions of the *Maintenance Testing Specifications* have been published in 1993, 1997, and 2001.

In 2005, this document was approved for the first time as an American National Standard. It was published as a revised American National Standard in 2011, 2015, and in 2019. The 2023 *Standard for Maintenance Testing Specifications for Electrical Power Equipment and Systems* is the most current revision of this document, and was approved as a revised American National Standard on March 6, 2023.

The ANSI/NETA *Standard for Maintenance Testing Specifications for Electrical Power Equipment and Systems* was developed for use by those responsible for the continued operation of existing electrical systems and equipment to guide them in specifying and performing the necessary tests to ensure that these systems and apparatus perform satisfactorily, minimizing downtime, and maximizing life expectancy. This document aids in ensuring safe, reliable operation of existing electrical power systems and equipment. Maintenance testing and understanding the condition of maintenance can identify potential problem areas before they become safety concerns or major problems requiring expensive and time-consuming solutions.

PREFACE

(This Preface is not part of American National Standard ANSI/NETA MTS-2023)

It is recognized by the Association that the needs for maintenance testing of commercial, industrial, governmental, and other electrical power systems vary widely. Many criteria are used in determining what equipment is to be tested and to what extent.

To help the user better understand and navigate more efficiently through this document, we offer the following information:

Notation of Changes
Material included in this edition of the document but not part of the previous edition is marked with a black vertical line to the left of the insertion of text, deletion of text, or alteration of text.

Document Structure
The document is divided into thirteen separate and defined sections:

Section	Description
Section 1	General Scope
Section 2	Applicable References
Section 3	Qualifications of Testing Organization and Personnel
Section 4	Division of Responsibility
Section 5	General
Section 6	Power System Studies
Section 7	Inspection and Test Procedures
Section 8	System Function Test
Section 9	Thermographic Survey
Section 10	Electromagnetic Field Survey
Section 11	Online Partial-Discharge Survey for Switchgear
Tables	Reference Tables
Appendices	Informational Documents

Section 7 Structure
Section 7 is the main body of the document with specific information on what to do relative to the inspection and maintenance testing of electrical power equipment and systems. It is not intended that this document explain how to test specific pieces of equipment or systems.

Sequence of Tests and Inspections
The tests and inspections specified in this document are not necessarily presented in chronological order and may be performed in a different sequence.

Expected Test Results
Section 7 consists of sections specific to each particular type of equipment. Within those sections there are, typically, four main bodies of information:

A. Visual and Mechanical Inspection
B. Electrical Tests
C. Test Values – Visual and Mechanical
D. Test Values – Electrical

PREFACE (*continued*)

Results of Visual and Mechanical Inspections

Some, but not all, visual and mechanical inspections have an associated test value or result. Those items with an expected result are referenced under Section C. Test Values – Visual and Mechanical. For example, Section 7.1 Switchgear and Switchboard Assemblies, item 7.1.A.8.2 calls for verifying tightness of connections using a calibrated torque wrench method. Under the Test Values – Visual and Mechanical Section 7.1.C.2, the expected results for that particular task are listed within Section C, with reference back to the original task description on item 7.1.A.8.2.

7. INSPECTION AND TEST PROCEDURES

7.1 Switchgear and Switchboard Assemblies

A. Visual and Mechanical Inspection

1. Inspect physical, electrical, and mechanical condition.
2. Inspect anchorage, alignment, grounding, and required area clearances.
*3. Prior to cleaning the unit, perform as-found tests.
4. Clean the unit.
5. Verify that fuse and/or circuit breaker sizes and types correspond to drawings and coordination study as well as to the circuit breaker address for microprocessor-communication packages.
6. Verify that current and voltage transformer ratios correspond to drawings.
7. Verify that wiring connections are tight and that wiring is secure to prevent damage during routine operation of moving parts.
8. Inspect bolted electrical connections for high resistance using one or more of the following:
    1. Use of a low-resistance ohmmeter in accordance with Section 7.1.B.1.
    2. Verify tightness of accessible bolted electrical connections by calibrated torque-wrench method in accordance with manufacturer's published data or Table 100.12.
    3. Perform a thermographic survey in accordance with Section 9.
9. Confirm correct operation and sequencing of electrical and mechanical interlock systems.
    1. Attempt closure on locked-open devices. Attempt to open locked-closed devices.
    2. Make key exchange with all devices included in the interlock scheme as applicable.
10. Use appropriate lubrication on moving current-carrying parts and on moving and sliding surfaces.
11. Inspect insulators for evidence of physical damage or contaminated surfaces.
12. Verify correct barrier and shutter installation and operation.
13. Exercise all active components.
14. Inspect mechanical indicating devices for correct operation.

* Optional

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7. INSPECTION AND TEST PROCEDURES

7.1 Switchgear and Switchboard Assemblies ***(continued)***

    1. Perform insulation-resistance tests. Perform measurements from winding-to-winding and each winding-to-ground. Test voltages shall be in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.1.
    2. Verify correct function of control transfer relays located in switchgear with multiple power sources.
9. Verify operation of switchgear/switchboard heaters and their controller.
10. Perform electrical tests of surge arresters in accordance with Section 7.19.
*11. Perform online partial-discharge survey in accordance with Section 11.
12. Perform system function tests in accordance with Section 8.

C. Test Values – Visual and Mechanical

1. Compare bolted connection resistance values to values of similar connections. Investigate values which deviate from those of similar bolted connections by more than 50 percent of the lowest value. (7.1.A.8.1)
2. Bolt-torque levels should be in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.12. (7.1.A.8.2)
3. Results of the thermographic survey shall be in accordance with Section 9. (7.1.A.8.3)

D. Test Values – Electrical

1. Compare bolted connection resistance values to values of similar connections. Investigate values which deviate from those of similar bolted connections by more than 50 percent of the lowest value.
2. Insulation-resistance values of bus insulation should be in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.1. Values of insulation resistance less than this table or manufacturer's recommendations should be investigated. Dielectric withstand voltage tests should not proceed until insulation-resistance levels are raised above minimum values.
3. If no evidence of distress or insulation failure is observed by the end of the total time of voltage application during the dielectric withstand test, the test specimen is considered to have passed the test.
4. Minimum insulation-resistance values of control wiring should be comparable to previously obtained results but not less than two megohms.

* Optional

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PREFACE (*continued*)

Results of Electrical Tests

Each electrical test has a corresponding expected result, and the test item and the expected result have identical item numbers in their section, that is, if the electrical test is item four, the expected result under the Test Values section is also item four. For example, under Section 7.15.1 Rotating Machinery, AC Induction Motors and Generators, item 7.15.1.B.2 (item 2 within the Electrical Tests section) calls for performing an insulation-resistance test in accordance with IEEE Standard 43. Under the Test Values – Electrical section, the expected results for that particular task are listed in the Test Values section under item 2.

7. INSPECTION AND TEST PROCEDURES

7.15.1 Rotating Machinery, AC Induction Motors and Generators

A. Visual and Mechanical Inspection

1. Inspect physical and mechanical condition.
2. Inspect anchorage, alignment, and grounding.
3. Inspect air baffles, filter media, stator winding, stator core, rotor, cooling fans, slip rings, brushes, brush rigging, and bearings.
4. Inspect bolted electrical connections for high resistance using one or more of the following:
    1. Use of low-resistance ohmmeter in accordance with Section 7.15.1.B.1.
    2. Verify tightness of accessible bolted electrical connections by calibrated torque-wrench method in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.12.
    3. Perform thermographic survey in accordance with Section 9.

*5. Perform special tests such as air-gap spacing and machine alignment.

6. Verify the application of appropriate lubrication and lubrication systems.
7. Verify the absence of unusual mechanical or electrical noise or signs of overheating.
8. Verify that resistance temperature detector (RTD) circuits conform to drawings.

B. Electrical Tests – AC Induction

1. Perform resistance measurements through bolted connections with a low-resistance ohmmeter in accordance with Section 7.15.1.A.4.1.
2. Perform insulation-resistance tests in accordance with IEEE 43.
    1. Machines larger than 200 horsepower (150 kilowatts):
       Test duration shall be for ten minutes. Calculate polarization index.
    2. Machines 200 horsepower (150 kilowatts) and less:
       Test duration shall be for one minute. Calculate the dielectric-absorption ratio.
3. Perform dielectric withstand voltage tests on machines rated at 2300 volts and greater in accordance with:

* Optional

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7. INSPECTION AND TEST PROCEDURES

7.15.1 Rotating Machinery, AC Induction Motors and Generators (*continued*)

5. Rotor should be clean and show no evidence of overheating or cracked bars or end rings and should show no evidence of rubs from contact with the stator.
6. Cooling fans should operate, not be loose on the rotor shaft, and have no cracks in them.
7. Slip ring to brush alignment shall be within manufacturer's published tolerances.
8. Brush rigging clearance shall be in accordance with manufacturer's published data.
9. Bearings should be inspected for evidence of overheating, rubs, rolling element damage, contamination, electrical damage, and lack of lubrication.
10. Slip ring wear should be within manufacturer's tolerances for continued use.
11. Brushes should be within manufacturer's tolerances for continued use.
12. Brush rigging should be intact.

2. Compare bolted connection resistance values to values of similar connections. Investigate values which deviate from those of similar bolted connections by more than 50 percent of the lowest value. (7.15.1.A.4.1)
3. Bolt-torque levels should be in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.12. (7.15.1.A.4.2)
4. Results of the thermographic survey shall be in accordance with Section 9. (7.15.1.A.4.3)

D. Test Values – Electrical Tests

1. Compare bolted connection resistance values to values of similar connections. Investigate any values that deviate from similar bolted connections by more than 50 percent of the lowest value.
2. The dielectric absorption ratio or polarization shall be compared to previously obtained results and should not be less than 1.0. The recommended minimum insulation resistance ($IR_{1\,min}$) test results in megohms should be corrected to 40° C and read as follows:
    1. $IR_{1\,min}$ = kV + 1 for most windings made before 1970, all field windings, and others not described in 2.2 and 2.3.

       (kV is the rated machine terminal-to-terminal voltage in rms kV)
    2. $IR_{1\,min}$ = 100 megohms for ac windings built after 1970 (form-wound coils).

* Optional

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Optional Tests

The purpose of these specifications is to assure that all tested electrical equipment and systems supplied by either contractor or owner are operational and within applicable standards and manufacturer's published tolerances and that equipment and systems are installed in accordance with design specifications.

Certain tests are assigned an optional classification. The following considerations are used in determining the use of the optional classification:

1. Does another listed test provide similar information?
2. How does the cost of the test compare to the cost of other tests providing similar information?
3. How commonplace is the test procedure? Is it new technology?

Optional Tests are denoted by an Asterisk (*) on each relevant page, i.e., *Optional.

If/When Applicable

The phrases "if applicable", "when applicable", and any variation thereof do not occur in this standard. This standard assumes that if devices or pieces of equipment are not present, they will not be subject to testing or verification.

Manufacturer's Instruction Manuals

It is important to follow the recommendations contained in the manufacturer's published data. Many of the details of a complete and effective testing procedure can be obtained from this source.

Summary

The guidance of an experienced testing professional should be sought when making decisions concerning the extent of testing. It is necessary to make an informed judgment for each particular system regarding how extensive a procedure is justified. The approach taken in these specifications is to present a comprehensive series of tests applicable to most industrial and larger commercial systems. In smaller systems, some of the tests can be deleted. In other cases, a number of the tests indicated as optional should be performed. In all instances, the condition of maintenance should be understood so that risk factors associated with safety should be part of the decision-making process.

Likewise, guidance of an experienced testing professional should also be sought when making decisions concerning the results of test data and their significance to the overall analysis of the device or system under test. Careful consideration of all aspects of test data and condition of maintenance, including manufacturer's published data and recommendations, must be included in the overall assessment of the device or system under test.

The Association encourages comment from users of this document. Please contact the NETA office.

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CONTENTS (*continued*)

TABLES

CONTENTS (*continued*)

TABLES (*Continued*)

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APPENDICES

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1. GENERAL SCOPE

1.1 Maintenance Testing Specifications

1. These specifications incorporate comprehensive field tests and inspections to assess the suitability for continued service, condition of maintenance, and reliability of electrical power distribution equipment and systems.

2. The purpose of these specifications is to assure tested electrical equipment and systems are operational, are within applicable standards and manufacturer's tolerances, and are suitable for continued service.

3. The work specified in these specifications may involve hazardous voltages, materials, operations, and equipment. These specifications do not purport to address all of the safety concerns, if any, associated with their use. It is the responsibility of the user of this standard to establish appropriate safety, health, and environmental practices and determine the applicability of regulatory limitations prior to use of this specification.

2. APPLICABLE REFERENCES

2.1 Codes, Standards, and Specifications

All inspections and field tests shall be in accordance with the latest edition of the following codes, standards, and specifications except as provided otherwise herein.

1. American National Standards Institute – ANSI

2. ASTM International – ASTM

ASTM D92	*Standard Test Method for Flash and Fire Points by Cleveland Open Cup Tester*
ASTM D445	*Standard Test Method for Kinematic Viscosity of Transparent and Opaque Liquids (and Calculation of Dynamic Viscosity)*
ASTM D664	*Standard Test Method for Acid Number of Petroleum Products by Potentiometric Titration*
ASTM D877/877M	*Standard Test Method for Dielectric Breakdown Voltage of Insulating Liquids using Disk Electrodes*
ASTM D923	*Standard Practices for Sampling Electrical Insulating Liquids*
ASTM D924	*Standard Test Method for Dissipation Factor (or Power Factor) and Relative Permittivity (Dielectric Constant) of Electrical Insulating Liquids*
ASTM D971	*Standard Test Method for Interfacial Tension of Oil against Water by the Ring Method*
ASTM D974	*Standard Test Method for Acid and Base Number by Color-Indicator Titration*
ASTM D1298	*Standard Test Method for Density, Relative Density, or API Gravity of Crude Petroleum and Liquid Petroleum Products by Hydrometer Method*
ASTM D1500	*Standard Test Method for ASTM Color of Petroleum Products (ASTM Color Scale)*
ASTM D1524	*Standard Test Method for Visual Examination of Used Electrical Insulating Liquids in the Field*
ASTM D1533	*Standard Test Method for Water in Insulating Liquids by Coulometric Karl Fischer Titration*

2. APPLICABLE REFERENCES

2.1 Codes, Standards, and Specification *(continued)*

ASTM D1816	*Standard Test Method for Dielectric Breakdown Voltage of Insulating Liquids Using VDE Electrodes*
ASTM D2029	*Standard Test Methods for Water Vapor Content of Electrical Insulating Gases by Measurement of Dew Point*
ASTM D2129	*Standard Test Method for Color of Clear Electrical Insulating Liquids (Platinum-Cobalt Scale)*
ASTM D2284	*Standard Test Method of Acidity of Sulfur Hexafluoride*
ASTM D2472	*Standard Specification for Sulphur Hexafluoride*
ASTM D2477	*Standard Test Method for Dielectric Breakdown Voltage and Dielectric Strength of Insulating Gases at Commercial Power Frequencies*
ASTM D2685	*Standard Test Method for Air and Carbon Tetrafluoride in Sulfur Hexafluoride by Gas Chromatography*
ASTM D2759	*Standard Practice for Sampling Gas from a Transformer under Positive Pressure*
ASTM D3284	*Standard Practice for Combustible Gases in the Gas Space of Electrical Apparatus Using Portable Meters*
ASTM D3612	*Standard Test Method for Analysis of Gases Dissolved in Electrical Insulating Oil by Gas Chromatography*
ASTM D5837	*Standard Test Method for Furanic Compounds in Electrical Insulating Liquids by High-Performance Liquid Chromatography (HPLC)*
ASTM D6786	*Standard Test Method for Particle Count in Mineral Insulating Oil Using Automatic Optical Particle Counters*
ASTM D6871	*Standard Specification for Natural Ester Fluids Used in Electrical Apparatus*

3. Association of Edison Illuminating Companies – AEIC

4. Canadian Standards Association - CSA

CSA Z462	*Workplace Electrical Safety Standard*
CSA Z463	*Maintenance of Electrical Systems*

2. APPLICABLE REFERENCES

2.1 Codes, Standards, and Specification *(continued)*

5. Electrical Apparatus Service Association – EASA

EASA AR100	*Recommended Practice for the Repair of Rotating Electrical Apparatus*

6. Institute of Electrical and Electronic Engineers – IEEE

IEEE C2	*National Electrical Safety Code*
IEEE C37 Compilation	*Guides and Standards for Circuit Breakers, Switchgear, Relays, Substations, and Fuses*
IEEE C57 Compilation	*Guides and Standards for Distribution, Power, and Regulating Transformers*
IEEE C62 Compilation	*Guides and Standards for Surge Protection*
IEEE C93.1	*Requirements for Power-Line Carrier Coupling Capacitors and Coupling Capacitor Voltage Transformers (CCVT)*
IEEE 43	*IEEE Recommended Practice for Testing Insulation Resistance of Electric Machinery*
IEEE 48	*IEEE Standard for Test Procedures and Requirements for Alternating-Current Cable Terminations Used on Shielded Cables having Laminated Insulation Rated 2.5 kV through 765 kV or Extruded Insulation Rated 2.5 kV through 500 kV*
IEEE 81	*IEEE Guide for Measuring Earth Resistivity, Ground Impedance, and Earth Surface Potentials of a Ground System*
IEEE 95	*IEEE Recommended Practice for Insulation Testing of AC Electric Machinery (2300 V and Above) with High Direct Voltage*
IEEE 100	*The Authoritative Dictionary of IEEE Standards Terms*
IEEE 141	*IEEE Recommended Practice for Electrical Power Distribution for Industrial Plants (Red Book)*
IEEE 142	*IEEE Recommended Practice for Grounding of Industrial and Commercial Power Systems (Green Book)*
IEEE 241	*IEEE Recommended Practice for Electric Power Systems in Commercial Buildings (Gray Book)*

IEEE 242	*IEEE Recommended Practice for Protection and Coordination of Industrial and Commercial Power Systems (Buff Book)*
IEEE 286	*IEEE Recommended Practice for Measurement Of Power Factor Tip-Up Of Electric Machinery Stator Coil Insulation*
IEEE 386	*IEEE Standard for Separable Insulated Connector Systems for Power Distribution Systems Rated 2.5 kV through 35 kV*
IEEE 399	*IEEE Recommended Practice for Industrial and Commercial Power Systems Analysis (Brown Book)*
IEEE 400	*IEEE Guide for Field Testing and Evaluation of the Insulation of Shielded Power Cable Systems Rated 5 kV and Above*
IEEE 400.1	*IEEE Guide for Field Testing of Laminated Dielectric, Shielded Power Cable Systems Rated 5 kV and Above with High Direct Current Voltage*
IEEE 400.2	*IEEE Guide for Field Testing of Shielded Power Cable Systems Using Very Low Frequency (VLF)(less than 1 Hz)*
IEEE 400.3	*IEEE Guide for Partial Discharge Testing of Shielded Power Cable Systems in a Field Environment*
IEEE 400.4	*IEEE Guide for Field Testing of Shielded Power Cable Systems Rated 5 kV and Above with Damped Alternating Current (DAC) Voltage*
IEEE 400.5	*Guide for Field Testing of Shielded DC Power Cable Systems Using High Voltage Direct Current (HVDC)*
IEEE 404	*IEEE Standard for Extruded and Laminated Dielectric Shielded Cable Joints Rated 2.5 kV to 500 kV*
IEEE 421.3	*IEEE Standard for High-Potential-Test Requirements for Excitation Systems for Synchronous Machines*
IEEE 446	*IEEE Recommended Practice for Emergency and Standby Power Systems for Industrial and Commercial Applications (Orange Book)*
IEEE 450	*IEEE Recommended Practice for Maintenance, Testing, and Replacement of Vented Lead-Acid Batteries for Stationary Applications*

2. APPLICABLE REFERENCES

2.1 Codes, Standards, and Specification *(continued)*

IEEE 493	*IEEE Recommended Practice for the Design of Reliable Industrial and Commercial Power Systems (Gold Book)*
IEEE 519	*IEEE Recommended Practice and Requirements for Harmonic Control in Electrical Power Systems*
IEEE 551	*IEEE Recommended Practice for Calculating AC Short-Circuit Currents in Industrial and Commercial Power Systems*
IEEE 602	*IEEE Recommended Practice for Electric Systems in Health Care Facilities (White Book)*
IEEE 637	*IEEE Guide for Reclamation of Insulating Oil and Criteria for its Use*
IEEE 644	*Procedures for Measurement of Power Frequency Electric and Magnetic Fields from AC Power Lines*
IEEE 739	*IEEE Recommended Practice for Energy Management in Industrial and Commercial Facilities (Bronze Book)*
IEEE 1015	*IEEE Recommended Practice for Applying Low-Voltage Circuit Breakers Used in Industrial and Commercial Power Systems (Blue Book)*
IEEE 1100	*IEEE Recommended Practice for Powering and Grounding Electronic Equipment (Emerald Book)*
IEEE 1106	*IEEE Recommended Practice for Installation, Maintenance, Testing, and Replacement of Vented Nickel-Cadmium Batteries for Stationary Applications*
IEEE 1159	*IEEE Recommended Practice on Monitoring Electrical Power Quality*
IEEE 1188	*IEEE Recommended Practice for Maintenance, Testing, and Replacement of Valve-Regulated Lead-Acid (VRLA) Batteries for Stationary Applications*
IEEE 1584	*IEEE Guide for Performing Arc-Flash Hazard Calculations*
IEEE 3007.3	*IEEE Recommended Practice for Electrical Safety in Industrial and Commercial Power Systems*

2. APPLICABLE REFERENCES

2.1 Codes, Standards, and Specification *(continued)*

7. Insulated Cable Engineers Association – ICEA

ICEA S-93-639/NEMA WC 74	*5-46 kV Shielded Power Cable for Use in the Transmission and Distribution of Electric Energy*
ICEA S-94-649	*Standard for Concentric Neutral Cables Rated 5kV through 46kV*
ICEA S-97-682	*Standard for Utility Shielded Power Cables Rated 5 kV through 46kV*

8. InterNational Electrical Testing Association – NETA

ANSI/NETA ATS	*Standard for Acceptance Testing Specifications for Electrical Power Equipment and Systems*
ANSI/NETA ECS	*Standard for Electrical Commissioning Specifications for Electrical Power Equipment and Systems*
ANSI/NETA ETT	*Standard for Certification of Electrical Testing Technicians*

9. National Electrical Manufacturers Association – NEMA

NEMA AB4	*Guidelines for Inspection and Preventive Maintenance of Molded-Case Circuit Breakers Used in Commercial and Industrial Applications*
NEMA C84.1	*Electrical Power Systems and Equipment – Voltage Ratings (60 Hertz)*
NEMA MG1	*Motors and Generators*

10. National Fire Protection Association – NFPA

NFPA 70	*National Electrical Code*
NFPA 70B	*Recommended Practice for Electrical Equipment Maintenance*
NFPA 70E	*Standard for Electrical Safety in the Workplace*
NFPA 99	*Health Care Facilities Code*
NFPA 101	*Life Safety Code*
NFPA 110	*Standard for Emergency and Standby Power Systems*

2. APPLICABLE REFERENCES

2.1 Codes, Standards, and Specification *(continued)*

NFPA 111	*Standard of Stored Electrical Energy Emergency and Standby Power Systems*
NFPA 780	*Standard for the Installation of Lightning Protection Systems*

11. Occupational Safety and Health Administration – OSHA

12. State and local codes and ordinances

13. Underwriters Laboratories, Inc. – UL

2.2 Other References

Manufacturers' instruction manuals for the equipment to be tested.

Megger; *A Stitch in Time…The Complete Guide to Electrical Insulation Testing*.

Paul Gill; *Electrical Power Equipment Maintenance and Testing*

2.3 Contact Information

American National Standards Institute – ANSI
www.ansi.org

ASTM International – ASTM
www.astm.org

Association of Edison Illuminating Companies – AEIC
www.aeic.org

Canadian Standards Association – CSA
www.csa.ca

Electrical Apparatus Service Association – EASA
www.easa.com

Institute of Electrical and Electronic Engineers – IEEE
www.ieee.org

Insulated Cable Engineers Association – ICEA
www.icea.net

International Electrotechnical Commission – IEC
Contact through American National Standards Institute

2. APPLICABLE REFERENCES

2.1 Codes, Standards, and Speficiation *(continued)*

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National Electrical Manufacturers Association– NEMA
www.nema.org

National Institute of Standards and Technology
www.nist.gov

National Fire Prevention Association – NFPA
www.nfpa.org

Occupational Safety and Health Administration – OSHA
www.osha.gov

Underwriters Laboratories, Inc. – UL
www.ul.com

3. QUALIFICATIONS OF TESTING ORGANIZATION AND PERSONNEL

3.1 Testing Organization

1. The testing organization shall be an independent, third-party entity which can function as an unbiased testing authority, professionally independent of the manufacturers, suppliers, and installers of equipment or systems being evaluated.

2. The testing organization shall be regularly engaged in the testing of electrical equipment devices, installations, and systems.

3. The testing organization shall use technicians who are regularly employed for testing services.

4. The testing organization shall submit appropriate documentation to demonstrate that it satisfactorily complies with these requirements.

3.2 Testing Personnel

1. Technicians performing these electrical tests and inspections shall be trained and experienced concerning the apparatus and systems being evaluated. These individuals shall be capable of conducting the tests in a safe manner and with complete knowledge of the hazards involved. They must evaluate the test data and make a judgment on the serviceability of the specific equipment.

2. Technicians shall be certified in accordance with ANSI/NETA ETT, Standard for Certification of Electrical Testing Technicians. Each on site crew leader shall hold a current certification, Level 3 or higher, in electrical testing.

4. DIVISION OF RESPONSIBILITY

4.1 The Owner's Representative

The owner's representative shall provide the testing organization with the following:

1. A short-circuit analysis, a coordination study, and a protective device setting sheet as described in Section 6.

2. A complete set of electrical plans and specifications, including all change orders.

3. Drawings and instruction manuals applicable to the scope of work.

4. An itemized description of equipment to be inspected and tested.

5. A determination of who shall provide a suitable and stable source of electrical power to each test site.

6. A determination of who shall perform certain preliminary low-voltage insulation-resistance, continuity, and low-voltage motor rotation tests prior to and in addition to tests specified herein.

7. Notification when equipment becomes available for maintenance tests. Work shall be coordinated to expedite project scheduling.

8. Site-specific hazard notification and safety training.

4.2 The Testing Organization

The testing organization shall provide the following:

1. All field technical services, tooling, equipment, instrumentation, and technical supervision to perform such tests and inspections.

2. Specific power requirements for test equipment.

3. Notification to the owner's representative prior to commencement of any testing.

4. A timely notification of any system, material, or workmanship that is found deficient based on the results of the maintenance tests.

5. A written record of all tests and a final report.

5. GENERAL

5.1 Safety and Precautions

All parties involved must be cognizant of industry-standard safety procedures. This document does not contain any specific safety procedures. It is recognized that an overwhelming majority of the tests and inspections recommended in these specifications are potentially hazardous. Individuals performing these tests shall be qualified and capable of conducting the tests in a safe manner and with complete knowledge of the hazards involved.

1. Safety practices shall include, but are not limited to, the following requirements:
    1. All applicable provisions of the Occupational Safety and Health Act, particularly OSHA 29 CFR 1910 and 29 CFR Part 1926.
    2. NFPA 70E, *Standard for Electrical Safety in the Workplace* and CSA Z462, *Workplace Electrical Safety*.
    3. Applicable state and local safety operating procedures.
    4. Owner’s safety practices.
2. A safety lead person shall be identified prior to commencement of work.
3. A safety briefing shall be conducted prior to the commencement of work.
4. All tests shall be performed with the apparatus de-energized and temporary protective grounds applied except where otherwise specifically required to be ungrounded or energized for certain tests.
5. The testing organization shall have a designated safety representative on the project to supervise operations with respect to safety. This individual may be the same person described in 5.1.2.

5.2 Suitability of Test Equipment

1. All test equipment shall meet the requirements in Section 5.3 and be in good mechanical and electrical condition.
2. Field test metering used to check power system meter calibration must be more accurate than the instrument being tested.
3. Accuracy of metering in test equipment shall be appropriate for the test being performed.
4. Waveshape and frequency of test equipment output waveforms shall be appropriate for the test and the tested equipment.

5. GENERAL

5.3 Test Instrument Calibration

1. The testing organization shall have a calibration program which assures that all applicable test instruments are maintained within rated accuracy for each test instrument calibrated.

2. The firm providing calibration service shall maintain up-to-date instrument calibration instructions and procedures for each test instrument calibrated.

3. The accuracy shall be directly traceable to the National Institute of Standards and Technology (NIST) or ISO/IEC 17025, *General Requirements for the Competence of Testing and Calibration Laboratories*.

4. All test instruments shall be calibrated within 12 months of the date of test.

5. Dated calibration labels shall be visible on all test equipment.

6. Records, which show date and results of instruments calibrated or tested, must be kept up-to-date.

7. Calibrating standard shall be of higher accuracy than that of the instrument tested.

5. GENERAL

5.4 Test Report

1. The test report shall include the following:

    1. Summary of project.
    2. Description of equipment tested.
    3. Description of tests.
    4. Device settings.
    5. Inspection and test data.
    6. Analysis and recommendations.

2. Test data records shall include the following minimum requirements:

    1. Identification of the testing organization.
    2. Equipment identification.
    3. Nameplate data.
    4. Conditions that may affect the results of the tests/calibrations such as humidity, temperature, and other conditions.
    5. Date of inspections, tests, maintenance, and/or calibrations.
    6. Identification of the testing technician.
    7. Indication of inspections, tests, maintenance, and/or calibrations to be performed and recorded.
    8. Indication of expected results when calibrations are to be performed.
    9. Indication of as-found and as-left results.
    10. Sufficient spaces to allow all results and comments to be indicated.

3. The testing organization shall furnish a copy or copies of the complete project report as specified in the maintenance testing contract.

5. GENERAL

5.5 Test Decal

1. The testing organization shall affix a test decal on the exterior of equipment or equipment enclosure of protective devices after performing electrical tests.

2. The test decal shall be color-coded to communicate the condition of maintenance for the protective device. Color scheme for condition of maintenance of overcurrent protective device shall be:

    1. White: electrically and mechanically acceptable.

    2. Yellow: minor deficiency not affecting fault detection and operation, but minor electrical or mechanical condition exists.

    3. Red: deficiency exists affecting performance, not suitable for service.

3. The decal shall include:

    1. Testing organization.

    2. Project identifier.

    3. Test date.

    4. Technician identifier.

6. POWER SYSTEM STUDIES

6.1 Short-Circuit Studies

1. Scope of Study.

   Determine the short-circuit current available at each component of the electrical system and the ability of the component to withstand and/or interrupt the current. Provide an analysis of possible operating scenario[s] which will be or have been influenced by the proposed or completed additions or changes to the subject system.

2. Procedure.

   The short-circuit study shall be performed in accordance with the recommended practices and procedures set forth in IEEE 3002.3.

3. Study Report.

   Results of the short-circuit study shall be summarized in a final report containing the following items:

   1. Basis, description, purpose, and scope of the study.

   2. Tabulation of the data used to model the system components and a corresponding one-line diagram.

   3. Descriptions of the scenario[s] evaluated and identification of the scenario used to evaluate equipment short-circuit ratings.

   4. Tabulation of equipment short-circuit ratings versus available fault duties. The tabulation shall identify percentage of rated short circuit and clearly note equipment with insufficient ratings.

   5. Conclusions and recommendations.

6. POWER SYSTEM STUDIES

6.2 Coordination Studies

1. Scope of Study.

    1. The extent of overcurrent protective device coordination shall include:

        1. Selective coordination: determine the protective device types, characteristics, settings, or ampere ratings which provide selective coordination, equipment protection, and correct interrupting ratings for the full range of available short-circuit currents at points of application for each overcurrent protective device.

        2. Compromised coordination: determine protective device types, characteristics, settings, or ampere ratings which permit ranges of non-coordination of overcurrent protective devices.

        3. Provide an analysis of possible operating scenario[s] which will be or have been influenced by the proposed or completed additions or changes to the subject system.

2. Procedure.

    The coordination study shall be performed in accordance with the recommended practices and procedures set forth in IEEE 399 and IEEE 242. Protective device selection and settings shall comply with requirements of NFPA 70 *National Electrical Code.*

3. Study Report.

    Results of the coordination study shall be summarized in a final report containing the following items:

    1. Basis, description, purpose, and scope of the study and a corresponding one-line diagram.

    2. Time-current curves, selective coordination ratios of fuses, or selective coordination tables of circuit breakers demonstrating the coordination of overcurrent protective devices to the scope.

    3. Tabulations of protective devices identifying circuit location, manufacturer, type, range of adjustment, IEEE device number, current transformer ratios, recommended settings or device size, and referenced time-current curve.

    4. Conclusions and recommendations.

* Optional

6.2 Coordination Studies (*continued*)

4. Implementation

   The owner shall engage qualified testing personnel for the purpose of inspecting, setting, testing, and calibrating the protective relays, circuit breakers, fuses, and other applicable devices as outlined in this specification.

* Optional

6. POWER SYSTEM STUDIES

6.3 Incident Energy Analysis

1. Scope of Study.

   Determine arc-flash incident energy levels and flash protection boundary distances based on the results of the short-circuit and coordination studies. Perform the analysis under worst-case arc-flash conditions for all modes of operation. Provide an analysis of possible operating scenarios which will be or have been influenced by the proposed or completed additions to the subject system.

2. Procedure.

   Identify locations and equipment to be included in the incident energy analysis.

   1. Prepare a one-line diagram of the power system.
   2. Perform a short-circuit study in accordance with Section 6.1.
   3. Perform a coordination study in accordance with Section 6.2.
   4. Identify the possible system operating modes, including tie-breaker positions, parallel generation, etc.
   5. Calculate the arcing fault current flowing through each circuit branch for each fault location using empirical formula in accordance with NFPA 70E, OSHA 1910.269, IEEE 1584, or other applicable standards.
   6. Determine the time required to clear the arcing fault current using the protective device settings and associated trip curves.
   7. Select the working distances based on system voltage and equipment class.
   8. Calculate the incident energy at each fault location at the prescribed working distance.

   *9. Determine the arc-flash hazard PPE category for the calculated incident energy level.

   10. Calculate the flash protection boundary at each fault location.
   11. Document the assessment in reports and one-line diagrams.
   12. Place appropriate labels on the equipment.

* Optional

6.3 Incident Energy Analysis (*continued)*

3. Study Report.

   Results of the incident-energy analysis shall be summarized in a final report containing the following items:

   1. Basis, method of hazard assessment, description, purpose, scope, and date of the study.
   2. Tabulations of the data used to model the system components and a corresponding one-line diagram.
   3. Descriptions of the scenario[s] evaluated and identification of the scenario used to develop incident-energy levels and arc-flash boundaries.
   4. Tabulations of equipment incident energies, arc-flash hazard PPE categories, and arc-flash boundaries. The tabulation shall identify and clearly note equipment that exceeds 40 cal/cm^2.
   5. List of required arc-flash labels and locations.
   6. Conclusions and recommendations.

6. POWER SYSTEM STUDIES

6.4 Load-Flow Studies

1. Scope of Study.

   Determine active and reactive power, voltage, current, and power factor throughout the electrical system. Provide an analysis of possible operating scenarios which will be or have been influenced by the proposed or completed additions or changes to the subject system.

2. Procedure.

   The load-flow study shall be performed in accordance with the recommended practices and procedures set forth in IEEE 3002.2.

3. Study Report.

   Results of the load-flow study shall be summarized in a final report containing the following items:

   1. Basis, description, purpose, and scope of the study.

   2. Tabulations of the data used to model the system components and a corresponding one-line diagram.

   3. Descriptions of the scenarios evaluated and the basis for each.

   4. Tabulation of power and current flow versus equipment ratings. The tabulation shall identify percentage of rated load and the scenario for which the percentage is based. Overloaded equipment shall be clearly noted.

   5. Tabulation of system voltages versus equipment ratings. The tabulation shall identify percentage of rated voltage and the scenario for which the percentage is based. Voltage levels outside the ranges recommended by equipment manufacturers, IEEE C84.1, or other appropriate standards shall be clearly noted.

   6. Tabulation of system real and reactive power losses with areas of concern clearly noted.

   7. Conclusions and recommendations.

* Optional

6. POWER SYSTEM STUDIES

6.5 Stability Studies

1. Scope of Study.

   Determine the ability of the electrical system’s synchronous machines to remain in step with one another following a disturbance. Provide an analysis of disturbances for possible operating scenarios which will be or have been influenced by the proposed or completed additions or changes to the subject system.

2. Procedure.

   The stability study shall be performed in accordance with the recommended practices and procedures set forth in IEEE 399.

3. Study Report.

   Results of the stability study shall be summarized in a final report containing the following items:

   1. Basis, description, purpose, and scope of the study.

   2. Tabulations of the data used to model the system components and a corresponding one-line diagram.

   3. Descriptions of the scenario[s] evaluated and tabulations or graphs showing the calculation results.

   4. Conclusions and recommendations.

* Optional

6.6 Harmonic-Analysis Studies

1. Scope of Study

   Determine the impact of nonlinear loads and their associated harmonic contributions on the voltage and current throughout the electrical system. Provide an analysis of possible operating scenarios which will be or have been influenced by the proposed or completed additions or changes to the subject system.

2. Procedure

   The harmonic-analysis study shall be performed in accordance with the recommended practices and procedures set forth in IEEE3002.8.

3. Study Report

   Results of the harmonic-analysis study shall be summarized in a final report containing the following items:

   1. Basis, description, purpose, and scope of the study.
   2. Tabulations of the data used to model the system components and a corresponding one-line diagram.
   3. Descriptions of the scenario[s] evaluated and the basis for each.
   4. Tabulation of rms voltage, peak voltage, rms current, and total capacitor bank loading versus associated equipment ratings. Equipment with insufficient ratings shall be clearly identified for each of the scenarios evaluated.
   5. Tabulation of calculated voltage distortion factor, current distortion factor, and individual harmonics versus the limits specified by IEEE 519. Calculated values exceeding the limits specified in the standard shall be clearly noted.
   6. Plots of impedance versus frequency showing resonant frequencies to be avoided.
   7. Tabulations of the system transformer capabilities based on the calculated nonsinusoidal load current and the procedures set forth in IEEE C57.110. Overloaded transformers shall be clearly noted.
   8. Conclusions and recommendations.

* Optional

7. INSPECTION AND TEST PROCEDURES

7.1 Switchgear and Switchboard Assemblies

A. Visual and Mechanical Inspection

1. Inspect physical, electrical, and mechanical condition.

2. Inspect anchorage, alignment, grounding, and required area clearances.

*3. Prior to cleaning the unit, perform as-found tests.

4. Clean the unit.

5. Verify that fuse and/or circuit breaker sizes and types correspond to drawings and coordination study as well as to the circuit breaker address for microprocessor-communication packages.

6. Verify that current and voltage transformer ratios correspond to drawings.

7. Verify that wiring connections are tight and that wiring is secure to prevent damage during routine operation of moving parts.

8. Inspect bolted electrical connections for high resistance using one or more of the following:

    1. Use of a low-resistance ohmmeter in accordance with Section 7.1.B.1.

    2. Verify tightness of accessible bolted electrical connections by calibrated torque-wrench method in accordance with manufacturer's published data or Table 100.12.

    3. Perform a thermographic survey in accordance with Section 9.

9. Confirm correct operation and sequencing of electrical and mechanical interlock systems.

    1. Attempt closure on locked-open devices. Attempt to open locked-closed devices.

    2. Make key exchange with all devices included in the interlock scheme as applicable.

10. Use appropriate lubrication on moving current-carrying parts and on moving and sliding surfaces.

11. Inspect insulators for evidence of physical damage or contaminated surfaces.

12. Verify correct barrier and shutter installation and operation.

13. Exercise all active components.

14. Inspect mechanical indicating devices for correct operation.

* Optional

7. INSPECTION AND TEST PROCEDURES

7.1 Switchgear and Switchboard Assemblies *(continued)*

15. Verify that filters are in place, filters are clean and free from debris, and vents are clear

16. Perform visual and mechanical inspection of instrument transformers in accordance with Section 7.10.

17. Perform visual and mechanical inspection of surge arresters in accordance with Section 7.19.

18. Inspect control power transformers.

    1. Inspect for physical damage, cracked insulation, broken leads, tightness of connections, defective wiring, and overall general condition.

    2. Verify that primary and secondary fuse or circuit breaker ratings match drawings.

    3. Verify correct functioning of drawout disconnecting contacts, grounding contacts, and interlocks.

19. Perform as-left tests.

B. Electrical Tests

1. Perform resistance measurements through bolted electrical connections with a low-resistance ohmmeter in accordance with Section 7.1.A.8.1.

2. Perform insulation-resistance tests for one minute on each bus section, phase-to-phase and phase-to-ground. Apply voltage in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.1.

*3. Perform a dielectric withstand voltage test on each bus section, each phase-to-ground with phases not under test grounded, in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.2. The test voltage shall be applied for one minute. Refer to Section 7.1.3 before performing test.

*4. Perform insulation-resistance tests on control wiring with respect to ground. The applied potential shall be 500 volts dc for 300-volt rated cable and 1000 volts dc for 600-volt rated cable. Test duration shall be one minute. For units with solid-state components or control devices that cannot tolerate the applied voltage, follow manufacturer's recommendation.

5. Perform electrical tests on instrument transformers in accordance with Section 7.10.

6. Perform ground-resistance tests in accordance with Section 7.13.

7. Test metering devices in accordance with Section 7.11.

8. Control Power Transformers.

* Optional

1. Perform insulation-resistance tests. Perform measurements from winding-to-winding and each winding-to-ground. Test voltages shall be in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.1.

2. Verify correct function of control transfer relays located in switchgear with multiple power sources.

9. Verify operation of switchgear/switchboard heaters and their controller.

10. Perform electrical tests of surge arresters in accordance with Section 7.19.

*11. Perform online partial-discharge survey in accordance with Section 11.

12. Perform system function tests in accordance with Section 8.

C. Test Values – Visual and Mechanical

1. Compare bolted connection resistance values to values of similar connections. Investigate values which deviate from those of similar bolted connections by more than 50 percent of the lowest value. (7.1.A.8.1)

2. Bolt-torque levels should be in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.12. (7.1.A.8.2)

3. Results of the thermographic survey shall be in accordance with Section 9. (7.1.A.8.3)

D. Test Values – Electrical

1. Compare bolted connection resistance values to values of similar connections. Investigate values which deviate from those of similar bolted connections by more than 50 percent of the lowest value.

2. Insulation-resistance values of bus insulation should be in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.1. Values of insulation resistance less than this table or manufacturer's recommendations should be investigated. Dielectric withstand voltage tests should not proceed until insulation-resistance levels are raised above minimum values.

3. If no evidence of distress or insulation failure is observed by the end of the total time of voltage application during the dielectric withstand test, the test specimen is considered to have passed the test.

4. Minimum insulation-resistance values of control wiring should be comparable to previously obtained results but not less than two megohms.

* Optional

7. INSPECTION AND TEST PROCEDURES

7.1 Switchgear and Switchboard Assemblies *(continued)*

5. Results of electrical tests on instrument transformers should be in accordance with Section 7.10.

6. Results of ground resistance tests should be in accordance with Section 7.13.

7. Accuracy of metering devices should be in accordance with Section 7.11.

8. Control Power Transformers

    1. Insulation-resistance values of control power transformers should be in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.5. Values of insulation resistance less than this table or manufacturer's recommendations should be investigated.

    2. Control transfer relays should perform as designed.

9. Heaters should be operational.

10. Results of electrical tests on surge arresters shall be in accordance with Section 7.19.

11. Results of online partial-discharge survey should be in accordance with manufacturer's published data. In the absence of manufacturer's published data, refer to Table 100.23.

12. Results of system function tests shall be in accordance with Section 8.

* Optional

7. INSPECTION AND TEST PROCEDURES

7.2.1.1 Transformers, Dry Type, Air-Cooled, Low-Voltage, Small

NOTE: This category consists of power transformers with windings rated 600 volts or less and sizes equal to or less than 167 kVA single-phase or 500 kVA three-phase.

A. Visual and Mechanical Inspection

1. Inspect physical and mechanical condition.

2. Inspect anchorage, alignment, and grounding.

*3. Prior to cleaning the unit, perform as-found tests.

4. Clean the unit.

5. Inspect bolted electrical connections for high resistance using one or more of the following:

    1. Use of a low-resistance ohmmeter in accordance with Section 7.2.1.1.B.1.

    2. Verify tightness of accessible bolted electrical connections by calibrated torque-wrench method in accordance with manufacturer's published data or Table 100.12.

    3. Perform a thermographic survey in accordance with Section 9.

6. Perform as-left tests.

7. Verify that as-left tap connections are as specified.

B. Electrical Tests

1. Perform resistance measurements through bolted connections with a low-resistance ohmmeter in accordance with Section 7.2.1.1.A.5.1.

2. Perform insulation-resistance tests winding-to-winding and each winding-to-ground. Apply voltage in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.5. Calculate the dielectric absorption ratio or polarization index.

*3. Perform turns-ratio tests at the designated tap position.

* Optional

7.2.1.1 Transformers, Dry Type, Air-Cooled, Low-Voltage, Small (*continued*)

C. Test Values – Visual and Mechanical

1. Compare bolted connection resistance values to values of similar connections. Investigate values which deviate from those of similar bolted connections by more than 50 percent of the lowest value. (7.2.1.1.A.5.1)

2. Bolt-torque levels should be in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.12. (7.2.1.1.A.5.2)

3. Results of the thermographic survey shall be in accordance with Section 9. (7.2.1.1.A.5.3)

4. Tap connections are left as found unless otherwise specified. (7.2.1.1.A.7)

D. Test Values – Electrical

1. Compare bolted electrical connection resistances to values of similar connections. Investigate values which deviate from those of similar bolted connections by more than 50 percent of the lowest value.

2. Minimum insulation-resistance values of transformer insulation should be in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.5. Values of insulation resistance less than this table or manufacturer's recommendations should be investigated. The dielectric absorption ratio or polarization index shall be compared to previously obtained results and should not be less than 1.0.

3. Turns-ratio test results should not deviate more than one-half percent from either the adjacent coils or the calculated ratio.

* Optional

7. INSPECTION AND TEST PROCEDURES

7.2.1.2 Transformers, Dry Type, Air-Cooled, Large

NOTE: This category consists of power transformers with windings rated higher than 600 volts and low-voltage transformers larger than 167 kVA single-phase or 500 kVA three-phase.

A. Visual and Mechanical Inspection

1. Inspect physical and mechanical condition.

2. Inspect anchorage, alignment, and grounding.

*3. Prior to cleaning the unit, perform as-found tests.

4. Clean the unit.

*5. Verify that control and alarm settings on temperature indicators are as specified.

6. Verify that cooling fans operate correctly.

7. Inspect bolted electrical connections for high resistance using one or more of the following:

    1. Use of a low-resistance ohmmeter in accordance with Section 7.2.1.2.B.1.

    2. Verify tightness of accessible bolted electrical connections by calibrated torque-wrench method in accordance with manufacturer's published data or Table 100.12.

    3. Perform a thermographic survey in accordance with Section 9.

8. Perform specific inspections and mechanical tests as recommended by the manufacturer.

9. Perform as-left tests.

10. Verify that as-left tap connections are as specified.

11. Verify the presence of surge arresters.

* Optional

7.2.1.2 Transformers, Dry Type, Air-Cooled, Large (*continued*)

B. Electrical Tests

1. Perform resistance measurements through bolted connections with a low-resistance ohmmeter in accordance with Section 7.2.1.2.A.7.1.

2. Perform insulation-resistance tests winding-to-winding and each winding-to-ground. Apply voltage in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.5. Calculate the dielectric absorption ratio and polarization index.

3. Perform insulation power-factor or dissipation-factor tests on all windings in accordance with the test equipment manufacturer's published data.

*4. Perform a power-factor or dissipation-factor tip-up test on windings rated greater than 2.5 kV.

5. Perform turns-ratio tests at the designated tap position.

6. Perform an excitation-current test on each phase.

*7. Measure the resistance of each winding at the designated tap position. The test current should not exceed 10% of rated current.

8. Measure core insulation resistance at 500 volts dc if the core is insulated and the core ground strap is removable.

*9. Perform an applied voltage test on all high- and low-voltage windings-to-ground. See IEEE C57.12.91, Sections 10.2 and 10.3.

10. Verify correct secondary voltage phase-to-phase and phase-to-neutral after energization and prior to loading.

11. Test surge arresters in accordance with Section 7.19.

*12. Perform online partial-discharge survey on winding rated higher than 1000 volts in accordance with Section 11.

* Optional

7.2.1.2 Transformers, Dry Type, Air-Cooled, Large (*continued*)

C. Test Values – Visual and Mechanical

1. Control and alarm settings on temperature indicators should operate within manufacturer's recommendations for specified settings. (7.2.1.2.A.5)

2. Cooling fans should operate. (7.2.1.2.A.6)

3. Compare bolted connection resistance values to values of similar connections. Investigate values which deviate from those of similar bolted connections by more than 50 percent of the lowest value. (7.2.1.2.A.7.1)

4. Bolt-torque levels should be in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.12. (7.2.1.2.A.7.2)

5. Results of the thermographic survey shall be in accordance with Section 9. (7.2.1.2.A.7.3)

6. Tap connections shall be left as found unless otherwise specified. (7.2.1.2.A.10)

D. Test Values – Electrical

1. Compare bolted connection resistance values to values of similar connections. Investigate values which deviate from those of similar bolted connections by more than 50 percent of the lowest value.

2. Minimum insulation-resistance values of transformer insulation should be in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.5. Values of insulation resistance less than this table or manufacturer's recommendations should be investigated. The dielectric absorption ratio and polarization index shall be compared to previously obtained results and should not be less than 1.0.

3. The following values are typical for insulation power-factor tests:

    1. C_{HL} Power transformers: 2.0 percent or less.

    2. C_{HL} Distribution transformers: 5.0 percent or less.

    3. C_H and C_L Power-factor or dissipation-factor values will vary due to support insulators and bus work utilized on dry transformers. Consult transformer manufacturer's or test equipment manufacturer's data for additional information.

4. Power-factor or dissipation-factor tip-up exceeding 1.0 percent should be investigated.

5. Turns-ratio test results should not deviate more than one-half percent from either the adjacent coils or the calculated ratio.

* Optional

7.2.1.2 Transformers, Dry Type, Air-Cooled, Large (*continued*)

6. The typical excitation current test data pattern for a three-legged core transformer is two similar current readings and one lower current reading. The test should produce a pattern typical for the specific transformer type and configuration.

7. Temperature corrected winding-resistance values should not deviate more than two percent from previously obtained results. Consult the manufacturer if winding-resistance values vary by more than two percent from measurements of adjacent phases.

8. Core insulation-resistance values should be comparable to previously-obtained results but not less than one megohm at 500 volts dc.

9. AC dielectric withstand test voltage shall not exceed 65 percent of factory acceptance test voltage for one minute duration. DC dielectric withstand test voltage shall not exceed 100 percent of the ac rms test voltage specified in IEEE C57.12.91, Section 10.2 for one minute duration. If no evidence of distress or insulation failure is observed by the end of the total time of voltage application during the dielectric withstand voltage test, the test specimen is considered to have passed the test.

10. Phase-to-phase and phase-to-neutral secondary voltages should be in agreement with nameplate data.

11. Test results for surge arresters shall be in accordance with Section 7.19.

12. Results of online partial-discharge survey should be in accordance with manufacturer’s published data. In the absence of manufacturer’s published data, refer to Table 100.23.

* Optional

7.2.2 Transformers, Liquid-Filled

A. Visual and Mechanical Inspection

1. Inspect physical and mechanical condition.

2. Inspect anchorage, alignment, and grounding.

3. Verify the presence of PCB labeling.

*4. Prior to cleaning the unit, perform as-found tests.

5. Clean bushings and control cabinets.

6. Verify operation of alarm, control, and trip circuits from temperature and level indicators, pressure-relief device, gas accumulator, and fault-pressure relay.

7. Verify that cooling fans and pumps operate correctly.

8. Inspect bolted electrical connections for high resistance using one or more of the following:

    1. Use of a low-resistance ohmmeter in accordance with Section 7.2.2.B.1.

    2. Verify tightness of accessible bolted electrical connections by calibrated torque-wrench method in accordance with manufacturer's published data or Table 100.12.

    3. Perform a thermographic survey in accordance with Section 9.

9. Verify correct liquid level in tanks and bushings.

10. Verify that positive pressure is maintained on gas-blanketed transformers.

11. Perform inspections and mechanical tests as recommended by the manufacturer.

12. Test load tap-changer in accordance with Section 7.12.

13. Verify the presence of transformer surge arresters.

14. Perform as-left tests.

15. Verify de-energized tap-changer position is left as specified.

B. Electrical Tests

1. Perform resistance measurements through bolted connections with a low-resistance ohmmeter in accordance with Section 7.2.2.A.8.1.

* Optional

7.2.2 Transformers, Liquid-Filled (*continued*)

2. Perform insulation-resistance tests, winding-to-winding and each winding-to-ground. Apply voltage in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.5. Calculate the dielectric absorption ratio and polarization index.

3. Perform turns-ratio tests at the designated tap position.

4. Perform insulation power-factor or dissipation-factor tests on all windings in accordance with test equipment manufacturer's published data.

5. Perform power-factor or dissipation-factor tests on each bushing equipped with a power-factor/ capacitance tap. In the absence of a power-factor/capacitance tap, perform hot-collar tests. These tests shall be in accordance with the test equipment manufacturer's published data.

6. Perform excitation-current tests in accordance with the test equipment manufacturer's published data.

*7. Perform sweep frequency response analysis tests.

8. Measure the resistance of each winding at the designated tap position.

*9. Perform dynamic resistance measurement for load tap changer.

*10. Perform leakage reactance three phase equivalent and per phase tests.

*11. If the core ground strap is accessible, remove and measure the core insulation resistance at 500 volts dc.

*12. Measure the percentage of oxygen in the gas blanket.

*13. Perform dielectric frequency response test in accordance with test equipment manufacturer's published data.

14. Remove a sample of insulating liquid in accordance with ASTM D923. The sample shall be tested for the following.

    1. Dielectric-breakdown voltage: ASTM D1816

    2. Acid neutralization number: ASTM D974

    *3. Specific gravity: ASTM D1298

    4. Interfacial tension: ASTM D971

    5. Color: ASTM D1500

* Optional

7.2.2 Transformers, Liquid-Filled (*continued*)

6. Visual condition: ASTM D1524

7. Water in insulating liquids: ASTM D1533

*8. Measure power factor or dissipation factor in accordance with ASTM D924

15. Remove a sample of insulating liquid in accordance with ASTM D3613 and perform dissolved-gas analysis (DGA) in accordance with IEEE C57.104 or ASTM D3612.

16. Test the instrument transformers in accordance with Section 7.10.

17. Test the surge arresters in accordance with Section 7.19.

18. Test the transformer neutral grounding impedance devices.

19. Verify operation of cubicle or air terminal compartment space heaters.

C. Test Values – Visual and Mechanical

1. Alarm, control, and trip circuits from temperature and level indicators as well as pressure relief device and fault pressure relay should operate within manufacturer's recommendations for their specified settings. (7.2.2.A.6)

2. Cooling fans and pumps should operate. (7.2.2.A.7)

3. Compare bolted connection resistance values to values of similar connections. Investigate values which deviate from those of similar bolted connections by more than 50 percent of the lowest value. (7.2.2.A.8.1)

4. Bolt-torque levels should be in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.12. (7.2.2.A.8.2)

5. Results of the thermographic survey shall be in accordance with Section 9. (7.2.2.A.8.3)

6. Liquid levels in the transformer tanks and bushings should be within indicated tolerances. (7.2.2.A.9)

7. Positive pressure should be indicated on pressure gauge for gas-blanketed transformers. (7.2.2.A.10)

D. Test Values – Electrical

1. Compare bolted connection resistance values to values of similar connections. Investigate values which deviate from those of similar bolted connections by more than 50 percent of the lowest value.

* Optional

7.2.2 Transformers, Liquid-Filled (*continued*)

2. Minimum insulation-resistance values of transformer insulation should be in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.5. Values of insulation resistance less than this table or manufacturer's recommendations should be investigated. The polarization index shall be compared to previously obtained results and should not be less than 1.0.

3. Turns-ratio test results should not deviate by more than one-half percent from either the adjacent coils or the calculated ratio.

4. Maximum power-factor/dissipation-factor values of liquid-filled transformers corrected to 20° C should be in accordance with the transformer manufacturer's published data. In the absence of manufacturer's published data, use Table 100.3. Distribution transformer power factor results shall compare to previously obtained results.

5. Investigate bushing power-factor values that vary from nameplate values by more than 50 percent. Investigate bushing capacitance values that vary from nameplate values by more than five percent. Investigate bushing hot-collar test values that exceed 0.1 Watts.

6. Typical excitation-current test data pattern for a three-legged core transformer is two similar current readings and one lower current reading. The tests should produce a pattern typical for the specific transformer type and configuration.

7. Sweep frequency response analysis test results should compare to factory acceptance and previous test results.

8. Temperature corrected winding-resistance values should not deviate more than two percent of previously obtained results. Consult the manufacturer if winding-resistance values vary by more than two percent from measurements of adjacent phases.

9. Dynamic resistance measurement results should compare to factory acceptance and previous test results.

10. Investigate leakage reactance three-phase equivalent test results that deviate from the nameplate by more than 3%. The per phase test results serve as a benchmark for future tests and should not deviate from the average of the three readings by more than 3%.

11. Core insulation values should be comparable to previously obtained results but not less than one megohm at 500 volts dc.

12. Investigate the presence of oxygen in the nitrogen gas blanket.

13. Dielectric frequency response analysis should compare to factory acceptance and previous test results.

14. Insulating liquid values should be in accordance with Table 100.4.

* Optional

7.2.2 Transformers, Liquid-Filled (*continued*)

15. Evaluate results of dissolved-gas analysis in accordance with IEEE C57.104.

16. Results of electrical tests on instrument transformers shall be in accordance with Section 7.10.

17. Results of surge arrester tests shall be in accordance with Section 7.19.

18. Compare grounding impedance device values to previously obtained results. In the absence of previously obtained values, compare obtained values to manufacturer's published data.

19. Heaters should be operational.

* Optional

7. INSPECTION AND TEST PROCEDURES

7.3.1 Cables, Low-Voltage, Low-Energy

– RESERVED –

* Optional

7. INSPECTION AND TEST PROCEDURES

7.3.2 Cables, Low-Voltage, 1,000 Volt Maximum

A. Visual and Mechanical Inspection

1. Inspect exposed sections of cables and connectors for physical damage and evidence of degradation.

2. Inspect bolted electrical connections for high resistance using one or more of the following:

    1. Use of a low-resistance ohmmeter in accordance with Section 7.3.2.B.1.

    2. Verify tightness of accessible bolted electrical connections by calibrated torque-wrench method in accordance with manufacturer's published data or Table 100.12.

    3. Perform a thermographic survey in accordance with Section 9.

3. Inspect cable tray and cable supports.

4. If cables are terminated through window-type current transformers, inspect to verify that neutral and ground conductors are correctly placed for operation of protective devices.

*5. Compare cable data with drawings and cable schedule.

B. Electrical Tests

1. Perform resistance measurements through bolted connections with a low-resistance ohmmeter in accordance with Section 7.3.2.A.2.1.

2. Perform an insulation-resistance test on each phase/conductor with all other phases/conductors grounded. The applied potential shall be 500 volts dc for 300-volt rated cable and 1000 volts dc for greater than 300-volt rated cable. The test duration shall be one minute.

*3. Verify uniform resistance of parallel conductors.

C. Test Values – Visual and Mechanical

1. Compare bolted connection resistance values to values of similar connections. Investigate values which deviate from those of similar bolted connections by more than 50 percent of the lowest value. (7.3.2.A.2.1)

2. Bolt-torque/connector-torque levels should be in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.12. (7.3.2.A.2.2)

3. Results of the thermographic survey shall be in accordance with Section 9. (7.3.2.A.2.3)

* Optional

7.3.2 Cables, Low-Voltage, 1,000 Volt Maximum (*continued*)

D. Test Values – Electrical

1. Compare bolted connection resistance values to values of similar connections. Investigate values which deviate from those of similar bolted connections by more than 50 percent of the lowest value.

2. Insulation-resistance values should be comparable to previously obtained results and similar circuits but not less than two megohms.

3. Deviations in resistance between parallel conductors shall be investigated.

* Optional

7.3.3 Shielded Cables, Medium- and High-Voltage

A. Visual and Mechanical Inspection

1. Inspect exposed sections of cables and connectors for physical damage and evidence of degradation and corona.

2. Inspect terminations and splices for physical damage, evidence of overheating, and corona.

3. Inspect bolted electrical connections for high resistance using one or more of the following:

    1. Use of a low-resistance ohmmeter in accordance with Section 7.3.3.B.1.

    2. Verify tightness of accessible bolted electrical connections by calibrated torque-wrench method in accordance with manufacturer's published data or Table 100.12.

    3. Perform a thermographic survey in accordance with Section 9.

4. Inspect shield grounding and cable support.

5. Verify that visible cable bends are not less than ICEA and/or manufacturer's minimum allowable bending radius.

*6. Inspect fireproofing in common cable areas.

7. If cables are terminated through window-type current transformers, inspect to verify that neutral and ground conductors are correctly placed and that shields are correctly terminated for operation of protective devices.

*8. Compare cable data with drawings and cable schedule.

B. Electrical Tests

1. Perform resistance measurements through bolted connections with a low-resistance ohmmeter in accordance with Section 7.3.3.A.3.1.

2. Perform an insulation-resistance test individually between each phase/conductor and its shield with all other phase/conductors and shields grounded. Apply voltage in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.1.

3. Perform a shield-continuity test on each power cable by ohmmeter method.

* Optional

7.3.3 Shielded Cables, Medium- and High-Voltage (*continued*)

*4. Perform a jacket integrity test for one minute. Use voltage in accordance with manufacturer's published data. In the absence of manufacturer's data use a minimum of 2 kV for cables up to 35 kV and minimum of 5 kV for cables rated higher than 35 kV.

5. In accordance with ICEA, IEC, IEEE and other power cable consensus standards, and manufacturer's published data, testing can be performed by means of direct current, power frequency alternating current, very low frequency alternating current, or damped alternating current. These sources may be used to perform insulation withstand tests and diagnostic tests such as partial discharge analysis and tan delta (or power factor). The selection can only be made after an evaluation of the available test methods, manufacturer's published data, and a review of the installed cable system.

6. Perform a dielectric withstand test using one or more of the following methods. Note the voltage may need to be limited due to power supply chosen and cable terminations, splices, and accessories.

    1. Direct current (DC) dielectric withstand test at maximum voltage on individual conductors with all shields and all conductors not under test grounded. Follow manufacturer's published data. In the absence of manufacturer's data see Table 100.6.1 for extruded dielectric insulation less than 5 years in service (DC potential testing is not recommended for extruded dielectric cables. Check manufacturer's published data) and Table 100.6.2 for laminated dielectric insulation.

    2. Very low frequency (VLF) dielectric withstand test – minimum 30-minute test on individual or grouped conductors with all shields and all conductors not under test grounded. Follow manufacturer's published data for voltage level. Should the circuit be considered of critical importance, longer time is recommend (see Appendix D). Follow manufacturer's published data. In the absence of manufacturer's published data see Table 100.6.3.

    3. Power frequency (20 Hz -300 Hz) dielectric withstand test – 60 minutes at maximum voltage on individual or grouped cables in accordance with manufacturer's published data. In the absence of manufacturer's published data see Table 100.6.5.

*7. Perform a tan-delta (insulation power factor/dissipation factor) test on each individual conductor with all shields and all conductors not under test grounded. In the absence of manufacturer's published data see Table 100.6.6 and Table 100.6.7. Record stability and mean tan-delta values at each step.

*8. Perform an online partial discharge (PD) test. Record ambient noise floor, waveform, and phase-resolved PD pattern of any PD detected.

* Optional

7.3.3 Shielded Cables, Medium- and High-Voltage (*continued*)

*9. Perform an offline partial discharge (PD) test. Record average or peak ambient background noise floor, PD calibration pulse, and any relevant PD, notating location, amplitude, repetition, PD inception, PD extinction voltage thresholds for cable and accessories, and phase resolved pattern. Follow manufacturer's published data for cable accessories, any connected equipment, and test equipment. In the absence of manufacturer's published data see Table 100.6.3, 100.6.4, and 100.6.5 for maximum voltage levels.

C. Test Values – Visual and Mechanical

1. Compare bolted connection resistance values to values of similar connections. Investigate values which deviate from those of similar bolted connections by more than 50 percent of the lowest value. (7.3.3.A.3.1)

2. Bolt-torque levels should be in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.12. (7.3.3.A.3.2)

3. Results of the thermographic survey shall be in accordance with Section 9. (7.3.3.A.3.3)

4. Cable bends should not be less than ICEA and/or manufacturer's minimum allowable bending radius. In the absence of manufacturer's data refer to Table 100.22.

D. Test Values – Electrical

1. Compare bolted connection resistance values to values of similar connections. Investigate values which deviate from those of similar bolted connections by more than 50 percent of the lowest value.

2. Insulation-resistance values should be in accordance with manufacturer's published data. Conductor and insulation geometry, temperature, and overall cable length need to be taken into account in accordance with manufacturer's published data for definitive minimum insulation resistance criteria.

3. Shielding shall exhibit continuity. Investigate resistance values in excess of ten ohms per 1000 feet of cable.

4. If no evidence of distress (distortion of applied waveform, overload/trip) of insulation failure is observed by the end of the total time of voltage application during the test, the test specimen is considered to have passed the test.

5. If no evidence of distress (distortion of applied waveform, overload/trip) of insulation failure is observed by the end of the total time of voltage application during the test, the test specimen is considered to have passed test.

6. For evaluation of tan-delta results see Table 100.6.7 or IEEE 400.2.

* Optional

7.3.3 Shielded Cables, Medium- and High-Voltage (*continued*)

7. There should be no evidence of partial discharge during online testing. Any environmental background noise should be filtered out or eliminated from the reported PD measurements. Record any PD originating from the cable, including amplitude, repetition, and phase pattern of PD relative to the system voltage, to help determine internal, surface, and corona type PD.

8. There should be no evidence of offline partial discharge at or below the PD voltage threshold limits in accordance with manufacturer's published data or IEEE 400.3.

7. INSPECTION AND TEST PROCEDURES

7.4 Metal-Enclosed Busways, Low- and Medium-Voltage

A. Visual and Mechanical Inspection

1. Inspect physical and mechanical condition.

2. Inspect anchorage, alignment, and grounding.

3. Inspect bolted electrical connections for high resistance using one or more of the following:

    1. Use of a low-resistance ohmmeter in accordance with Section 7.4.B.1.

    2. Verify tightness of accessible bolted electrical connections and bus joints by calibrated torque-wrench method in accordance with manufacturer's published data or Table 100.12.

    3. Perform a thermographic survey in accordance with Section 9.

4. Confirm physical orientation in accordance with manufacturer's labels to ensure adequate cooling.

5. Verify weep or drain plugs are in accordance with manufacturer's published data.

6. Verify correct installation of joint shield.

7. Inspect and clean ventilating openings.

B. Electrical Tests

1. Perform resistance measurements through bolted connections and bus joints with a low-resistance ohmmeter in accordance with Section 7.4.A.3.1.

2. Perform insulation resistance tests on each busway for one minute, phase-to-phase and phase-to-ground. Apply voltage in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.1.

3. Perform a dielectric withstand voltage test on each busway, phase-to-ground with phases not under test grounded, in accordance with manufacturer's published data. If manufacturer has no recommendation for this test, it shall be in accordance with Table 100.17. Where no DC test value is shown in Table 100.17, an AC value shall be used. The test voltage shall be applied for one minute.

*4. Perform a contact-resistance test on each accessible connection point of uninsulated busway and compare values with the adjacent phases.

* Optional

7.4 Metal-Enclosed Busways, Low- and Medium-Voltage (*continued*)

*5. Perform a contact-resistance test on assembled sections of insulated busways and compare values with the adjacent phases.

6. Verify operation of busway space heaters

*7. Perform online partial-discharge survey in accordance with Section 11.

C. Test Values – Visual and Mechanical

1. Compare bolted connection resistance values to values of similar connections. Investigate values which deviate from those of similar bolted connections by more than 50 percent of the lowest value. (7.4.A.3.1)

2 Bolt-torque levels should be in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.12. (7.4.A.3.2)

3. Results of the thermographic survey shall be in accordance with Section 9. (7.4.A.3.3)

D. Test Values – Electrical

1. Compare bolted connection resistance values to values of similar connections. Investigate values which deviate from those of similar bolted connections by more than 50 percent of the lowest value.

2.1 Low-Voltage Metal-Enclosed Busways - Insulation-resistance test voltages and minimum insulation resistance values shall be in accordance with manufacturer's published data. In the absence of manufacturer's published data, insulation resistance values are to be corrected to a nominal 100-foot busway run. Use the following formula to determine the minimum insulation resistance value, in Megohms, to a 100-foot nominal value:

Minimum Megohm Value = 100 ÷ [busway length in feet]

Adjusted 1000 ft Nominal Value = measured insulation resistance x (Length of Run/1000)

Converted values of insulation resistance less than manufacturer's minimum or less than the corrected minimum Megohm value should be investigated. Dielectric withstand voltage tests should not proceed until insulation-resistance levels are raised above minimum values.

* Optional

7.4 Metal-Enclosed Busways, Low- and Medium-Voltage (*continued*)

2.2 Medium-Voltage Metal-Enclosed Busways – Insulation-resistance test voltages and resistance values shall be in accordance with the manufacturer's published data or Table 100.1.

3. If no evidence of distress or insulation failure is observed by the end of the total time of voltage application during the dielectric withstand voltage test, the test specimen is considered to have passed the test.

4. Microhm or dc millivolt drop values should not exceed the high levels of the normal range as indicated in the manufacturer's published data. If manufacturer's data is not available, investigate values which deviate from those of similar bus connections and sections by more than 50 percent of the lowest value.

5. Microhm or dc millivolt drop values should not exceed the high levels of the normal range as indicated in the manufacturer's published data. If manufacturer's data is not available, investigate values which deviate from those of similar bus connections and sections by more than 50 percent of the lowest value.

6. Heaters should be operational.

7. Results of online partial-discharge survey should be in accordance with manufacturer's published data. In the absence of manufacturer's published data, refer to Table 100.23.

* Optional

7. INSPECTION AND TEST PROCEDURES

7.5.1.1 Switches, Air, Low-Voltage

A. Visual and Mechanical Inspection

1. Inspect physical and mechanical condition.

2. Inspect anchorage, alignment, grounding, and required clearances.

*3. Prior to cleaning the unit, perform as-found tests.

4. Clean the unit.

5. Verify correct blade alignment, blade penetration, travel stops, and mechanical operation.

6. Verify that fuse sizes and types are in accordance with drawings, short-circuit study, and coordination study.

7. Verify that each fuse has adequate mechanical support and contact integrity.

8. Inspect bolted electrical connections for high resistance using one or more of the following:

    1. Use of a low-resistance ohmmeter in accordance with Section 7.5.1.1.B.1.

    2. Verify tightness of accessible bolted electrical connections by calibrated torque-wrench method in accordance with manufacturer's published data or Table 100.12.

    3. Perform a thermographic survey in accordance with Section 9.

9. Verify operation and sequencing of interlocking systems.

10. Verify phase-barrier mounting is intact.

11. Verify correct operation of indicating and control devices.

12. Use appropriate lubrication on moving current-carrying parts and on moving and sliding surfaces.

13. Perform as-left tests.

* Optional

7.5.1.1 Switches, Air, Low-Voltage (*continued*)

B. Electrical Tests

1. Perform resistance measurements through bolted connections with a low-resistance ohmmeter in accordance with Section 7.5.1.1.A.8.1.

2. Measure contact resistance across each switchblade and fuseholder.

3. Perform insulation-resistance tests for one minute on each pole, phase-to-phase and phase-to-ground with switch closed and across each open pole. Apply voltage in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.1.

4. Measure fuse resistance.

5. Verify cubicle space heater operation.

6. Perform a ground-fault test in accordance with Section 7.14.

7. Perform tests on other protective devices in accordance with Section 7.9.

C. Test Values – Visual and Mechanical

1. Compare bolted connection resistance values to values of similar connections. Investigate values which deviate from those of similar bolted connections by more than 50 percent of the lowest value. (7.5.1.1.A.8.1)

2. Bolt-torque levels should be in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.12. (7.5.1.1.A.8.2) 7.4.B*7

3. Results of the thermographic survey shall be in accordance with Section 9. (7.5.1.1.A.8.3)

D. Test Values – Electrical

1. Compare bolted connection resistance values to values of similar connections. Investigate values which deviate from those of similar bolted connections by more than 50 percent of the lowest value.

2. Microhm or dc millivolt drop values should not exceed the high levels of the normal range as indicated in the manufacturer's published data. If manufacturer's data is not available, investigate values that deviate from adjacent poles or similar switches by more than 50 percent of the lowest value.

3. Insulation-resistance values should be in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.1. Values of insulation resistance less than this table or manufacturer's recommendations should be investigated.

* Optional

7.5.1.1 Switches, Air, Low-Voltage (*continued*)

4. Investigate fuse-resistance values that deviate from each other by more than 15 percent.

5. Heaters should be operational.

6. Ground fault tests should be in accordance with Section 7.14.

7. Results of protective device tests should be in accordance with Section 7.9.

* Optional

7.5.1.2 Switches, Air, Medium-Voltage, Metal-Enclosed

A. Visual and Mechanical Inspection

1. Inspect physical and mechanical condition.

2. Inspect anchorage, alignment, grounding, and required clearances.

*3. Prior to cleaning the unit, perform as-found tests.

4. Clean the unit.

5. Verify correct blade alignment, blade penetration, travel stops, arc interrupter operation, and mechanical operation.

6. Verify that fuse sizes and types are in accordance with drawings, short-circuit studies, and coordination study.

7. Verify that expulsion-limiting devices are in place on all fuses having expulsion-type elements.

8. Verify that each fuseholder has adequate mechanical support and contact integrity.

9. Inspect bolted electrical connections for high resistance using one or more of the following:

    1. Use of a low-resistance ohmmeter in accordance with Section 7.5.1.2.B.1.

    2. Verify tightness of accessible bolted electrical connections by calibrated torque-wrench method in accordance with manufacturer's published data or Table 100.12.

    3. Perform a thermographic survey in accordance with Section 9.

10. Verify operation and sequencing of interlocking systems.

11. Verify that phase-barrier mounting is intact.

12. Verify correct operation of all indicating and control devices.

13. Use appropriate lubrication on moving current-carrying parts and on moving and sliding surfaces.

14. Perform as-left tests.

* Optional

7.5.1.2 Switches, Air, Medium-Voltage, Metal-Enclosed (*continued*)

B. Electrical Tests

1. Perform resistance measurements through bolted connections with a low-resistance ohmmeter in accordance with Section 7.5.1.2.A.9.1.

2. Measure contact resistance across each switchblade assembly and fuseholder.

3. Perform insulation-resistance tests for one minute on each pole, phase-to-phase and phase-to-ground with switch closed and across each open pole. Apply voltage in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.1.

4. Perform a dielectric withstand voltage test on each pole with switch closed. Test each pole-to-ground with all other poles grounded. Test voltage shall be in accordance with manufacturer's published data or Table 100.2.

5. Measure fuse resistance.

6. Verify cubicle space heater operation.

*7. Perform online partial-discharge survey in accordance with Section 11.

C. Test Values – Visual and Mechanical

1. Compare bolted connection resistance values to values of similar connections. Investigate values which deviate from those of similar bolted connections by more than 50 percent of the lowest value. (7.5.1.2.A.9.1)

2. Bolt-torque levels should be in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.12. (7.5.1.2.A.9.2)

3. Results of the thermographic survey shall be in accordance with Section 9. (7.5.1.2.A.9.3)

D. Test Values – Electrical

1. Compare bolted connection resistance values to values of similar connections. Investigate values which deviate from those of similar bolted connections by more than 50 percent of the lowest value.

2. Microhm or dc millivolt drop values should not exceed the high levels of the normal range as indicated in the manufacturer's published data. If manufacturer's data is not available, investigate values that deviate from adjacent poles or similar switches by more than 50 percent of the lowest value.

* Optional

7.5.1.2 Switches, Air, Medium-Voltage, Metal-Enclosed (*continued*)

3. Insulation-resistance values should be in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.1. Values of insulation resistance less than this table or manufacturer's recommendations should be investigated. Dielectric withstand voltage tests shall not proceed until insulation-resistance levels are raised above minimum values.

4. If no evidence of distress or insulation failure is observed by the end of the total time of voltage application during the dielectric withstand voltage test, the test specimen is considered to have passed the test.

5. Investigate fuse resistance values that deviate from each other by more than 15 percent.

6. Heaters should be operational.

7. Results of online partial-discharge survey should be in accordance with manufacturer's published data. In the absence of manufacturer's published data, refer to Table 100.23.

* Optional

7.5.1.3 Switches, Air, Medium- and High-Voltage, Open

A. Visual and Mechanical Inspection

1. Inspect physical and mechanical condition.

2. Inspect anchorage, alignment, grounding, and required clearances.

*3. Prior to cleaning insulators, perform as-found tests.

4. Clean the insulators.

5. Verify correct blade alignment, blade penetration, travel stops, arc interrupter operation, and mechanical operation.

6. Verify that fuse sizes and types are in accordance with drawings, short-circuit studies, and coordination study.

7. Verify that each fuseholder has adequate mechanical support and contact integrity.

8. Inspect bolted electrical connections for high resistance using one or more of the following:

    1. Use of a low-resistance ohmmeter in accordance with Section 7.5.1.3.B.1.

    2. Verify tightness of accessible bolted electrical connections by calibrated torque-wrench method in accordance with manufacturer's published data or Table 100.12.

    3. Perform a thermographic survey in accordance with Section 9.

9. Verify operation and sequencing of interlocking systems.

10. Perform mechanical operator tests in accordance with manufacturer's published data.

11. Verify correct operation and adjustment of motor operator limit switches and mechanical interlocks.

12. Use appropriate lubrication on moving current-carrying parts and on moving and sliding surfaces.

13. Perform as-left tests.

14. Record as-found and as-left operation counter readings.

* Optional

7.5.1.3 Switches, Air, Medium- and High-Voltage, Open (*continued*)

B. Electrical Tests

1. Perform resistance measurements through bolted connections with a low-resistance ohmmeter in accordance with Section 7.5.1.3.A.8.1.

2. Perform a contact-resistance test across each switchblade and fuseholder.

3. Perform insulation-resistance tests for one minute on each pole, phase-to-phase and phase-to-ground with switch closed and across each open pole. Apply voltage in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.1.

*4. Perform insulation-resistance tests on all control wiring with respect to ground. The applied potential shall be 500 volts dc for 300-volt rated cable and 1000 volts dc for 600-volt rated cable. Test duration shall be one minute. For units with solid-state components or control devices that cannot tolerate the applied voltage, follow manufacturer's recommendation.

5. Perform a dielectric withstand voltage test on each pole with switch closed. Test each pole-to-ground with all other poles grounded. Test voltage shall be in accordance with manufacturer's published data or Table 100.19.

6. Measure fuse resistance.

C. Test Values – Visual and Mechanical

1. Compare bolted connection resistance values to values of similar connections. Investigate values which deviate from those of similar bolted connections by more than 50 percent of the lowest value. (7.5.1.3.A.8.1)

2. Bolt-torque levels should be in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.12. (7.5.1.3.A.8.2)

3. Results of the thermographic survey shall be in accordance with Section 9. (7.5.1.3.A.8.3)

4. Operation counter should advance one digit per close-open cycle. (7.5.1.3.A.14)

D. Test Values – Electrical

1. Compare bolted connection resistance values to values of similar connections. Investigate values which deviate from those of similar bolted connections by more than 50 percent of the lowest value.

* Optional

7.5.1.3 Switches, Air, Medium- and High-Voltage, Open (*continued*)

2. Microhm or dc millivolt drop values should not exceed the high levels of the normal range as indicated in the manufacturer's published data. If manufacturer's data is not available, investigate values that deviate from adjacent poles or similar switches by more than 50 percent of the lowest value.

3. Insulation-resistance values should be in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.1. Values of insulation resistance less than this table or manufacturer's recommendations should be investigated. Dielectric withstand voltage tests should not proceed until insulation-resistance levels are raised above minimum values.

4. Minimum insulation-resistance values of control wiring should be comparable to previously obtained results but not less than two megohms.

5. If no evidence of distress or insulation failure is observed by the end of the total time of voltage application during the dielectric withstand voltage test, the test specimen is considered to have passed the test.

6. Investigate fuse resistance values that deviate from each other by more than 15 percent.

* Optional

7. INSPECTION AND TEST PROCEDURES

7.5.2 Switches, Oil, Medium-Voltage

A. Visual and Mechanical Inspection

1. Inspect physical and mechanical condition.

2. Inspect anchorage, alignment, grounding, and required clearances.

*3. Prior to cleaning the unit, perform as-found tests.

4. Clean the unit.

5. Perform mechanical operator tests in accordance with manufacturer’s published data.

6. Verify correct operation and adjustment of motor operator limit switches and mechanical interlocks.

*7. Verify correct blade alignment, blade penetration, travel stops, arc interrupter operation, and mechanical operation.

8. Verify that each fuseholder has adequate mechanical support and contact integrity.

9. Verify that fuse sizes and types are in accordance with drawings, short circuit studies, and coordination study.

10. Test all electrical and mechanical interlock systems for correct operation and sequencing.

11. Inspect bolted electrical connections for high resistance using one or more of the following:

    1. Use of a low-resistance ohmmeter in accordance with Section 7.5.2.B.1.

    2. Verify tightness of accessible bolted electrical connections by calibrated torque-wrench method in accordance with manufacturer’s published data or Table 100.12.

    3. Perform a thermographic survey in accordance with Section 9.

12. Verify that insulating oil level is correct.

13. Inspect and/or replace gaskets as recommended by the manufacturer.

14. Use appropriate lubrication on moving current-carrying parts and on moving and sliding surfaces.

* Optional

7.5.2 Switches, Oil, Medium-Voltage (*continued*)

15. Perform as-left tests.

16. Record as-found and as-left operation counter readings.

B. Electrical Tests

1. Perform resistance measurements through bolted connections with a low-resistance ohmmeter in accordance with Section 7.5.2.A.11.1.

2. Perform a contact/pole-resistance test.

3. Perform insulation-resistance tests for one minute on each pole, phase-to-phase and phase-to-ground with the switch closed, and across each open pole. Apply voltage in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.1.

*4. Perform insulation-resistance tests on all control wiring with respect to ground. The applied potential shall be 500 volts dc for 300-volt rated cable and 1000 volts dc for 600-volt rated cable. Test duration shall be one minute. For units with solid-state components or control devices that cannot tolerate the applied voltage, follow manufacturer's recommendation.

*5. Perform a dielectric withstand voltage test on each pole with switch closed. Test each pole-to-ground with all other poles grounded. Test voltage shall be in accordance with manufacturer's published data or Table 100.19.

*6 Remove a sample of insulating liquid in accordance with ASTM D923. Sample shall be tested in accordance with the referenced standard.

    1. Dielectric breakdown voltage: ASTM D877

    2. Color: ANSI/ASTM D1500

    3. Visual condition: ASTM D1524

    4. Moisture test: ASTM D1533

7. Measure fuse resistance.

C. Test Values – Visual and Mechanical

1. Compare bolted connection resistance values to values of similar connections. Investigate values which deviate from those of similar bolted connections by more than 50 percent of the lowest value. (7.5.2.A.11.1)

* Optional

7.5.2 Switches, Oil, Medium-Voltage (*continued*)

2. Bolt-torque levels should be in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.12. (7.5.2.A.11.2)

3. Results of the thermographic survey shall be in accordance with Section 9. (7.5.2.A.11.3)

4. Operations counter should advance one digit per close-open cycle. (7.5.2.A.16)

D. **Test Values – Electrical**

1. Compare bolted connection resistance values to values of similar connections. Investigate values which deviate from those of similar bolted connections by more than 50 percent of the lowest value.

2. Microhm or dc millivolt drop values should not exceed the high levels of the normal range as indicated in the manufacturer's published data. If manufacturer's data is not available, investigate values that deviate from adjacent poles or similar switches by more than 50 percent of the lowest value.

3. Insulation-resistance values should be in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.1. Values of insulation resistance less than this table or manufacturer's recommendations should be investigated. Dielectric withstand voltage tests should not proceed until insulation-resistance levels are raised above minimum values.

4. Insulation-resistance values of control wiring should be comparable to previously obtained results but not less than two megohms.

5. If no evidence of distress or insulation failure is observed by the end of the total time of voltage application during the dielectric withstand voltage test, the test specimen is considered to have passed the test.

6. Insulating liquid test results should be in accordance with Table 100.4.

7. Investigate fuse resistance values that deviate from each other by more than 15 percent.

* Optional

7. INSPECTION AND TEST PROCEDURES

7.5.3 Switches, Vacuum, Medium-Voltage

A. Visual and Mechanical Inspection

1. Inspect physical and mechanical condition.

2. Inspect anchorage, alignment, grounding, and required clearances.

*3. Prior to cleaning the unit, perform as-found tests.

4. Clean the unit.

5. Perform mechanical operator tests in accordance with manufacturer's published data.

6. Verify correct operation and adjustment of motor operator limit switches and mechanical interlocks.

7. Measure critical distances on operating mechanism as recommended by the manufacturer.

8. Inspect bolted electrical connections for high resistance using one or more of the following:

    1. Use of a low-resistance ohmmeter. See Section 7.5.3.B.1.

    2. Verify tightness of accessible bolted electrical connections by calibrated torque-wrench method in accordance with manufacturer's published data or Table 100.12.

    3. Perform a thermographic survey in accordance with Section 9.

9. Inspect insulating assemblies for evidence of physical damage or contaminated surfaces.

10. Verify that each fuseholder has adequate support and contact integrity.

11. Verify that fuse sizes and types are in accordance with drawings, short-circuit studies, and coordination study.

12. Test all electrical and mechanical interlock systems for correct operation and sequencing.

13. Use appropriate lubrication on moving current-carrying parts and on moving and sliding surfaces.

14. Perform as-left tests.

15. Record as-found and as-left operation counter readings.

* Optional

7.5.3 Switches, Vacuum, Medium-Voltage (*continued*)

B. Electrical Tests

1. Perform resistance measurements through bolted electrical connections with a low-resistance ohmmeter in accordance with section 7.5.3.A.8.1.

2. Perform a contact/pole-resistance test.

3. Perform insulation-resistance tests for one minute on each pole, phase-to-phase and phase-to-ground with switch closed, and across each open pole. Apply voltage in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.1.

*4. Perform magnetron atmospheric condition (MAC) test on each vacuum interrupter.

5. Perform a vacuum bottle integrity (dielectric withstand voltage) test across each vacuum bottle with the switch in the open position in strict accordance with manufacturer's published data.

*6. Perform insulation-resistance tests on all control wiring with respect to ground. The applied potential shall be 500 volts dc for 300-volt rated cable and 1000 volts dc for 600-volt rated cable. Test duration shall be one minute. For units with solid-state components, follow the manufacturer's recommendation.

*7. Perform a dielectric withstand voltage test in accordance with manufacturer's published data.

8. Verify open and close operation from control devices.

9. Measure fuse resistance.

C. Test Values – Visual and Mechanical

1. Critical distances of operating mechanism should be in accordance with manufacturer's published data. (7.5.3.A.7)

2. Compare bolted connection resistance values to values of similar connections. Investigate values which deviate from those of similar bolted connections by more than 50 percent of the lowest value. (7.5.3.A.8.1)

3. Bolt-torque levels should be in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.12. (7.5.3.A.8.2)

4. Results of the thermographic survey shall be in accordance with Section 9. (7.5.3.A.8.3)

5. Operation counter should advance one digit per close-open cycle. (7.5.3.A.16)

* Optional

7.5.3 Switches, Vacuum, Medium-Voltage (*continued*)

D. Test Values – Electrical

1. Compare bolted connection resistance values to values of similar connections. Investigate values which deviate from those of similar bolted connections by more than 50 percent of the lowest value.

2. Microhm or dc millivolt drop values should not exceed the high levels of the normal range as indicated in the manufacturer's published data. If manufacturer's data is not available, investigate values that deviate from adjacent poles or similar switches by more than 50 percent of the lowest value.

3. Insulation-resistance values should be in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.1. Values of insulation resistance less than this table or manufacturer's recommendations should be investigated. Dielectric withstand voltage tests should not proceed until insulation-resistance levels are raised above minimum values.

4. Evaluate each vacuum interrupter in accordance with test equipment manufacturer's instructions.

5. If no evidence of distress or insulation failure is observed by the end of the total time of voltage application during the vacuum bottle integrity test, the test specimen is considered to have passed the test.

6. Insulation-resistance values of control wiring should be comparable to previously obtained results but not less than two megohms.

7. If no evidence of distress or insulation failure is observed by the end of the total time of voltage application during the dielectric withstand voltage test, the test specimen is considered to have passed the test.

8. Results of open and close operation from control devices should be in accordance with system design.

9. Investigate fuse resistance values that deviate from each other by more than 15 percent.

* Optional

7.5.4 Switches, SF_6, Medium-Voltage

A. Visual and Mechanical Inspection

1. Inspect physical and mechanical condition.

2. Inspect anchorage, alignment, grounding, and required clearances.

*3. Prior to cleaning the unit, perform as-found tests.

4. Clean the unit.

5. Inspect and service mechanical operator and SF_6 gas insulated system in accordance with the manufacturer's published data.

6. Verify correct operation of SF_6 gas pressure alarms and limit switches as recommended by the manufacturer.

7. Measure critical distances as recommended by the manufacturer.

8. Inspect bolted electrical connections for high resistance using one or more of the following:

    1. Use of a low-resistance ohmmeter in accordance with Section 7.5.4.B.1.

    2. Verify tightness of accessible bolted electrical connections by calibrated torque-wrench method in accordance with manufacturer's published data. In the absence of manufacturer's data, use Table 100.12.

    3. Perform a thermographic survey in accordance with Section 9.

9. Verify that fuse sizes and types are in accordance with drawings, short-circuit studies, and coordination study.

10. Verify that each fuseholder has adequate mechanical support and contact integrity.

11. Verify operation and sequencing of interlocking systems.

12. Use appropriate lubrication on moving current-carrying parts and on moving and sliding surfaces.

13. Test for SF_6 gas leaks in accordance with manufacturer's published data.

14. Perform as-left tests.

15. Record as-found and as-left operation counter readings.

* Optional

7.5.4 Switches, SF_6, Medium-Voltage (*continued*)

B. Electrical Tests

1. Perform resistance measurements through accessible bolted electrical connections with a low-resistance ohmmeter. See Section 7.5.4.A.8.1.

2. Perform a contact-resistance test.

3. Perform insulation-resistance tests for one minute on each pole, phase-to-phase and phase-to-ground with switch closed, and across each open pole. Apply voltage in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.1.

*4. Remove a sample of SF_6 gas and test in accordance with Table 100.13.

5. Perform a dielectric withstand voltage test across each gas bottle with the switch in the open position in accordance with manufacturer's published data.

*6. Perform insulation-resistance tests on all control wiring with respect to ground. The applied potential shall be 500 volts dc for 300-volt rated cable and 1000 volts dc for 600-volt rated cable. Test duration shall be one minute. For units with solid-state components, follow manufacturer's recommendation.

*7. Perform a dielectric withstand voltage test in accordance with manufacturer's published data.

8. Verify open and close operation from control devices.

9. Measure fuse resistance.

C. Test Values – Visual and Mechanical

1. Critical distances of operating mechanism should be in accordance with manufacturer's published data. (7.5.4.A.7)

2. Compare bolted connection resistance values to values of similar connections. Investigate values which deviate from those of similar bolted connections by more than 50 percent of the lowest value. (7.5.4.A.8.1)

3. Bolt-torque levels should be in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.12. (7.5.4.A.8.2)

4. Results of the thermographic survey shall be in accordance with Section 9. (7.5.4.A.8.3)

5. Operations counter should advance one digit per close-open cycle. (7.5.4.A.15)

* Optional

7.5.4 Switches, SF_6, Medium-Voltage (*continued*)

D. Test Values – Electrical

1. Compare bolted connection resistance values to values of similar connections. Investigate values which deviate from those of similar bolted connections by more than 50 percent of the lowest value.

2. Microhm or dc millivolt drop values should not exceed the high levels of the normal range as indicated in the manufacturer's published data. If manufacturer's data is not available, investigate values that deviate from adjacent poles or similar switches by more than 50 percent of the lowest value.

3. Insulation-resistance values should be in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.1. Values of insulation resistance less than this table or manufacturer's recommendations should be investigated. Dielectric withstand voltage tests should not proceed until insulation-resistance levels are raised above minimum values.

4. Results of SF_6 gas tests should be in accordance with Table 100.13.

5. If no evidence of distress or insulation failure is observed by the end of the total time of voltage application during the dielectric withstand voltage test, the test specimen is considered to have passed the test.

6. Insulation-resistance values of control wiring should be comparable to previously obtained results but not less than two megohms.

7. If no evidence of distress or insulation failure is observed by the end of the total time of voltage application during the dielectric withstand voltage test, the test specimen is considered to have passed the test.

8. Results of open and close operation from control devices should be in accordance with system design.

9. Investigate fuse resistance values that deviate from each other by more than 15 percent.

7.5.5 Switches, Cutouts

A. Visual and Mechanical Inspection

1. Inspect physical and mechanical condition.

2. Inspect anchorage, alignment, and grounding.

*3. Prior to cleaning the unit, perform as-found tests.

4. Clean the unit.

5. Inspect bolted electrical connections for high resistance using one or more of the following:

    1. Use of a low-resistance ohmmeter in accordance with Section 7.5.5.B.1.

    2. Verify tightness of accessible bolted electrical connections by calibrated torque-wrench method in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.12.

    3. Perform a thermographic survey in accordance with Section 9.

6. Verify correct blade alignment, blade penetration, travel stops, latching mechanism, and mechanical operation.

7. Verify that fuse size and types are in accordance with drawings, short-circuit study, and coordination study.

8. Verify that each fuseholder has adequate mechanical support and contact integrity.

9. Perform as-left tests.

B. Electrical Tests

1. Perform resistance measurements through bolted connections with a low-resistance ohmmeter in accordance with Section 7.5.5.A.5.1.

2. Measure contact resistance across each cutout.

3. Perform insulation-resistance tests for one minute on each pole, phase-to-phase and phase-to-ground with switch closed and across each open pole. Apply voltage in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.1.

* Optional

7.5.5 Switches, Cutouts *(continued)*

4. Perform a dielectric withstand voltage test on each pole, phase to ground with cutout closed. Ground adjacent cutouts. Test voltage shall be in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.1.

5. Measure fuse resistance.

C. Test Values – Visual and Mechanical

1. Compare bolted connection resistance values to values of similar connections. Investigate values which deviate from those of similar bolted connections by more than 50 percent of the lowest value. (7.5.5.A.5.1)

2. Bolt-torque levels should be in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.12. (7.5.5.A.5.2)

3. Results of the thermographic survey shall be in accordance with Section 9. (7.5.5.A.5.3)

D. Test Values – Electrical

1. Compare bolted connection resistance values to values of similar connections. Investigate values which deviate from those of similar bolted connections by more than 50 percent of the lowest value.

2. Microohm or dc millivolt drop values should not exceed the high levels of the normal range as indicated in the manufacturer's published data. If manufacturer's data is not available, investigate values which deviate from adjacent poles or similar switches by more than 50 percent of the lowest value.

3. Insulation-resistance values should be in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.1. Values of insulation resistance less than this table or manufacturer's recommendations should be investigated. Dielectric withstand voltage tests should not proceed until insulation-resistance levels are raised above minimum values.

4. If no evidence of distress or insulation failure is observed by the end of the total time of voltage application during the dielectric withstand voltage test, the test specimen is considered to have passed the test.

5. Investigate fuse resistance values that deviate from each other by more than 15 percent.

* Optional

7. INSPECTION AND TEST PROCEDURES

7.6.1.1 Circuit Breakers, Air, Insulated-Case/Molded-Case

A. Visual and Mechanical Inspection

1. Inspect physical and mechanical condition.

2. Inspect anchorage and alignment.

*3. Prior to cleaning the unit, perform as-found tests.

4. Clean the unit.

5. Operate the circuit breaker to ensure smooth operation.

6. Inspect bolted electrical connections for high resistance using one or more of the following:

    1. Use of a low-resistance ohmmeter in accordance with Section 7.6.1.1.B.1.

    2. Verify tightness of accessible bolted electrical connections by calibrated torque-wrench method in accordance with manufacturer’s published data or Table 100.12.

    3. Perform a thermographic survey in accordance with Section 9.

7. Inspect operating mechanism, contacts, and arc chutes in unsealed units.

8. Perform adjustments for final protective device settings in accordance with coordination study provided by end user.

9. Perform as-left tests.

B. Electrical Tests

1. Perform resistance measurements through bolted connections with a low-resistance ohmmeter in accordance with Section 7.6.1.1.A.6.1.

2. Perform insulation-resistance tests for one minute on each pole, phase-to-phase and phase-to-ground with the circuit breaker closed, and across each open pole. Apply voltage in accordance with manufacturer’s published data. In the absence of manufacturer’s published data, use Table 100.1.

3. Perform a static contact/pole-resistance test.

* Optional

7.6.1.1 Circuit Breakers, Air, Insulated-Case/Molded-Case (*continued*)

*4. Perform insulation-resistance tests on all control wiring with respect to ground. The applied potential shall be 500 volts dc for 300-volt rated cable and 1000 volts dc for 600-volt rated cable. Test duration shall be one minute. For units with solid-state components, follow manufacturer's recommendation.

5. Determine long-time pickup and delay by primary current injection.

6. Determine short-time pickup and delay by primary current injection.

7. Determine ground-fault pickup delay by primary current injection.

8. Determine instantaneous pickup current by primary current injection.

*9. Test functions of the trip unit by means of secondary injection.

10. Perform minimum pickup voltage test on shunt trip and close coils in accordance with Table 100.20.

11. Verify correct operation of auxiliary features such as trip and pickup indicators, zone interlocking, electrical close and trip operation, trip-free, antipump function, and trip unit battery condition.

12. Reset all trip logs and indicators.

13. Verify operation of charging mechanism.

C. Test Values – Visual and Mechanical

1. Compare bolted connection resistance values to values of similar connections. Investigate values which deviate from those of similar bolted connections by more than 50 percent of the lowest value. (7.6.1.1.A.6.1)

2. Bolt-torque levels should be in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.12. (7.6.1.1.A.6.2)

3. Results of the thermographic survey shall be in accordance with Section 9. (7.6.1.1.A.6.3)

4. Settings shall comply with coordination study recommendations. (7.6.1.1.A.8)

* Optional

7.6.1.1 Circuit Breakers, Air, Insulated-Case/Molded-Case (*continued*)

D. Test Values – Electrical

1. Compare bolted connection resistance values to values of similar connections. Investigate values which deviate from those of similar bolted connections by more than 50 percent of the lowest value.

2. Insulation-resistance values should be in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.1. Values of insulation resistance less than this table or manufacturer's recommendations should be investigated.

3. Microhm or dc millivolt drop values should not exceed the high levels of the normal range as indicated in the manufacturer's published data. If manufacturer's data is not available, investigate values that deviate from adjacent poles or similar breakers by more than 50 percent of the lowest value.

4. Insulation-resistance values of control wiring should be comparable to previously obtained results but not less than two megohms.

5. Long-time pickup values should be as specified, and the trip characteristic should not exceed manufacturer's published time-current characteristic tolerance band, including adjustment factors. If manufacturer's curves are not available, trip times should not exceed the value shown in Table 100.7.

6. Short-time pickup values should be as specified, and the trip characteristic should not exceed manufacturer's published time-current tolerance band.

7. Ground fault pickup values should be as specified, and the trip characteristic should not exceed manufacturer's published time-current tolerance band.

8. Instantaneous pickup values of molded-case circuit breakers should fall within manufacturer's published tolerances and/or Table 100.8.

9. Pickup values and trip characteristics should be within manufacturer's published tolerances.

10. Minimum pickup voltage on shunt trip and close coils should be in accordance with manufacturer's published data. In the absence of manufacturer's published data, refer to Table 100.20.

11. Breaker open, close, trip, trip-free, antipump, and auxiliary features should function as designed.

12. Trip logs and indicators are reset.

13. The charging mechanism should operate in accordance with manufacturer's published data.

* Optional

7. INSPECTION AND TEST PROCEDURES

7.6.1.2 Circuit Breakers, Air, Low-Voltage Power

A. Visual and Mechanical Inspection

1. Inspect physical and mechanical condition.

2. Inspect anchorage, alignment, and grounding.

3. Verify that all maintenance devices are available for servicing and operating the breaker.

*4. Prior to cleaning the unit, perform as-found tests.

5. Clean the unit.

6. Inspect arc chutes.

7. Inspect moving and stationary contacts for condition, wear, and alignment.

8. Verify that primary and secondary contact wipe and other dimensions vital to satisfactory operation of the breaker are in accordance with manufacturer's published data.

9. Perform all mechanical operator and contact alignment tests on both the breaker and its operating mechanism in accordance with manufacturer's published data.

10. Inspect bolted electrical connections for high resistance using one or more of the following:

    1. Use of a low-resistance ohmmeter in accordance with Section 7.6.1.2.B.1.

    2. Verify tightness of accessible bolted electrical connections by calibrated torque-wrench method in accordance with manufacturer's published data or Table 100.12.

    3. Perform a thermographic survey in accordance with Section 9.

11. Verify cell fit and element alignment.

12. Verify racking mechanism operation.

13. Use appropriate lubrication on moving current-carrying parts and on moving and sliding surfaces.

14. Perform adjustments for final protective device settings in accordance with coordination study provided by end user.

15. Perform as-left tests.

16. Record as-found and as-left operation counter readings.

* Optional

7.6.1.2 Circuit Breakers, Air, Low-Voltage Power (*continued*)

B. Electrical Tests

1. Perform resistance measurements through bolted connections with a low-resistance ohmmeter in accordance with Section 7.6.1.2.A.10.1.

2. Perform insulation-resistance tests for one minute on each pole, phase-to-phase and phase-to-ground with the circuit breaker closed, and across each open pole. Apply voltage in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.1.

3. Perform a static contact/pole-resistance test.

*4. Perform insulation-resistance tests on all control wiring with respect to ground. The applied potential shall be 500 volts dc for 300-volt rated cable and 1000 volts dc for 600-volt rated cable. Test duration shall be one minute. For units with solid-state components or control devices that cannot tolerate the applied voltage, follow manufacturer's recommendation.

5. Determine long-time pickup and delay by primary current injection.

6. Determine short-time pickup and delay by primary current injection.

7. Determine ground-fault pickup and delay by primary current injection.

8. Determine instantaneous pickup current by primary current injection.

*9. Test functions of the trip unit by means of secondary injection.

10. Perform minimum pickup voltage test on shunt trip and close coils in accordance with manufacturer's published data. In the absence of manufacturer's published data, refer to Table 100.20.

11. Verify correct operation of auxiliary features such as trip and pickup indicators, zone interlocking, electrical close and trip operation, trip-free, antipump function, and trip unit battery condition.

12. Reset all trip logs and indicators.

13. Verify operation of charging mechanism.

* Optional

7.6.1.2 Circuit Breakers, Air, Low-Voltage Power (*continued*)

C. Test Values – Visual and Mechanical

1. Compare bolted connection resistance values to values of similar connections. Investigate values which deviate from those of similar bolted connections by more than 50 percent of the lowest value. (7.6.1.2.A.10.1)

2. Bolt-torque levels should be in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.12. (7.6.1.2.A.10.2)

3. Results of the thermographic survey shall be in accordance with Section 9. (7.6.1.2.A.10.3)

4. Settings shall comply with coordination study recommendations. (7.6.1.2.A.15)

5. Operations counter should advance one digit per close-open cycle. (7.6.1.2.A.16)

D. Test Values – Electrical

1. Compare bolted connection resistance values to values of similar connections. Investigate values which deviate from those of similar bolted connections by more than 50 percent of the lowest value.

2. Insulation-resistance values of breakers should be in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.1. Values of insulation resistance less than this table or manufacturer's recommendations should be investigated.

3. Microhm or dc millivolt drop values should not exceed the high levels of the normal range as indicated in the manufacturer's published data. If manufacturer's data is not available, investigate values that deviate from adjacent poles or similar breakers by more than 50 percent of the lowest value.

4. Insulation-resistance values of control wiring should be comparable to previously obtained results but not less than two megohms.

5. Long-time pickup values should be as specified, and the trip characteristic shall not exceed manufacturer's published time-current characteristic tolerance band.

6. Short-time pickup values should be as specified, and the trip characteristic should not exceed manufacturer's published time-current tolerance band.

7. Ground fault pickup values should be as specified, and the trip characteristic should not exceed manufacturer's published time-current tolerance band.

8. Instantaneous pickup values should be within the tolerances of manufacturer's published data.

* Optional

7.6.1.2 Circuit Breakers, Air, Low-Voltage Power (*continued*)

9. Pickup values and trip characteristic should be as specified and within manufacturer's published tolerances.

10. Minimum pickup voltage on shunt trip and close coils should be in accordance with manufacturer's published data. In the absence of manufacturer's published data, refer to Table 100.20.

11. Auxiliary features should operate in accordance with manufacturer's published data.

12. Trip logs and indicators are reset.

13. The charging mechanism should operate in accordance with manufacturer's published data.

* Optional

7.6.1.3 Circuit Breakers, Air, Medium-Voltage

A. Visual and Mechanical Inspection

1. Inspect physical and mechanical condition.

2. Inspect anchorage, alignment, and grounding.

3. Verify that all maintenance devices are available for servicing and operating the breaker.

*4. Perform operator analysis (first-trip) test.

*5. Prior to cleaning the unit, perform as-found tests.

6. Clean the unit.

7. Inspect arc chutes.

8. Inspect moving and stationary contacts for condition, wear, and alignment.

9. If recommended by manufacturer, slow close/open breaker and check for binding, friction, contact alignment, contact sequence, and penetration. Verify that contact sequence is in accordance with manufacturer's published data. In the absence of manufacturer's published data, refer to IEEE C37.04.

10. Perform mechanical maintenance and tests on the operating mechanism in accordance with manufacturer's published data.

11. Inspect bolted electrical connections for high resistance using one or more of the following:

    1. Use of a low-resistance ohmmeter in accordance with Section 7.6.1.3.B.1.

    2. Verify tightness of accessible bolted electrical connections by calibrated torque-wrench method in accordance with manufacturer's published data or Table 100.12.

    3. Perform a thermographic survey in accordance with Section 9.

12. Verify cell fit and element alignment.

13. Verify racking mechanism operation.

14. Inspect puffer operation.

15. Use appropriate lubrication on moving current-carrying parts and on moving and sliding surfaces.

* Optional

7.6.1.3 Circuit Breakers, Air, Medium-Voltage (*continued*)

16. Perform contact timing test.

*17. Perform mechanism-motion analysis.

*18. Perform trip/close coil current signature analysis.

19. Perform as-left tests.

20. Record as-found and as-left operation-counter readings.

B. Electrical Tests

1. Perform resistance measurements through bolted connections with a low-resistance ohmmeter. See Section 7.6.1.3.A.11.1.

2. Perform insulation-resistance tests for one minute on each pole, phase-to-phase and phase-to-ground with the circuit breaker closed, and across each open pole. Apply voltage in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.1.

*3. Perform insulation-resistance tests on all control wiring with respect to ground. The applied potential shall be 500 volts dc for 300-volt rated cable and 1000 volts dc for 600-volt rated cable. Test duration shall be one minute. For units with solid-state components or control devices that cannot tolerate the applied voltage, follow manufacturer's recommendation.

4. Perform a static contact/pole-resistance test.

5. With the breaker in a test position, perform the following tests:

    1. Trip and close breaker with the control switch.

    2. Trip breaker by operating each of its protective relays.

    3. Verify mechanism charge, trip-free, and antipump functions.

*6. Perform minimum pickup voltage tests on trip and close coils in accordance with manufacturer's published data. In the absence of manufacturer's published data refer to Table 100.20.

*7. Perform power-factor or dissipation-factor tests with breaker in both the open and closed positions.

*8. Perform power-factor or dissipation-factor tests on each bushing equipped with a power-factor/ capacitance tap. In the absence of a power-factor/ capacitance tap, perform hot-collar

* Optional

7.6.1.3 Circuit Breakers, Air, Medium-Voltage (*continued*)

tests. These tests shall be in accordance with the test equipment manufacturer's published data.

*9. Perform a dielectric withstand voltage test on each phase with the circuit breaker closed and the poles not under test grounded. Test voltage should be in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.19.

10. Verify blowout coil circuit continuity.

11. Verify operation of heaters.

*12. Test instrument transformers in accordance with Section 7.10.

C. Test Values – Visual and Mechanical

1. Compare first-trip operation time and trip-coil current waveform to manufacturer's published data. In the absence of manufacturer's published data, compare first-trip operation time and trip-coil current waveform to previously obtained results. (7.6.1.3.A.4)
2. Compare bolted connection resistance values to values of similar connections. Investigate values which deviate from those of similar bolted connections by more than 50 percent of the lowest value. (7.6.1.3.A.11.1)
3. Bolt-torque levels should be in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.12. (7.6.1.3.A.11.2)
4. Results of the thermographic survey shall be in accordance with Section 9. (7.6.1.3.A.11.3)
5. Contact timing values shall be in accordance with manufacturer's published data and previous test data. (7.6.1.3.A.16)
6. Travel and velocity values shall be in accordance with manufacturer's published data and previous test data. (7.6.1.3.A.17)
7. Trip/close coil current values shall be in accordance with manufacturer's published data and previous test data. (7.6.1.3.A.18)
8. Operations counter should advance one digit per close-open cycle. (7.6.1.2.A.20)

D. Test Values – Electrical

1. Compare bolted connection resistance values to values of similar connections. Investigate values which deviate from those of similar bolted connections by more than 50 percent of the lowest value.

* Optional

7.6.1.3 Circuit Breakers, Air, Medium-Voltage (*continued*)

2. Insulation-resistance values of circuit breakers should be in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.1. Values of insulation resistance less than this table or manufacturer's recommendations should be investigated.

3. Insulation-resistance values of control wiring should be comparable to previously obtained results but not less than two megohms.

4. Microhm or dc millivolt drop values shall not exceed the high levels of the normal range as indicated in the manufacturer's published data. If manufacturer's data is not available, investigate values that deviate from adjacent poles or similar breakers by more than 50 percent of the lowest value.

5. Breaker mechanism charge, close, open, trip, trip-free, and antipump features shall function as designed.

6. Minimum pickup for trip and close coils shall be in accordance with manufacturer's published data. In the absence of manufacturer's data, refer to Table 100.20.

7. Power-factor or dissipation-factor values shall be compared with previous test results of similar breakers or manufacturer's published data.

8. Power-factor or dissipation-factor and capacitance values should be within ten percent of nameplate rating for bushings. Hot collar tests are evaluated on a milliampere/milliwatt loss basis, and the results should be compared to values of similar bushings.

9. If no evidence of distress or insulation failure is observed by the end of the total time of voltage application during the dielectric withstand voltage test, the circuit breaker is considered to have passed the test.

10. The blowout coil circuit should exhibit continuity.

11. Heaters should be operational.

12. The results of instrument transformer tests shall be in accordance with Section 7.10.

7.6.2 Circuit Breakers, Oil, Medium- and High-Voltage

A. Visual and Mechanical Inspection

1. Inspect physical and mechanical condition.

2. Inspect anchorage, alignment, and grounding.

3. Verify that maintenance devices are available for servicing and operating the breaker.

*4. Perform operator analysis (first-trip) test.

5. Verify correct oil level in tanks and bushings.

6. Verify that breather vents are clear.

*7. Prior to cleaning the unit, perform as-found tests.

8. Clean the unit.

9. Inspect and service hydraulic or pneumatic system and/or air compressor in accordance with manufacturer's published data.

10. Test alarms, pressure switches, and limit switches for pneumatic and/or hydraulic operators as recommended by the manufacturer.

11. Perform all maintenance and tests on the operating mechanism in accordance with manufacturer's published data.

*12. Perform internal inspection:

    1. Remove oil. Lower tanks or remove manhole covers as necessary. Inspect bottom of tank for broken parts and debris and clean carbon residue from tank.

    2. Inspect lift rod and toggle assemblies, contacts, interrupters, bumpers, dashpots, bushing current transformers, tank liners, and gaskets.

    3. If recommended by manufacturer, slow close/open breaker and check for binding, friction, contact alignment, and penetration. Verify that contact sequence is in accordance with manufacturer's published data. In the absence of manufacturer's published data, refer to IEEE C37.04.

    4. Refill tank(s) with new or filtered oil to correct levels.

* Optional

7.6.2 Circuit Breakers, Oil, Medium- and High-Voltage (*continued*)

13. Inspect all bolted electrical connections for high resistance using one or more of the following:

    1. Use of a low-resistance ohmmeter in accordance with Section 7.6.2.B.1.

    2. Verify tightness of accessible bolted electrical connections by calibrated torque-wrench method in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.12.

    3. Perform a thermographic survey in accordance with Section 9.

14. Verify cell fit and element alignment.

15. Verify racking mechanism operation.

16. Perform contact timing test.

17. Perform mechanism-motion analysis.

*18. Perform trip/close coil current signature analysis.

19. Use appropriate lubrication on moving current-carrying parts and on moving and sliding surfaces.

20. Perform as-left tests.

21. Record as-found and as-left operation counter readings.

B. Electrical Tests

1. Perform resistance measurements through all bolted connections with a low-resistance ohmmeter in accordance with Section 7.6.2.A.13.1.

2. Perform insulation-resistance tests for one minute on each pole, phase-to-phase and phase-to-ground with circuit breaker closed, and across each open pole. Apply voltage in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.1.

3. Perform a static contact/pole-resistance test.

*4. Perform a dynamic contact/pole resistance test.

*5. Perform insulation-resistance tests on all control wiring with respect to ground. The applied potential shall be 500 volts dc for 300-volt rated cable and 1000 volts dc for 600-volt rated

* Optional

7.6.2 Circuit Breakers, Oil, Medium- and High-Voltage (*continued*)

cable. Test duration shall be one minute. For units with solid-state components or for control devices that cannot tolerate the voltage, follow manufacturer's recommendation.

6. Remove a sample of insulating liquid in accordance with ASTM D 923. The sample shall be tested in accordance with the referenced standard.

   1. Dielectric breakdown voltage: ASTM D877
   2. Color: ASTM D1500
   *3. Power factor: ASTM D924
   *4. Interfacial tension: ASTM D971
   *5. Visual condition: ASTM D1524
   6. Neutralization number (acidity): ASTM D974
   7. Water content: ASTM D1533

7. With breaker in a test position, perform the following tests:

   1. Trip and close breaker with the control switch.
   2. Trip breaker by operating each of its protective relays.
   3. Verify trip-free and antipump functions.
   *4. Perform minimum pickup voltage tests on trip and close coils in accordance with manufacturer's published data. In the absence of manufacturer's published data, refer to Table 100.20.

8. Perform power-factor or dissipation-factor tests on each pole with the breaker open and each phase with the breaker closed. Determine tank loss index.

9. Perform power-factor or dissipation-factor tests on each bushing equipped with a power-factor/ capacitance tap. In the absence of a power-factor/ capacitance tap, perform hot-collar tests. These tests shall be in accordance with the test equipment manufacturer's published data.

*10. Perform a dielectric withstand voltage test on each phase with the circuit breaker closed and the poles not under test grounded. Test voltage should be in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.19.

11. Verify operation of heaters.

* Optional

7.6.2 Circuit Breakers, Oil, Medium- and High-Voltage (*continued*)

*12. Test instrument transformers in accordance Section 7.10

C. Test Values – Visual and Mechanical

1. Compare first-trip operation time and trip-coil current waveform to manufacturer's published data. In the absence of manufacturer's published data, compare first-trip operation time and trip-coil current waveform to previously obtained results. (7.6.2.A.4)

2. Settings for alarm, pressure, and limit switches should be in accordance with manufacturer's published data. (7.6.2.A.10)

3. Compare bolted connection resistance values to values of similar connections. Investigate values which deviate from those of similar bolted connections by more than 50 percent of the lowest value. (7.6.2.A.13.1)

4. Bolt-torque levels should be in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.12. (7.6.2.A.13.2)

5. Results of the thermographic survey shall be in accordance with Section 9. (7.6.2.A.13.3)

6. Contact timing values shall be in accordance with manufacturer's published data and previous test data. (7.6.2.A.16)

7. Travel and velocity values shall be in accordance with manufacturer's published data and previous test data. (7.6.2.A.17)

8. Trip/close coil current values shall be in accordance with manufacturer's published data and previous test data. (7.6.2.A.18)

9. Operations counter should advance one digit per close-open cycle. (7.6.2.A.21)

D. Test Values – Electrical

1. Compare bolted connection resistance values to values of similar connections. Investigate values which deviate from those of similar bolted connections by more than 50 percent of the lowest value.

2. Insulation-resistance values of circuit breakers should be in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.1. Values of insulation resistance less than this table or manufacturer's recommendations should be investigated.

3. Microhm or dc millivolt drop values should not exceed the high levels of the normal range as indicated in the manufacturer's published data. If manufacturer's data is not available,

* Optional

7.6.2 Circuit Breakers, Oil, Medium- and High-Voltage (*continued*)

investigate values that deviate from adjacent poles or similar breakers by more than 50 percent of the lowest value.

4. Dynamic contact resistance values shall be in accordance with manufacturer's published data.

5. Insulation-resistance values of control wiring should be comparable to previously obtained results but not less than two megohms.

6. Insulating liquid test results should be in accordance with Table 100.4.

7. Open, close, trip-free, and antipump features should function in accordance with manufacturer's design. Minimum pickup for trip and close coils should conform to manufacturer's published data. In the absence of manufacturer's published data, refer to Table 100.20.

8. Power-factor or dissipation-factor values and tank loss index shall be compared to manufacturer's published data. In the absence of manufacturer's published data, the comparison shall be made to test data from similar breakers or data from test equipment manufacturers.

9. Power-factor or dissipation-factor and capacitance values should be within ten percent of nameplate rating for bushings. Hot collar tests are evaluated on a milliampere/milliwatt loss basis, and the results should be compared to values of similar bushings.

10. If no evidence of distress or insulation failure is observed by the end of the total time of voltage application during the dielectric withstand voltage test, the test specimen is considered to have passed the test.

11. Heaters should be operational.

12. Results of electrical tests on instrument transformers should be in accordance with Section 7.10.

* Optional

7.6.3 Circuit Breakers, Vacuum, Medium-Voltage

A. Visual and Mechanical Inspection

1. Inspect physical and mechanical condition.

2. Inspect anchorage, alignment, and grounding.

3. Verify that all maintenance devices are available for servicing and operating the breaker.

*4. Perform operator analysis (first-trip) test.

*5. Prior to cleaning the unit, perform as-found tests.

6. Clean the unit.

7. Inspect vacuum bottle assemblies.

8. Measure critical distances such as contact gap as recommended by the manufacturer.

9. If recommended by the manufacturer, slow close/open the breaker and check for binding, friction, contact alignment, contact sequence, and penetration.

10. Perform all mechanical maintenance and tests on the operating mechanism in accordance with manufacturer’s published data.

11. Inspect bolted electrical connections for high resistance using one or more of the following:

    1. Use of a low-resistance ohmmeter in accordance with Section 7.6.3.B.1.

    2. Verify tightness of accessible bolted electrical connections by calibrated torque-wrench method in accordance with manufacturer’s published data. In the absence of manufacturer’s published data, use Table 100.12.

    3. Perform a thermographic survey in accordance with Section 9.

12. Verify cell fit and element alignment.

13. Verify racking mechanism operation.

14. Inspect vacuum bellows operation.

15. Use appropriate lubrication on moving current-carrying parts and on moving and sliding surfaces.

* Optional

7.6.3 Circuit Breakers, Vacuum, Medium-Voltage (*continued*)

16. Perform contact timing test.

*17. Perform mechanism-motion analysis.

*18. Perform trip/close coil current signature analysis.

19. Perform as-left tests.

20. Record as-found and as-left operation counter readings.

B. Electrical Tests

1. Perform resistance measurements through bolted connections with a low-resistance ohmmeter in accordance with Section 7.6.3.A.11.1.

2. Perform insulation-resistance tests for one minute on each pole, phase-to-phase and phase-to-ground with circuit breaker closed and across each pole with the breaker open. Apply voltage in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.1.

*3. Perform insulation-resistance tests on all control wiring with respect to ground. The applied potential shall be 500 volts dc for 300-volt rated cable and 1000 volts dc for 600-volt rated cable. Test duration shall be one minute. For units with solid-state components or control devices that cannot tolerate the applied voltage, follow manufacturer's recommendation.

4. Perform a static contact/pole-resistance test.

5. With breaker in a test position, perform the following tests:

    1. Trip and close breaker with the control switch.

    2. Trip breaker by operating each of its protective relays.

    3. Verify mechanism charge, trip-free, and antipump functions.

*6. Perform minimum pickup voltage tests on trip and close coils in accordance with manufacturer's published data. In the absence of manufacturer's published data refer to Table 100.20.

7. Perform a vacuum bottle integrity (dielectric withstand voltage) test across each vacuum bottle with the breaker in the open position in strict accordance with manufacturer's published data.

*8. Perform magnetron atmospheric condition (MAC) test on each vacuum interrupter.

* Optional

7.6.3 Circuit Breakers, Vacuum, Medium-Voltage (*continued*)

*9. Perform a dielectric withstand voltage test on each phase with the circuit breaker closed and the poles not under test grounded. Test voltage should be in accordance with manufacturer’s published data. In the absence of manufacturer’s published data, use Table 100.19.

10. Verify operation of heaters.

*11. Test instrument transformers in accordance with Section 7.10.

C. Test Values – Visual and Mechanical

1. Compare first-trip operation time and trip-coil current waveform to manufacturer’s published data. In the absence of manufacturer's published data, compare first-trip operation time and trip-coil current waveform to previously obtained results. (7.6.3.A.4)

2. Mechanical operation and contact alignment should be in accordance with manufacturer’s published data. (7.6.3.A.9)

3. Compare bolted connection resistance values to values of similar connections. Investigate values which deviate from those of similar bolted connections by more than 50 percent of the lowest value. (7.6.3.A.11.1)

4. Bolt-torque levels should be in accordance with manufacturer’s published data. In the absence of manufacturer’s published data, use Table 100.12. (7.6.3.A.11.2)

5. Results of the thermographic survey shall be in accordance with Section 9. (7.6.3.A.11.3)

6. Contact timing values shall be in accordance with manufacturer’s published data and previous test data. (7.6.3.A.16)

7. Travel and velocity values shall be in accordance with manufacturer’s published data and previous test data. (7.6.3.A.17)

8. Trip/close coil current values shall be in accordance with manufacturer’s published data and previous test data. (7.6.3.A.18)

9. Operation counter should advance one digit per close-open cycle. (7.6.3.A.20)

D. Test Values – Electrical

1. Compare bolted connection resistance values to values of similar connections. Investigate values which deviate from those of similar bolted connections by more than 50 percent of the lowest value.

2. Insulation-resistance values of circuit breakers should be in accordance with manufacturer’s published data. In the absence of manufacturer’s published data, use Table 100.1. Values of

* Optional

7.6.3 Circuit Breakers, Vacuum, Medium-Voltage (*continued*)

insulation resistance less than this table or manufacturer's recommendations should be investigated.

3. Insulation-resistance values of control wiring should be comparable to previously obtained results but not less than two megohms.

4. Microhm or dc millivolt drop values shall not exceed the high levels of the normal range as indicated in the manufacturer's published data. If manufacturer's data is not available, investigate values that deviate from adjacent poles or similar breakers by more than 50 percent of the lowest value.

5. Breaker mechanism charge, close, open, trip, trip-free, and antipump features shall function as designed.

6. Minimum pickup for trip and close coils shall be in accordance with manufacturer's published data. In the absence of manufacturer's data, refer to Table 100.20.

7. If no evidence of distress or insulation failure is observed by the end of the total time of voltage application during the vacuum bottle integrity test, the test specimen is considered to have passed the test.

8. Evaluate each vacuum interrupter in accordance with test equipment manufacturer's instructions.

9. If no evidence of distress or insulation failure is observed by the end of the total time of voltage application during the dielectric withstand voltage test, the test specimen is considered to have passed the test.

10. Heaters should be operational.

11. Results of instrument transformer tests shall be in accordance with Section 7.10.

7. INSPECTION AND TEST PROCEDURES

7.6.4 Circuit Breakers, SF_6

A. Visual and Mechanical Inspection

1. Inspect physical and mechanical condition.

2. Inspect anchorage, alignment, and grounding.

3. Verify that all maintenance devices are available for servicing and operating the breaker.

*4. Perform operator analysis (first-trip) test.

*5. Prior to cleaning the unit, perform as-found tests.

6. Clean the unit.

*7. When provisions are made for sampling, remove a sample of SF_6 gas and test in accordance with Table 100.13. Do not break seal or distort sealed-for-life interrupters.

8. Inspect and service operating mechanism and/or hydraulic or pneumatic system and SF_6 gas-insulated system in accordance with manufacturer's published data.

9. Test for SF_6 gas leaks in accordance with manufacturer's published data.

10. Test alarms, pressure switches, and limit switches for pneumatic and/or hydraulic operators and SF_6 gas pressure in accordance with manufacturer's published data.

11. If recommended by manufacturer, slow close/open breaker and check for binding, friction, contact alignment, and penetration. Verify that contact sequence is in accordance with manufacturer's published data. In the absence of manufacturer's published data, refer to IEEE C37.04.

12. Inspect all bolted electrical connections for high resistance using one or more of the following:

    1. Use of a low-resistance ohmmeter in accordance with Section 7.6.4.B.1.

    2. Verify tightness of accessible bolted electrical connections by calibrated torque-wrench method in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.12.

    3. Perform a thermographic survey in accordance with Section 9.

13. Verify cell fit and element alignment.

* Optional

7.6.4 Circuit Breakers, SF_6 (*continued)*

14. Verify racking mechanism operation.

15. Use appropriate lubrication on moving current-carrying parts and on moving and sliding surfaces.

16. Perform contact timing test.

*17. Perform mechanism-motion analysis.

*18. Perform trip/close coil current signature analysis.

19. Perform as-left tests.

20. Record as-found and as-left operation counter readings.

B. Electrical Tests

1. Perform resistance measurements through all bolted connections with a low-resistance ohmmeter in accordance with Section 7.6.4.A.13.1.

2. Perform insulation-resistance tests for one minute on each pole, phase-to-phase and phase-to-ground with breaker closed, and across each open pole. For single-tank breakers, perform insulation resistance tests from pole-to-pole. Apply voltage in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.1.

3. Perform a static contact/pole-resistance test.

*4. Perform a dynamic contact/pole resistance test.

*5. Perform insulation-resistance tests on all control wiring with respect to ground. The applied potential shall be 500 volts dc for 300-volt rated cable and 1000 volts dc for 600-volt rated cable. Test duration shall be one minute. For units with solid-state components or for control devices that cannot tolerate the voltage, follow manufacturer's recommendation.

6. With breaker in a test position, perform the following tests:

    1. Trip and close breaker with the control switch.

    2. Trip breaker by operating each of its protective relays.

    3. Verify trip-free and antipump functions.

    *4. Perform minimum pickup voltage tests on trip and close coils in accordance with Table 100.20.

* Optional

7.6.4 Circuit Breakers, SF_6 (*continued)*

7. Perform power-factor or dissipation-factor tests on each pole with the breaker open and on each phase with breaker closed.

8. Perform power-factor or dissipation-factor tests on each bushing equipped with a power-factor/capacitance tap. In the absence of a power-factor/ capacitance tap, perform hot-collar tests. These tests shall be in accordance with the test equipment manufacturer’s published data.

*9. Perform a dielectric withstand voltage test in accordance with manufacturer’s published data.

10. Verify operation of heaters.

*11. Test instrument transformers in accordance with Section 7.10.

C. Test Values – Visual and Mechanical

1. Compare first-trip operation time and trip-coil current waveform to manufacturer’s published data. In the absence of manufacturer's published data, compare first-trip operation time and trip-coil current waveform to previously obtained results. (7.6.4.A.4)

2. SF_6 gas should have values in accordance with Table 100.13. (7.6.4.A.7)

3. Results of the SF_6 gas leak test should confirm that no SF_6 gas leak exists. (7.6.4.A.9)

4. Settings for alarm, pressure, and limit switches should be in accordance with manufacturer’s published data. (7.6.4.A.10)

5. Compare bolted connection resistance values to values of similar connections. Investigate values which deviate from those of similar bolted connections by more than 50 percent of the lowest value. (7.6.4.A.12.1)

6. Bolt-torque levels should be in accordance with manufacturer’s published data. In the absence of manufacturer’s published data, use Table 100.12. (7.6.4.A.12.2)

7. Results of the thermographic survey shall be in accordance with Section 9. (7.6.4.A.12.3)

8. Contact timing values shall be in accordance with manufacturer’s published data and previous test data. (7.6.4.A.16)

9. Travel and velocity values shall be in accordance with manufacturer’s published data and previous test data. (7.6.4.A.17)

10. Trip/close coil current values shall be in accordance with manufacturer’s published data and previous test data. (7.6.4.A.18)

* Optional

7.6.4 Circuit Breakers, SF_6 (*continued)*

11. Operations counter should advance one digit per close-open cycle. (7.6.4.A.20)

D. Test Values – Electrical

1. Compare bolted connection resistance values to values of similar connections. Investigate values which deviate from those of similar bolted connections by more than 50 percent of the lowest value.

2. Insulation-resistance values of circuit breakers should be in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.1.Values of insulation resistance less than this table or manufacturer's recommendations should be investigated.

3. Microhm or dc millivolt drop values should not exceed the high levels of the normal range as indicated in the manufacturer's published data. If manufacturer's data is not available, investigate values that deviate from adjacent poles or similar breakers by more than 50 percent of the lowest value.

4. Dynamic contact resistance values shall be in accordance with manufacturer's published data.

5. Insulation-resistance values of control wiring should be comparable to previously obtained results but not less than two megohms.

6. Open, close, trip-free, and antipump features should function in accordance with the manufacturer's design. Minimum pickup for trip and close coils should conform to manufacturer's published data. In the absence of manufacturer's published data, refer to Table 100.20.

7. Power-factor or dissipation-factor values shall be compared to manufacturer's published data. In the absence of manufacturer's published data, the comparison shall be made to test data from similar breakers or data from test equipment manufacturers.

8. Power-factor or dissipation-factor and capacitance test values should be within ten percent of nameplate rating for bushings. Hot collar tests are evaluated on a milliampere/milliwatt loss basis, and the results should be compared to values of similar bushings.

9. If no evidence of distress or insulation failure is observed by the end of the total time of voltage application during the dielectric withstand voltage test, the test specimen is considered to have passed the test.

10. Heaters should be operational.

11. Results of electrical tests on instrument transformers should be in accordance with Section 7.10.

* Optional

7. INSPECTION AND TEST PROCEDURES

7.7 Circuit Switchers

A. Visual and Mechanical Inspection

1. Inspect physical and mechanical condition.

2. Inspect anchorage, alignment, and grounding.

*3. Prior to cleaning the unit, perform as-found tests.

4. Clean the unit.

5. Perform mechanical maintenance and tests on both the circuit switcher and its operating mechanism.

6. Inspect bolted electrical connections for high resistance using one or more of the following:

    1. Use of a low-resistance ohmmeter in accordance with Section 7.7.B.1.

    2. Verify tightness of accessible bolted electrical connections by calibrated torque-wrench method in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.12.

    3. Perform a thermographic survey in accordance with Section 9.

7. Verify operation of SF_6 interrupters is in accordance with manufacturer's published data.

8. Verify SF_6 pressure is in accordance with manufacturer's published data.

9. Verify operation of isolating switch is in accordance with manufacturer's published data.

10. Test all interlocking systems for correct operation and sequencing.

11. Verify all indicating and control devices for correct separation.

12. Use appropriate lubrication on moving current-carrying parts and on moving and sliding surfaces.

13. Perform as-left tests.

14. Record as-found and as-left operation counter readings.

* Optional

7.7 Circuit Switchers (*continued*)

B. Electrical Tests

1. Perform resistance measurements through all connections with a low-resistance ohmmeter in accordance with Section 7.7.A.6.1.

2. Perform contact-resistance tests of interrupters and isolating switches.

3. Perform insulation-resistance tests on each pole, phase-to-ground. Apply voltage in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.1.

*4. Perform insulation-resistance tests on all control wiring with respect to ground. The applied potential shall be 500 volts dc for 300-volt rated cable and 1000 volts dc for 600-volt rated cable. Test duration shall be one minute. For units with solid-state components, follow manufacturer's recommendation.

*5. Perform minimum pickup voltage tests on trip and close coils in accordance with Table 100.20.

6. Verify correct operation of auxiliary features such as electrical close and trip operation, trip-free, and antipump function. Reset all trip logs and indicators.

7. Trip circuit switcher by operation of each protective device.

8. Verify correct operation of electrical trip of interrupters.

9. Perform a dielectric withstand voltage test in accordance with manufacturer's published data.

10. Verify operation of heaters.

C. Test Values – Visual and Mechanical

1. Compare bolted connection resistance values to values of similar connections. Investigate values which deviate from those of similar bolted connections by more than 50 percent of the lowest value. (7.7.A.6.1)

2. Bolt-torque levels should be in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.12. (7.7.A.6.2)

3. Results of the thermographic survey shall be in accordance with Section 9. (7.7.A.6.3)

4. SF_6 interrupters should operate in accordance with manufacturer's published data. (7.7.A.7)

5. SF_6 pressure should be in accordance with manufacturer's published data. (7.7.A.8)

* Optional

7.7 Circuit Switchers (*continued*)

6. Isolating switch should operate in accordance with manufacturer’s design. (7.7.A.9)

7. Interlocking systems should operate per system lockout design. (7.7.A.10)

8. Results of open and close operation from control devices should be in accordance with system design. (7.7.A.11)

9. Operation counter should advance one digit per close-open cycle. (7.7.A.14)

D. Test Values – Electrical

1. Compare bolted connection resistance values to values of similar connections. Investigate values which deviate from those of similar bolted connections by more than 50 percent of the lowest value.

2. Microhm or dc millivolt drop values should not exceed the high levels of the normal range as indicated in the manufacturer’s published data. If manufacturer’s data is not available, investigate values that deviate from adjacent poles or similar breakers by more than 50 percent of the lowest value.

3. Insulation-resistance values of circuit switchers should be in accordance with manufacturer’s published data. In the absence of manufacturer’s published data, use Table 100.1.Values of insulation resistance less than this table or manufacturer’s recommendations should be investigated.

4. Insulation-resistance values of control wiring should be comparable to previously obtained results but not less than two megohms.

5. Minimum pickup for trip and close coils should conform to manufacturer’s published data. In the absence of manufacturer’s published data, refer to Table 100.20.

6. Switcher mechanism charge, close, open, trip, trip-free, and antipump features shall function as designed.

7. Protective device operation should function in accordance with the manufacturer’s published data.

8. Electrical trip interrupters shall function as designed.

9. If no evidence of distress or insulation failure is observed by the end of the total time of voltage application during the dielectric withstand voltage test, the circuit switcher is considered to have passed the test.

10. Heaters should be operational.

* Optional

7. INSPECTION AND TEST PROCEDURES

7.8. Network Protectors

A. Visual and Mechanical Inspection

1. Inspect physical and mechanical condition.

2. Inspect anchorage, alignment, and grounding.

*3. Prior to cleaning the unit, perform as-found tests.

4. Clean the unit.

5. Inspect arc chutes.

6. Inspect moving and stationary contacts for condition, wear, and alignment.

7. Verify that maintenance devices are available for servicing and operating the network protector.

8. Verify that primary and secondary contact wipe and other dimensions vital to satisfactory operation of the breaker are correct.

9. Perform mechanical operator and contact alignment maintenance and tests on both the protector and its operating mechanism in accordance with manufacturer's published data.

10. Inspect bolted electrical connections for high resistance using one or more of the following:

    1. Use of a low-resistance ohmmeter in accordance with Section 7.8.B.1.

    2. Verify tightness of accessible bolted electrical connections by calibrated torque-wrench method in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.12.

    3. Perform a thermographic survey in accordance with Section 9.

11. Verify cell fit and element alignment.

12. Verify racking mechanism operation.

13. Use appropriate lubrication on moving current-carrying parts and on moving and sliding surfaces.

14. Perform as-left tests.

* Optional

7.8. Network Protectors (*continued*)

15. Record the as-found and as-left operations counter readings.

16. Perform a leak test on submersible enclosures in accordance with manufacturer's published data.

B. Electrical Tests

1. Perform resistance measurements through bolted connections with a low-resistance ohmmeter in accordance with Section 7.8.A.10.1.

2. Perform insulation-resistance tests for one minute on each pole, phase-to-phase and phase-to-ground with network protector closed, and across each open pole. Apply voltage in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.1.

3. Perform a contact/pole-resistance test.

*4. Perform insulation-resistance test on all control wiring with respect to ground. The applied potential shall be 500 volts dc for 300-volt rated cable and 1000 volts dc for 600-volt rated cable. Test duration shall be one minute. For units with solid-state components or for control devices that cannot tolerate the voltage, follow manufacturer's recommendation.

*5. Verify current transformer ratios in accordance with Section 7.10.

6. Measure the resistance of each network protector power fuse.

7. Measure minimum pickup voltage of the motor control relay.

*8. Verify that the motor can charge the closing mechanism at the minimum voltage specified by the manufacturer.

9. Measure minimum pickup voltage of the trip actuator. Verify that the actuator resets correctly.

10. Calibrate the network protector relays in accordance with Section 7.9.

11. Perform operational tests.

    1. Verify correct operation of all mechanical and electrical interlocks.

    2. Verify trip-free operation.

    3. Verify correct operation of the auto-open-close control handle.

    4. Verify the protector will close with voltage on the transformer side only.

* Optional

7.8. Network Protectors (*continued*)

5. Verify the protector will open when the source feeder breaker is opened.

C. Test Values – Visual and Mechanical

1. Compare bolted connection resistance values to values of similar connections. Investigate values which deviate from those of similar bolted connections by more than 50 percent of the lowest value. (7.8.A.10.1)

2. Bolt-torque levels should be in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.12. (7.8.A.10.2)

3. Results of the thermographic survey shall be in accordance with Section 9. (7.8.A.10.3)

4. Operations counter should advance one digit per close-open cycle. (7.8.A.15)

D. Test Values – Electrical

1. Compare bolted connection resistance values to values of similar connections. Investigate values which deviate from those of similar bolted connections by more than 50 percent of the lowest value.

2. Insulation resistance of the network protector should be in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.1. Values of insulation resistance less than this table or manufacturer's recommendations should be investigated.

3. Microhm or dc millivolt drop values should not exceed the high levels of the normal range as indicated in the manufacturer's published data. If manufacturer's data is not available, investigate values that deviate from adjacent poles or similar protectors by more than 50 percent of the lowest value.

4. Insulation-resistance values of control wiring should be comparable to previously obtained results but not less than two megohms.

5. Results of current transformer ratios shall be in accordance with Section 7.10.

6. Investigate fuse resistance values that deviate from each other by more than 15 percent.

7. Minimum pickup voltage of the motor control relay should be in accordance with manufacturer's published data, but not more than 75 percent of rated control circuit voltage.

8. Minimum acceptable motor closing voltage should not exceed 75 percent of rated control circuit voltage.

7.8. Network Protectors (*continued*)

9. Trip actuator minimum pickup voltage should not exceed 75 percent of rated control circuit voltage.

10. Results of network protector relays should be in accordance with Section 7.9.

11. Network protector operation should be in accordance with design requirements.

* Optional

7. INSPECTION AND TEST PROCEDURES

7.9.1 Protective Relays, Electromechanical and Solid-State

A. Visual and Mechanical Inspection

1. Inspect relays and cases for physical damage.

*2. Prior to cleaning the unit, perform as-found tests.

3. Clean and inspect the unit.

    1. Relay Case

        1. Tighten case connections.

        2. Inspect cover for correct gasket seal.

        3. Clean cover glass.

        4. Inspect shorting hardware, connection paddles, and/or knife switches.

        5. Remove any foreign material from the case.

        6. Verify target reset.

    2. Relay

        1. Inspect relay for foreign material, particularly in disk slots of the damping and electromagnets.

        2. Verify disk clearance. Verify contact clearance and spring bias.

        3. Inspect spiral spring convolutions.

        4. Inspect disk and contacts for freedom of movement and correct travel.

        5. Verify tightness of mounting hardware and connections.

        6. Burnish contacts.

        7. Inspect bearings and pivots.

4. Verify that all settings are in accordance with coordination study or setting sheet supplied by owner.

* Optional

7.9.1 Protective Relays, Electromechanical and Solid-State (*continued)*

B. Electrical Tests

*1. Perform insulation-resistance test on each circuit-to-frame. Procedures for performing insulation-resistance tests on solid-state relays should be determined from the relay manufacturer's published data.

2. Test targets and indicators.

    1. Determine pickup and dropout of electromechanical targets.

    2. Verify operation of all light-emitting diode indicators.

    3. Set contrast for liquid-crystal display readouts.

3. Protection Elements (by ANSI device number)

    1. 2/62 Timing Relay

        1. Determine time delay.

        2. Verify operation of instantaneous contacts.

    2. 21 Distance Relay

        1. Determine maximum reach.

        2. Determine maximum torque angle and directional characteristic.

        3. Determine offset.

        *4. Plot impedance circle.

    3. 24 Volts/Hertz Relay

        1. Determine pickup frequency at rated voltage.

        2. Determine pickup frequency at a second voltage level.

        3. Determine time delay.

    4. 25 Sync Check Relay

        1. Determine closing zone at rated voltage.

        2. Determine maximum voltage differential and slip frequency that permits closing at zero degrees.

* Optional

7.9.1 Protective Relays, Electromechanical and Solid-State (*continued)*

3. Determine live line, live bus, dead line, and dead bus set points.

4. Determine time delay.

*5. Determine advanced closing angle.

6. Verify dead bus/live line, dead line/live bus and dead bus/dead line control functions.

5. 27 Undervoltage Relay

1. Determine pick up value of the setting

2. Determine time delay.

3. Determine the time delay at a second point on the timing curve for inverse time relays.

6. 32 Directional Power Relay

1. Determine minimum pickup at maximum torque angle.

2. Determine contact closing zone.

3. Determine maximum torque angle.

4. Determine time delay.

5. Verify the time delay at a second point on the timing curve for inverse time relays.

*6. Plot the operating characteristic.

7. 40 Loss of Field (Impedance) Relay

1. Determine maximum reach.

2. Determine maximum torque angle.

3. Determine offset.

*4. Plot impedance circle.

8. 46 Current Balance Relay

1. Determine pickup of each unit.

* Optional

7.9.1 Protective Relays, Electromechanical and Solid-State (*continued)*

2. Determine percent slope.

3. Determine time delay.

9. 46N Negative Sequence Current Relay

1. Determine negative sequence alarm level.

2. Determine negative sequence minimum trip level.

3. Determine maximum time delay.

4. Verify two points on the $(I_2)^2t$ curve.

10. 47 Phase Sequence or Phase Balance Voltage Relay

1. Determine positive sequence voltage to close the normally open contact.

2. Determine positive sequence voltage to open the normally closed contact (undervoltage trip).

3. Verify negative sequence trip.

4. Determine time delay to close the normally open contact with sudden application of 120 percent of pickup.

5. Determine time delay to close the normally closed contact upon removal of voltage when previously set to rated system voltage.

11. 49R Thermal Replica Relay

1. Determine time delay at 300 percent of setting.

2. Determine a second point on the operating curve.

*3. Determine minimum pickup.

12. 49T Temperature (RTD) Relay

1. Determine trip resistance.

2. Determine reset resistance.

13. 50 Instantaneous Overcurrent Relay

1. Determine pickup.

* Optional

7.9.1 Protective Relays, Electromechanical and Solid-State (*continued)*

2. Determine dropout.

*3. Determine time delay.

14. 50BF Breaker Failure

1. Determine current supervision pickup.

2. Determine time delays.

3. Test all used initiate inputs and all used outputs.

15. 51 Time Overcurrent

1. Determine minimum pickup.

2. Determine time delay at two points on the time current curve.

16. 55 Power Factor Relay

1. Determine tripping lead and lag angles.

2. Determine enable time delay.

3. Determine operate time delay.

17. 59 Overvoltage Relay

1. Determine overvoltage pickup.

2. Determine time delay to close the contact with sudden application of 120 percent of pickup.

18. 60 Voltage Balance Relay

1. Determine voltage difference to close the contacts with one source at rated voltage.

*2. Plot the operating curve for the relay.

19. 63 Transformer Sudden Pressure Relay

1. Determine rate-of-rise or the pickup level of suddenly applied pressure in accordance with manufacturer's published data.

2. Verify operation of the 63 FPX seal-in circuit.

* Optional

7.9.1 Protective Relays, Electromechanical and Solid-State (*continued)*

3. Verify trip circuit to remote operating device.

20. 64 Ground Detector Relay

1. Determine maximum impedance to ground causing relay pickup.

21. 67 Directional Overcurrent Relay

1. Determine directional unit minimum pickup at maximum torque angle.
2. Determine contact closing zone.

*3. Determine maximum torque angle.

*4. Plot operating characteristics.

5. Determine overcurrent unit pickup.
6. Determine overcurrent unit time delay at two points on the time current curve.

22. 79 Reclosing Relay

1. Determine time delay for each programmed reclosing interval.
2. Verify lockout for unsuccessful reclosing.
3. Determine reset time.

*4. Determine close pulse duration.

5. Verify instantaneous overcurrent lockout.

23. 81 Frequency Relay

1. Verify frequency set points.
2. Determine time delays.
3. Determine undervoltage cutoff.

24. 85 Pilot Wire Monitor

1. Determine overcurrent pickup.
2. Determine undercurrent pickup.

* Optional

3. Determine pilot wire ground pickup level.

25. 87 Differential

1. Determine operating unit pickup.

2. Determine the operation of each restraint unit.

3. Determine slope.

4. Determine harmonic restraint.

5. Determine instantaneous pickup.

*6. Plot operating characteristics for each restraint.

4. Control Verification

Verify that each of the relay contacts performs its intended function in the control scheme including breaker trip tests, close inhibit tests, 86 lockout tests, and alarm functions.

C. Test Values – Visual and Mechanical

1. Relay case

1. Case connections should be torqued in accordance with manufacturer's published data. (7.9.1.A.3.1.1)

2. Cover gasket should be intact and able to prevent foreign matter from entering the case. (7.9.1.A.3.1.2)

3. Cover glass should be clean. (7.9.1.A.3.1.3)

4. Connection paddles, and/or knife switches should be clean. (7.9.1.A.3.1.4)

5. Case should be free of foreign material. (7.9.1.A.3.1.5)

6. The target reset should be operational. (7.9.1.A.3.1.6)

2. Relay

1. Relay should be free of foreign material. (7.9.1.A.3.2.1)

2. Relay disc clearance, contact clearance, and spring bias should operate in accordance with manufacturer's published data. (7.9.1.A.3.2.2)

* Optional

7.9.1 Protective Relays, Electromechanical and Solid-State (*continued)*

3. Relay spiral spring should be concentric and should not show signs of overheating. (7.9.1.A.3.2.3)

4. Relay discs and contacts should have freedom of movement and correct travel distance in accordance with manufacturer's published data. (7.9.1.A.3.2.4)

5. Mounting hardware and connections shall be tightened to the manufacturer's recommended torque values. (7.9.1.A.3.2.5)

6. Contacts should be clean and make good contact. (7.9.1.A.3.2.6)

7. Bearings and pivots shall be clean and allow freedom of movement. (7.9.1.A.3.2.7)

3. As-left relay settings should match the most recent coordination and arc-flash study or engineered setting files. (7.9.1.A.4)

D. Test Values – Electrical

1. Insulation-resistance values should be in accordance with manufacturer's published data. Values of insulation resistance less than the manufacturer's recommendations should be investigated.

2. Targets and Indicators

   1. Pickup and dropout of electromechanical targets should be in accordance with manufacturer's published data.

   2. Light-emitting diodes should illuminate.

3. Operation of protection elements for devices listed in section 7.9.1.B, 1 through 25, should be calibrated using manufacturer's recommended tolerances unless critical test points are specified by the setting engineer.

4. Relay elements should operate relay contacts and operate or block end devices as per the design.

* Optional

7.9.2 Protective Relays, Microprocessor-Based

A. Visual and Mechanical Inspection

1. Record model number, style number, serial number, firmware revision, software revision, and rated control voltage.

*2. Download all events from the event recorder in filtered and unfiltered mode before performing any tests on the relay.

3. Download the sequence of events, maintenance data, and statistical data prior to testing the relay.

4. Verify operation of light-emitting diodes, display, and targets.

*5. Record passwords for all access levels.

6. Clean the front panel and remove foreign material from the case.

7. Check tightness of connections.

8. Verify that the frame is grounded in accordance with manufacturer's instructions.

9. Download settings and logic from the relay for the report and compare the settings to those specified in the coordination study.

10 Verify relay displays the correct date and time. Compare relay time to actual time and record the differential.

11. Check with owner for applicable firmware updates and product recalls.

12. Inspect, clean, and verify operation of shorting devices.

B. Electrical Tests

*1. Perform insulation-resistance tests from each circuit to the grounded frame in accordance with manufacturer's published data.

2. Apply voltage or current to all analog inputs and verify correct registration of the relay meter functions.

*3. Verify SCADA metering values at remote terminals.

*4. Protection Elements

Check functional operation of each element used in the protection scheme as described for electromechanical and solid-state relays in 7.9.1.B.3.

* Optional

7.9.2 Protective Relays, Microprocessor-Based (*continued*)

5. Control Verification

    1. Check operation of all active digital inputs.

    2. Check all relay outputs, preferably by operating the controlled device such as circuit breaker, auxiliary relay, or alarm.

    *3. Check all internal logic functions used in the protection scheme.

    4. For pilot schemes, perform protection system communication tests.

    5. Upon completion of testing, reset all min/max records and fault counters. Delete sequence-of-events records and all event records.

    6. Verify trip and close coil monitoring functions.

    *7. Verify setting change alarm to SCADA.

    *8. Verify relay SCADA communication and indications such as protection operate, protection fail, communication fail, fault recorder trigger.

    *9. Verify all communication links are operational.

C Test Values – Visual and Mechanical

1. As-left relay settings should match the most recent coordination and arc-flash study or engineered setting files. As-found relay settings that do not match the as-left settings should be reviewed. (7.9.2.A.9)

2. Relay should be clean and operational. (7.9.2.A.6)

3. Clear maintenance and statistical data. (7.9.2.A.3)

4. Light-emitting diodes, displays, and targets should illuminate. (7.9.2.A.4)

5. Downloaded settings and logic should agree with the most recent engineered setting files. (7.9.2.A.9)

6. Verify relay displays the correct date and time. (7.9.2.A.10)

D. Test Values – Electrical

1. Insulation-resistance values should be in accordance with manufacturer’s published data. Values of insulation resistance less that manufacturer’s recommendations should be investigated.

* Optional

7.9.2 Protective Relays, Microprocessor-Based (*continued*)

2. Voltage and current analog readings should be in accordance with manufacturer’s published tolerances.

3. SCADA metering values should function in accordance with the manufacturer’s published data.

4. Operation of protection elements for devices as listed in 7.9.1.B, items1 through 25, should be operational and within manufacturer’s recommended tolerances.

5. Control verification of inputs, outputs, and protection schemes, as listed in 7.9.2.B.5, items 1 through 9, should operate as per the design. Results should be within the manufacturer’s published tolerances.

* Optional

7. INSPECTION AND TEST PROCEDURES

7.10.1 Instrument Transformers, Current Transformers

A. Visual and Mechanical Inspection

1. Inspect physical and mechanical condition.

*2. Prior to cleaning the unit, perform as-found tests.

3. Clean the unit.

4. Inspect bolted electrical connections for high resistance using one or more of the following:

    1. Use of a low-resistance ohmmeter in accordance with Section 7.10.1.B.1.

    2. Verify tightness of accessible bolted electrical connections by calibrated torque-wrench method in accordance with manufacturer's published data or Table 100.12.

    3. Perform a thermographic survey in accordance with Section 9.

5. Verify that all required grounding and shorting connections provide contact.

6. Use appropriate lubrication on moving current-carrying parts and on moving and sliding surfaces.

7. Perform as-left tests.

B. Electrical Tests

1. Perform resistance measurements through bolted connections with a low-resistance ohmmeter in accordance with Section 7.10.1.A.4.1.

2. Perform insulation-resistance test of each current transformer and wiring-to-ground at 1000 volts dc for one minute. For units with solid-state components that cannot tolerate the applied voltage, follow manufacturer's recommendations.

*3. Perform a polarity test of each current transformer in accordance with IEEE C57.13.1.

*4. Perform a ratio-verification test using the voltage or current method in accordance with IEEE C57.13.1.

*5. Perform an excitation test on transformers used for relaying applications in accordance with IEEE C57.13.1.

*6. Measure current circuit burdens at transformer terminals.

* Optional

7.10.1 Instrument Transformers, Current Transformers (*continued*)

7. Perform insulation-resistance tests on the primary winding with the secondary grounded. Test voltages shall be in accordance with Table 100.5.

8. Perform dielectric withstand voltage tests on the primary winding with the secondary grounded. Test voltages shall be in accordance with Table100.9.

*9. Perform power-factor or dissipation-factor tests in accordance with manufacturer's published data.

10. Verify that current circuits are grounded and have only one grounding point in accordance with IEEE C57.13.3.

C. Test Values – Visual and Mechanical

1. Compare bolted connection resistance values to values of similar connections. Investigate values which deviate from those of similar bolted connections by more than 50 percent of the lowest value. (7.10.1.A.4.1)

2. Bolt-torque levels should be in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.12. (7.10.1.A.4.2)

3. Results of the thermographic survey shall be in accordance with Section 9. (7.10.1.A.4.3)

D. Test Values – Electrical

1. Compare bolted connection resistance values to values of similar connections. Investigate values which deviate from those of similar bolted connections by more than 50 percent of the lowest value.

2. Insulation-resistance values of instrument transformers should not be less than values shown in Table 100.5.

3. Polarity results should agree with transformer markings.

4. Ratio errors should not be greater than values shown in IEEE C57.13.

5. Excitation results should match the curve supplied by the manufacturer or be in accordance with IEEE C57.13.1.

6. Compare measured burdens to instrument transformer ratings.

7. Insulation-resistance values of instrument transformers should not be less than values shown in Table 100.5.

* Optional

7. INSPECTION AND TEST PROCEDURES

7.10.1 Instrument Transformers, Current Transformers (*continued*)

8. If no evidence of distress or insulation failure is observed by the end of the total time of voltage application during the dielectric withstand voltage test, the primary winding is considered to have passed the test.

9. Power-factor or dissipation-factor values shall be in accordance with manufacturer's published data. In the absence of manufacturer's published data, use test equipment manufacturer's published data.

10. Test results should indicate that the circuits have only one grounding point.

* Optional

7.10.2 Instrument Transformers, Voltage Transformers

A. Visual and Mechanical Inspection

1. Inspect physical and mechanical condition.

*2. Prior to cleaning the unit, perform as-found tests.

3. Clean the unit.

4. Inspect bolted electrical connections for high resistance using one or more of the following:

    1. Use of a low-resistance ohmmeter in accordance with Section 7.10.2.B.1.

    2. Verify tightness of accessible bolted electrical connections by calibrated torque-wrench method in accordance with manufacturer's published data or Table 100.12.

    3. Perform a thermographic survey in accordance with Section 9.

5. Verify that all required grounding and connections provide contact.

6. Verify correct operation of transformer withdrawal mechanism and grounding operation.

7. Verify correct primary and secondary fuse sizes for voltage transformers.

8. Use appropriate lubrication on moving current-carrying parts and on moving and sliding surfaces.

9. Perform as-left tests.

B. Electrical Tests

1. Perform resistance measurements through bolted connections with a low-resistance ohmmeter in accordance with Section 7.10.2.A.4.1

2. Perform insulation-resistance tests for one minute winding-to-winding and each winding-to-ground. Test voltages shall be applied in accordance with Table 100.5. For units with solid-state components that cannot tolerate the applied voltage, follow manufacturer's recommendations.

*3. Perform a polarity test on each transformer, as applicable, to verify the polarity marks or H1-X1 relationship.

*4. Perform a turns-ratio test on as-found tap position.

*5. Measure voltage circuit burdens at transformer terminals.

* Optional

7.10.2 Instrument Transformers, Voltage Transformers (*continued*)

*6. Perform a dielectric withstand voltage test on the primary windings with the secondary windings connected to ground. The dielectric voltage shall be in accordance with Table 100.9. The test voltage shall be applied for one minute.

*7. Perform power-factor or dissipation-factor tests in accordance with manufacturer's published data.

8. Verify that potential circuits are grounded and have only one grounding point in accordance with IEEE C57.13.3.

C. Test Values – Visual and Mechanical

1. Compare bolted connection resistance values to values of similar connections. Investigate values which deviate from those of similar bolted connections by more than 50 percent of the lowest value. (7.10.2.A.4.1)

2. Bolt-torque levels should be in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.12. (7.10.2.A.4.2)

3. Results of the thermographic survey shall be in accordance with Section 9. (7.10.2.A.4.3)

D. Test Values Electrical

1. Compare bolted connection resistance values to values of similar connections. Investigate values which deviate from those of similar bolted connections by more than 50 percent of the lowest value.

2. Insulation-resistance values of instrument transformers should not be less than values shown in Table 100.5.

3. Polarity results should agree with transformer markings.

4. In accordance with IEEE C57.13.8.1.1 the ratio error should be as follows:

    1. Revenue metering applications: equal to or less than ±0.1 percent for ratio and ±0.9 mrad (three minutes) for phase angle.

    2. Other applications: equal to or less than ±1.2 percent for ratio and ±17.5 mrad (one degree) for phase angle.

5. Compare measured burdens to instrument transformer ratings.

6. If no evidence of distress or insulation failure is observed by the end of the total time of voltage application during the dielectric withstand voltage test, the primary windings are considered to have passed the test.

* Optional

7.10.2 Instrument Transformers, Voltage Transformers (*continued*)

7. Power-factor or dissipation-factor values shall be in accordance with manufacturer's published data. In the absence of manufacturer's published data, use test equipment manufacturer's published data.

8. Test results should indicate that the circuits have only one grounding point.

* Optional

7.10.3 Instrument Transformers, Coupling-Capacitor Voltage Transformers

A. Visual and Mechanical Inspection

1. Inspect physical and mechanical condition.

*2. Prior to cleaning the unit, perform as-found tests.

3. Clean the unit.

4. Inspect bolted electrical connections for high resistance using one or more of the following:

    1. Use of a low-resistance ohmmeter in accordance with Section 7.10.3.B.1.

    2. Verify tightness of accessible bolted electrical connections by calibrated torque-wrench method in accordance with manufacturer's published data or Table 100.12.

    3. Perform a thermographic survey in accordance with Section 9.

5. Verify that all required grounding and connections provide contact.

6. Verify correct operation of the grounding switches.

7. Verify correct primary and secondary fuse sizes for voltage transformers.

8. Use appropriate lubrication on moving current-carrying parts and on moving and sliding surfaces.

9. Perform as-left tests.

B. Electrical Tests

1. Perform resistance measurements through bolted connections with a low-resistance ohmmeter in accordance with Section 7.10.3.A.4.1.

2. Perform insulation-resistance tests for one minute, winding-to-winding and each winding-to-ground. Test voltages shall be applied in accordance with Table 100.5. For units with solid-state components that cannot tolerate the applied voltage, follow manufacturer's recommendations.

*3. Perform a polarity test on each transformer, as applicable, to verify the polarity marks or H1-X1 relationship. See IEEE C93.1 for standard polarity marking.

4. Perform a ratio test on the designated tap position.

* Optional

7.10.3 Instrument Transformers, Coupling-Capacitor Voltage Transformers (*continued*)

*5. Measure voltage circuit burdens at transformer terminals.

*6. Perform a dielectric withstand voltage test on the primary windings with the secondary windings connected to ground. The dielectric withstand voltage shall be in accordance with Table 100.9. The test voltage shall be applied for one minute.

7. Measure capacitance of capacitor sections.

8. Measure insulation power factor or dissipation factor in accordance with test equipment manufacturer's published data

9. Verify that potential circuits are grounded and have only one grounding point in accordance with IEEE C57.13.3.

C. Test Values – Visual and Mechanical

1. Compare bolted connection resistance values to values of similar connections. Investigate values which deviate from those of similar bolted connections by more than 50 percent of the lowest value. (7.10.3.A.4.1)

2. Bolt-torque levels should be in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.12. (7.10.3.A.4.2)

3. Results of the thermographic survey shall be in accordance with Section 9. (7.10.3.A.4.3)

D. Test Values – Electrical

1. Compare bolted connection resistance values to values of similar connections. Investigate values which deviate from those of similar bolted connections by more than 50 percent of the lowest value.

2. Insulation-resistance values of instrument transformers should not be less than values shown in Table 100.5.

3. Polarity results should agree with transformer markings.

4. In accordance with IEEE C57.13; 8.1.1 the ratio error should be as follows:

    1. Revenue metering applications: equal to or less than ±0.1 percent for ratio and ±0.9 mrad (three minutes) for phase angle.

    2. Other applications: equal to or less than +1.2 percent for ratio and ±17.5 mrad (one degree) for phase angle.

* Optional

7.10.3 Instrument Transformers, Coupling-Capacitor Voltage Transformers (*continued*)

5. Compare measured burdens to instrument transformer ratings.

6. If no evidence of distress or insulation failure is observed by the end of the total time of voltage application during the dielectric withstand voltage test, the test specimen is considered to have passed the test.

7. Capacitance of capacitor sections of coupling-capacitance voltage transformers should be in accordance with manufacturer’s published data.

8. Power-factor or dissipation-factor values shall be in accordance with manufacturer’s published data. In the absence of manufacturer’s published data, use test equipment manufacturer’s published data.

9. Test results should indicate that the circuits have only one grounding point.

* Optional

7.10.4 Instrument Transformers, High-Accuracy Instrument Transformers

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7. INSPECTION AND TEST PROCEDURES

7.11.1 Metering Devices, Electromechanical and Solid-State

A. Visual and Mechanical Inspection

1. Inspect physical and mechanical condition.
2. Verify tightness of electrical connections
3. Inspect cover gasket, cover glass, condition of spiral spring, disk clearance, contacts, and case-shorting contacts, as applicable.

*4. Prior to cleaning the unit, perform as-found tests.

5. Clean the unit.
6. Verify freedom of movement, end play, and alignment of rotating disk(s).
7. Perform as-left tests.

B. Electrical Tests

1. Verify accuracy of meters per the manufacturers published data.
2. Calibrate meters in accordance with manufacturer’s published data.

*3. Verify all instrument multipliers.

* Optional

7.11.1 Metering Devices, Electromechanical and Solid-State (*continued*)

C. Test Values – Visual and Mechanical

1. Tightness of electrical connections shall assure a low resistance connection. (7.11.2.A.3)

2. Display and indicating devices shall operate per manufacturer's published data. (7.11.2.A.5)

D. Test Values – Electrical

1. Meter accuracy should be in accordance with manufacturer's published data.

2. Calibration results should be within manufacturer's specified tolerances.

3. Instrument multipliers should be in accordance with specified system design.

* Optional

7.11.2 Metering Devices, Microprocessor-Based

A. Visual and Mechanical Inspection

1. Inspect meters and cases for physical damage.
2. Clean front panel.
3. Verify tightness of electrical connections.
4. Record model number, serial number, firmware revision, software revision, and rated control voltage.
5. Verify operation of display and indicating devices.
6. Record passwords.
7. Verify unit is grounded in accordance with manufacturer's instructions.
8. Verify unit is connected in accordance with manufacturer's instructions and project drawings.
9. Download settings from the meter, print a copy of the settings for the report, and compare the settings to those specified.

B. Electrical Tests

1. Apply voltage or current as appropriate to each analog input and verify correct measurement and indication.
2. Confirm correct operation and setting of each auxiliary input/output feature in use, including mechanical relay, digital, and analog.
3. Confirm measurements and indications are consistent with loads present.

C. Test Values – Visual and Mechanical

1. Tightness of electrical connections shall assure a low resistance connection. (7.11.2.A.3)
2. Display and indicating devices shall operate per manufacturer's published data. (7.11.2.A.5)

7.11.2 Metering Devices, Microprocessor-Based (*continued*)

D. Test Values – Electrical

1. Measurement and indication of applied values of voltage and current should be within manufacturer's published tolerances for accuracy.

2. All auxiliary input/output features should operate per settings and manufacturer's published data.

3. Measurements and indications should be consistent with energized system loads.

7. INSPECTION AND TEST PROCEDURES

7.12.1.1 Regulating Apparatus, Voltage, Step-Voltage Regulators

A. Visual and Mechanical Inspection

1. Inspect physical and mechanical condition.

2. Inspect anchorage, alignment, and grounding.

3. Record position indicator as-found, maximum, and minimum values.

*4. Prior to cleaning the unit, perform as-found tests.

5. Clean bushings and control cabinets.

6. Verify auxiliary device operation.

7. Inspect bolted electrical connections for high resistance using one or more of the following:

    1. Use of a low-resistance ohmmeter in accordance with Section 7.12.1.1.B.1.

    2. Verify tightness of accessible bolted electrical connections by calibrated torque-wrench method in accordance with manufacturer's published data or Table 100.12.

    3. Perform a thermographic survey in accordance with Section 9.

8. Verify correct operation of motor and drive train and automatic motor cutoff at maximum lower and maximum raise positions.

9. Verify correct liquid level in all tanks and bushings.

10. Perform specific inspections and mechanical tests as recommended by the manufacturer.

*11. Perform an internal inspection and maintenance in accordance with manufacturer's data:

    1. Remove oil.

    2. Clean carbon residue and debris from compartment.

    3. Inspect contacts for wear and alignment.

    4. Inspect all electrical and mechanical connections for tightness using calibrated torque wrench method in accordance with manufacturer's published data or Table 100.12.

    5. Inspect tap-changer compartment terminal board, contact support boards, and insulating operating components for evidence of moisture, cracks, excessive wear, breakage, and/or signs of electrical tracking.

* Optional

7.12.1.1 Regulating Apparatus, Voltage, Step-Voltage Regulators (*continued*)

6. Electrically operate tap-changer through full range of taps.

7. Replace gaskets and seal compartment.

8. Fill with filtered oil.

12. Use appropriate lubrication on moving current-carrying parts and on moving and sliding surfaces.

13. Perform as-left tests.

14. Record as-found and as-left operation counter readings.

B. Electrical Tests

1. Perform resistance measurements through bolted connections with a low-resistance ohmmeter in accordance with Section 7.12.1.1.A.7.1.

2. Perform insulation-resistance tests on each winding-to-ground in any off-neutral position. Apply voltage in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.5. Calculate polarization index.

3. Perform insulation power-factor or dissipation-factor tests on windings in accordance with test equipment manufacturer's published data.

4. Perform power-factor or dissipation-factor tests on each bushing equipped with a power-factor/capacitance tap. In the absence of a power-factor/ capacitance tap, perform hot-collar tests. These tests shall be in accordance with the test equipment manufacturer's published data.

5. Measure winding resistance of source windings in the neutral position. Measure the resistance of all taps on load windings.

6. Perform special tests and adjustments as recommended by manufacturer.

*7. If the regulator has a separate tap-changer compartment, measure the percentage of oxygen of the nitrogen gas blanket in the main tank.

8. Perform turns-ratio test on each voltage step position. Verify that the tap position indicator correctly identifies all tap positions.

9. Verify accurate operation of voltage range limiter.

10. Verify operation and accuracy of bandwidth, time delay, voltage, and line-drop compensation adjustments of tap-changer control device

* Optional

7.12.1.1 Regulating Apparatus, Voltage, Step-Voltage Regulators (*continued*)

*11. If regulator has a separate tap-changer compartment, sample insulating liquid in the main tank in accordance with ASTM D923 and perform dissolved-gas analysis in accordance with IEEE C57.104 or ASTM D3612.

12. Remove a sample of insulating liquid from the main tank or common tank in accordance with ASTM D923. Sample shall be tested in accordance with the referenced standard.

    1. Dielectric breakdown voltage: ASTM D1816
    2. Acid neutralization number: ASTM D974
    3. Specific gravity: ASTM D1298
    4. Interfacial tension: ASTM D971
    5. Color: ASTM D1500
    6. Visual condition: ASTM D1524
    *7. Power factor: ASTM D924
    Required when the regulator voltage is 46 kV or higher.
    8. Water in insulating liquids: ASTM D1533.

13. Remove a sample of insulating liquid from the tap-changer tank in accordance with ASTM D 923. Sample shall be tested in accordance with the referenced standard.

    1. Dielectric breakdown voltage: ASTM D1816
    2. Color: ASTM D1500
    3. Visual condition: ASTM D1524

*14. Remove a sample of insulating liquid from the tap-changer compartment or common tank in accordance with ASTM D923 and perform dissolved-gas analysis (DGA) in accordance with IEEE C57.104 or ASTM D3612.

15. Verify operation of heaters.

* Optional

7.12.1.1 Regulating Apparatus, Voltage, Step-Voltage Regulators (*continued*)

C. Test Values – Visual and Mechanical

1. Auxiliary devices should operate in accordance with system design. (7.12.1.1.A.6)

2. Compare bolted connection resistance values to values of similar connections. Investigate values which deviate from those of similar bolted connections by more than 50 percent of the lowest value. (7.12.1.1.A.7.1)

3. Bolt-torque levels should be in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.12. (7.12.1.1.A.7.2)

4. Results of the thermographic survey shall be in accordance with Section 9. (7.12.1.1.A.7.3)

5. Motor, drive train, and automatic cutoff should operate in accordance with manufacturer's design. (7.12.1.1.A.8)

6. Liquid level in tanks and bushings should be within indicated tolerances. (7.12.1.1.A.9)

7. On internal inspection, contact alignment and wear should be within manufacturer's recommendations for continued service. (7.12.1.1.A.11.3)

8. Bolt torque levels should be in accordance with Table 100.12 unless otherwise specified by the manufacturer. (7.12.1.1.A.11.4)

9. Terminal board, contact support boards, and insulating operating components should not show signs of moisture, tracking, excessive wear, breakage, or electrical tracking. (7.12.1.1.A.11.5)

10. The tap-changer should operate electrically through the full range of taps. (7.12.1.1.A.11.6)

11. The operation counter should move incrementally for each operation performed. (7.12.1.1.A.14)

D. Test Values – Electrical

1. Compare bolted connection resistance values to values of similar connections. Investigate values which deviate from those of similar bolted connections by more than 50 percent of the lowest value.

2. Insulation-resistance values should be in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.5. Values of insulation resistance less than this table or manufacturer's recommendations should be investigated. Resistance values shall be temperature corrected in accordance with Table 100.14. The polarization index shall be compared to previously obtained results.

* Optional

7.12.1.1 Regulating Apparatus, Voltage, Step-Voltage Regulators (*continued*)

3. Maximum power-factor or dissipation-factor values of liquid-filled regulators should be in accordance with manufacturer's published data. In the absence of manufacturer's data, compare to test equipment manufacturer's published data. Representative values are indicated in Table 100.3.

4. Power-factor or dissipation-factor and capacitance values should be within ten percent of nameplate rating for bushings. Hot collar tests are evaluated on a milliampere/milliwatt loss basis, and the results should be compared to values of similar bushings.

5. Consult manufacturer if winding-resistance values vary by more than two percent from test results of adjacent phases.

6. Special tests and adjustments should meet manufacturer's published requirements.

7. Investigate presence of oxygen in nitrogen gas blanket.

8. Turns-ratio test results should maintain a normal deviation between each voltage step and should not deviate more than one-half percent from the calculated voltage ratio.

9. Voltage range limiter should operate within manufacturer's recommendations.

10. Accuracy of bandwidth, time-delay, voltage, and live drop compensation adjustments should be as specified.

11. Results of dissolved-gas analysis of insulating liquid on the main tank of regulators having a separate tap-changer compartment shall be evaluated in accordance with IEEE C57.104 or ASTM D3612.

12. Results of insulating liquid tests on the main tank or common tank of single tank voltage regulators should be in accordance with Table 100.4.

13. Results of insulating liquid tests on the tap-changer tank of regulators having a separate tap-changer compartment should be in accordance with Table 100.4.

14. Compare results of dissolved gas analysis to previous test results.

15. Heaters should be operational.

* Optional

7.12.1.2 Regulating Apparatus, Voltage, Induction Regulators

A. Visual and Mechanical Inspection

1. Inspect physical and mechanical condition.

2. Inspect anchorage, alignment, and grounding.

3. Record position indicator as-found, maximum, and minimum values.

*4. Prior to cleaning the unit, perform as-found tests.

5. Clean bushings and control cabinets.

6. Verify correct auxiliary device operation.

7. Inspect bolted electrical connections for high resistance using one or more of the following:

    1. Use of a low-resistance ohmmeter in accordance with Section 7.12.1.2.B.1.

    2. Verify tightness of accessible bolted electrical connections by calibrated torque-wrench method in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.12.

    3. Perform a thermographic survey in accordance with Section 9.

8. Check motor and drive train for correct operation and automatic motor cutoff at maximum lower and maximum raise.

9. Verify appropriate liquid level in all tanks and bushings.

10. Perform mechanical maintenance, inspections, and tests as recommended by the manufacturer.

11. Use appropriate lubrication on moving current-carrying parts and on moving and sliding surfaces.

12. Perform as-left tests.

B. Electrical Tests

1. Perform resistance measurements through bolted connections with a low-resistance ohmmeter in accordance with Section 7.12.1.2.A.7.1.

2. Perform insulation-resistance tests winding-to-winding and each winding-to-ground. Apply voltage in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.5. Calculate polarization index.

* Optional

7.12.1.2 Regulating Apparatus, Voltage, Induction Regulators (*continued*)

3. Perform power-factor or dissipation-factor tests on winding insulation in accordance with test equipment manufacturer's published data.

4. Perform power-factor or dissipation-factor tests on each bushing equipped with a power-factor/capacitance tap. In the absence of a power-factor/capacitance tap, perform hot-collar tests. These tests shall be in accordance with the test equipment manufacturer's published data.

5. Verify voltage regulation.

6. Verify that the indicator correctly identifies neutral position.

7. Perform winding resistance tests on each winding.

8. Sample insulating liquid in accordance with ASTM D923. Sample shall be tested in accordance with the referenced standard.

    1. Dielectric breakdown voltage: ASTM D1816

    2. Acid neutralization number: ASTM D974

    *3. Specific gravity: ASTM D1298

    4. Interfacial tension: ASTM D971

    5. Color: ASTM D1500

    6. Visual condition: ASTM D1524

    *7. Power factor: ASTM D924
    Required when the regulator voltage is 46 kV or higher.

    *8. Water content: ASTM D1533
    Required when the regulator voltage is 25 kV or higher.

*9. Remove a sample of insulating liquid in accordance with ASTM D923 and perform dissolved-gas analysis (DGA) in accordance with ASTM D3612 or IEEE C57.104.

*10. Test for the presence of oxygen in the gas blanket of liquid-filled regulators.

11. Verify operation of control cabinet space heater.

* Optional

7.12.1.2 Regulating Apparatus, Voltage, Induction Regulators (*continued*)

C. Test Values – Visual and Mechanical

1. Auxiliary devices should operate in accordance with system design. (7.12.1.2.A.6)

2. Compare bolted connection resistance values to values of similar connections. Investigate values which deviate from those of similar bolted connections by more than 50 percent of the lowest value. (7.12.1.2.A.7.1)

3. Bolt-torque levels should be in accordance with manufacturer’s published data. In the absence of manufacturer’s published data, use Table 100.12. (7.12.1.2.A.7.2)

4. Results of the thermographic survey shall be in accordance with Section 9. (7.12.1.2.A.7.3)

5. Motor, drive train, and automatic cutoff should operate in accordance with manufacturer’s design intent and automatic motor cutoff should operate at maximum lower and maximum raise positions. (7.12.1.2.A.8)

6. Liquid level in tanks and bushings should be within indicated tolerances. (7.12.1.2.A.9)

D. Test Values – Electrical

1. Compare bolted connection resistance values to values of similar connections. Investigate values which deviate from those of similar bolted connections by more than 50 percent of the lowest value.

2. Insulation-resistance values should be in accordance with manufacturer’s published data. In the absence of manufacturer’s published data, use Table 100.5. Values of insulation resistance less than this table or manufacturer’s recommendations should be investigated. The polarization index shall be compared to previously obtained results.

3. Maximum power-factor or dissipation-factor values of liquid-filled regulators should be in accordance with manufacturer’s published data. In the absence of manufacturer’s data, compare to test equipment manufacturer’s published data. Representative values are indicated in Table 100.3.

4. Power-factor or dissipation-factor and capacitance values should be within ten percent of nameplate rating for bushings. Hot collar tests are evaluated on a milliampere/milliwatt loss basis, and the results should be compared to values of similar bushings.

5. The regulation should be a linear ratio throughout the range between the maximum raise and the maximum lower positions.

6. Indicator should indicate neutral position correctly.

* Optional

7.12.1.2 Regulating Apparatus, Voltage, Induction Regulators (*continued*)

7. Consult the manufacturer if winding-resistance values vary by more than two percent from measurements of adjacent windings.

8. Results of insulating liquid tests should be in accordance with Table 100.4.

9. Evaluate results of dissolved-gas analysis in accordance with IEEE C57.104.

10. Investigate presence of oxygen in nitrogen gas blanket.

11. Heaters should be operational.

7.12.2 Regulating Apparatus, Current

– RESERVED –

* Optional

7. INSPECTION AND TEST PROCEDURES

7.12.3 Regulating Apparatus, Load Tap-Changers

A. Visual and Mechanical Inspection

1. Inspect physical and mechanical condition.

2. Inspect anchorage, alignment, and grounding.

3. Record position indicator as-found, maximum, and minimum values.

*4. Prior to cleaning the unit, perform as-found tests.

5. Clean bushings and control cabinets.

6. Inspect bolted electrical connections for high resistance using one or more of the following:

    1. Use of a low-resistance ohmmeter in accordance with Section 7.12.3.B.1.

    2. Verify tightness of accessible bolted electrical connections by calibrated torque-wrench method in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.12.

    3. Perform a thermographic survey in accordance with Section 9.

7. Verify correct auxiliary device operation.

8. Verify motor and drive train for correct operation and automatic motor cutoff at maximum lower and maximum raise.

9. Verify correct liquid level in all tanks.

10. Perform mechanical maintenance, inspections, and tests as recommended by the manufacturer.

*11. Visually inspect wear/erosion indicators on vacuum bottles.

*12. Perform an internal inspection:

    1. Remove oil.

    2. Clean carbon residue and debris from compartment.

    3. Inspect contacts for wear and alignment.

* Optional

7.12.3 Regulating Apparatus, Load Tap-Changers (*continued*)

4. Inspect all electrical and mechanical connections for tightness using calibrated torque-wrench method in accordance with manufacturer's published data or Table 100.12.

5. Inspect tap-changer compartment terminal board, contact support boards, and insulated operating components for evidence of moisture, cracks, excessive wear, breakage, and/or signs of electrical tracking.

6. Electrically operate tap-changer through full range of taps.

7. Replace gaskets and seal compartment.

8. Fill with filtered oil.

13. Use appropriate lubrication on moving current-carrying parts and on moving and sliding surfaces.

14. Perform as-left tests.

15. Record as-found and as-left operation counter readings.

B. Electrical Tests

1. Perform resistance measurements through bolted connections with low-resistance ohmmeter in accordance with Section 7.12.3.A.6.1.

2. Perform insulation-resistance tests in any off-neutral position in accordance with Section 7.2.2.B.2.

3. Perform insulation power-factor or dissipation-factor tests in off-neutral position in accordance with Section 7.2.2.B.4.

4. Perform winding-resistance tests at all tap positions.

*5. Perform dynamic resistance measurement for load tap changer.

6. Perform special tests and adjustments as recommended by the manufacturer.

7. Perform turns-ratio test at all tap positions.

8. Remove a sample of insulating liquid in accordance with ASTM D923. The sample shall be tested for the following in accordance with the referenced standard.

    1. Dielectric breakdown voltage: ASTM D1816.

* Optional

7.12.3 Regulating Apparatus, Load Tap-Changers (*continued*)

2. Color: ASTM D1500

3. Visual condition: ASTM D1524

8. Remove a sample of insulating liquid in accordance with ASTM D3613 and perform dissolved gas analysis (DGA) in accordance with IEEE C57.104 or ASTM D3612.

*9. Perform magnetron atmospheric condition (MAC) test on each vacuum interrupter.

10. Perform vacuum bottle integrity tests (dielectric withstand voltage) across each vacuum bottle with the contacts in the open position in strict accordance with manufacturer's published data.

11. Verify operation of heaters.

*12. Perform excitation test on maximum raise, lower, and neutral tap positions.

C. Test Values – Visual and Mechanical

1. Compare bolted connection resistance values to values of similar connections. Investigate values which deviate from those of similar bolted connections by more than 50 percent of the lowest value. (7.12.3.A.6.1)

2. Bolt-torque levels should be in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.12. (7.12.3.A.6.2)

3. Results of the thermographic survey shall be in accordance with Section 9. (7.12.3.A.6.3)

4. Auxiliary device operation should be in accordance with design intent. (7.12.3.A.7)

5. Motor, drive train, and automatic cutoff should operate in accordance with manufacturer's design intent and automatic motor cutoff should operate at maximum lower and maximum raise positions. (7.12.3.A.8)

6. Liquid level in tanks should be within indicated tolerances. (7.12.3.A.9)

7. Vacuum bottle wear/erosion indicators should be within manufacturer's recommended tolerances. (7.12.3.A.11)

8. Contact wear and alignment should be within manufacturer's recommended tolerances. (7.12.3.A.12.3)

9. Bolt-torque levels should be in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.12. (7.12.3.A.12.4)

* Optional

7.12.3 Regulating Apparatus, Load Tap-Changers (*continued*)

10. No evidence of moisture, cracks, excessive wear, breakage, or electrical tracking should be found. (7.12.3.A.12.5)

11. The tap-changer should operate through the full range of taps. (7.12.3.A.12.6)

12. Operations counter should have had an incremental change in accordance with tap-changer operation. (7.12.3.A.15)

D. Test Values – Electrical

1. Compare bolted connection resistance values to values of similar connections. Investigate values which deviate from those of similar bolted connections by more than 50 percent of the lowest value.

2. Insulation-resistance values should be in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.5. Values of insulation resistance less than this table or manufacturer's recommendations should be investigated.

3. Maximum winding insulation power-factor or dissipation-factor values of liquid-filled transformers should be in accordance with manufacturer's specifications. Representative values are indicated in Table 100.3. Also, compare with test equipment manufacturer's published data.

4. Consult the manufacturer if winding-resistance values vary by more than two percent from measurements of adjacent phases.

5. Special tests and adjustments should conform to manufacturer's specifications.

6. Turns-ratio test results should maintain a normal deviation between each voltage step and should not deviate more than one-half percent from the calculated voltage ratio.

7. Results of insulating liquid tests should be in accordance with Table 100.4.

8. Evaluate results of dissolved-gas analysis in accordance with ANSI/IEEE C57.104.

9. In the absence of manufacturer's published data, each vacuum interrupter pressure shall not be greater than 1x10-2 Pa and shall not deviate from adjacent poles by more than two orders of magnitude.

10. If no evidence of distress or insulation failure is observed by the end of the total time of voltage application during the dielectric withstand voltage test, the test specimen is considered to have passed the test.

11. Heaters should be operational.

* Optional

7.12.3 Regulating Apparatus, Load Tap-Changers (*continued*)

12. The typical excitation-current test data pattern for a three-legged core transformer is two similar current readings and one lower current reading. The test should produce a pattern typical for the specific transformer type and configuration.

* Optional

7. INSPECTION AND TEST PROCEDURES

7.13 Grounding Systems

A. Visual and Mechanical Inspection

1. Inspect physical and mechanical condition.

2. Inspect accessible electrical connections for high resistance using one or more of the following:

    1. Use of low-resistance ohmmeter in accordance with Section 7.13.B.1.

    2. Verify tightness of accessible bolted electrical connections by calibrated torque-wrench method in accordance with manufacturer's published data or Table 100.12.

3. Inspect anchorage.

B. Electrical Tests

1. Perform resistance measurements through bolted connections with a low-resistance ohmmeter in accordance with section 7.13.A.2.1.

2. Perform fall-of-potential or alternative test in accordance with IEEE 81 on the main grounding electrode or system.

3. Perform point-to-point tests to determine the resistance between the main grounding system and all major electrical equipment frames, system neutral, and derived neutral points.

C. Test Values – Visual and Mechanical

1. Grounding system electrical and mechanical connections should be free of corrosion. (7.13.A.1)

2. Compare bolted connection resistance values to values of similar connections. Investigate values which deviate from those of similar bolted connections by more than 50 percent of the lowest value. (7.13.A.2.1)

3. Bolt-torque levels should be in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.12. (7.13.A.2.2)

D. Test Values – Electrical

1. Compare bolted connection resistance values to values of similar connections. Investigate values which deviate from those of similar bolted connections by more than 50 percent of the lowest value.

2. The resistance between the main grounding electrode and ground should be no greater than five ohms for large commercial or industrial systems and one ohm or less for generating or

* Optional

7.13 Grounding Systems (*continued*)

transmission station grounds unless otherwise specified by the owner. (Reference IEEE 142)

3. Investigate point-to-point resistance values that exceed 0.5 ohm.

* Optional

7. INSPECTION AND TEST PROCEDURES

7.14 Ground-Fault Protection Systems, Low-Voltage

A. Visual and Mechanical Inspection

1. Inspect the components for damage and errors in polarity or conductor routing.

    1. Verify that the ground connection is made on the source side of the neutral disconnect link and also on the source side of any ground fault sensor.

    2. Verify that the neutral sensors are connected with correct polarity on both primary and secondary.

    3. Verify that all phase conductors and the neutral pass through the sensor in the same direction for zero sequence systems.

    4. Verify that grounding conductors do not pass through zero sequence sensors.

    5. Verify that the grounded conductor is solidly grounded.

*2. Prior to cleaning the unit, perform as-found tests.

3. Clean the unit.

4. Inspect bolted electrical connections for high resistance using one or more of the following:

    1. Use of a low-resistance ohmmeter in accordance with Section 7.14.B.1.

    2. Verify tightness of accessible bolted electrical connections by calibrated torque-wrench method in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.12.

5. Verify correct operation of all functions of the self-test panel.

6. Verify pickup and time-delay settings in accordance with the settings provided in the owner's specifications. Record appropriate operation and test sequences as required by NFPA 70 *National Electrical Code*, Article 230.95.

7. Perform as-left tests.

B. Electrical Tests

1. Perform resistance measurements through bolted connections with a low-resistance ohmmeter in accordance with Section 7.14.A.4.1.

2. Measure the system neutral-to-ground insulation resistance with the neutral disconnect link temporarily removed. Replace neutral disconnect link after testing.

* Optional

7. INSPECTION AND TEST PROCEDURES

7.14 Ground-Fault Protection Systems, Low-Voltage (*continued*)

*3. Perform insulation-resistance tests on all control wiring with respect to ground. The applied potential shall be 500 volts dc for 300-volt rated cable and 1000 volts dc for 600-volt rated cable. Test duration shall be one minute. For units with solid-state components or control devices that cannot tolerate the applied voltage, follow manufacturer's recommendation.

4. Perform ground fault protective device pickup tests using primary current injection.

5. For summation type systems utilizing phase and neutral current transformers, verify correct polarities by applying current to each phase-neutral current transformer pair. This test also applies to molded-case breakers utilizing an external neutral current transformer.

6. Measure time delay of the ground fault protective device at a value equal to or greater than 150 percent of the pickup value.

7. Verify that reduced control voltage tripping capability is 55 percent for ac systems and 80 percent for dc systems.

8. Verify blocking capability of zone interlock systems.

C. Test Values – Visual and Mechanical

1. Compare bolted connection resistance values to values of similar connections. Investigate values which deviate from those of similar bolted connections by more than 50 percent of the lowest value. (7.14.A.4.1)

2. Bolt-torque levels should be in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.12. (7.14.A.4.2)

D. Test Values – Electrical

1. Compare bolted connection resistance values to values of similar connections. Investigate values which deviate from those of similar bolted connections by more than 50 percent of the lowest value.

2. System neutral-to-ground insulation resistance should be a minimum of one megohm.

3. Insulation-resistance values of control wiring should be comparable to previously obtained results but not less than two megohms.

4. Results of pickup test should be greater than 90 percent of the ground fault protection device pickup setting and less than 1200 amperes or 125 percent of the pickup setting, whichever is smaller.

* Optional

7.14 Ground-Fault Protection Systems, Low-Voltage (*continued*)

5. The ground fault protective device should operate when current direction is the same relative to polarity marks in the two current transformers. The ground fault protective device should not operate when current direction is opposite relative to polarity marks in the two current transformers.

6. Relay timing should be in accordance with manufacturer's specifications but must be no longer than one second at 3000 amperes in accordance with NFPA 70, *National Electrical Code*, Article 230.95.

7. The circuit interrupting device should operate when control voltage is 55 percent of nominal voltage for ac circuits and 80 percent of nominal voltage for dc circuits.

8. Results of zone-blocking tests should be in accordance with manufacturer's published data and design specifications.

* Optional

7. INSPECTION AND TEST PROCEDURES

7.15.1 Rotating Machinery, AC Induction Motors and Generators

A. Visual and Mechanical Inspection

1. Inspect physical and mechanical condition.

2. Inspect anchorage, alignment, and grounding.

3. Inspect air baffles, filter media, stator winding, stator core, rotor, cooling fans, slip rings, brushes, brush rigging, and bearings.

4. Inspect bolted electrical connections for high resistance using one or more of the following:

    1. Use of low-resistance ohmmeter in accordance with Section 7.15.1.B.1.

    2. Verify tightness of accessible bolted electrical connections by calibrated torque-wrench method in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.12.

    3. Perform thermographic survey in accordance with Section 9.

*5. Perform special tests such as air-gap spacing and machine alignment.

6. Verify the application of appropriate lubrication and lubrication systems.

7. Verify the absence of unusual mechanical or electrical noise or signs of overheating.

8. Verify that resistance temperature detector (RTD) circuits conform to drawings.

B. Electrical Tests – AC Induction

1. Perform resistance measurements through bolted connections with a low-resistance ohmmeter in accordance with Section 7.15.1.A.4.1.

2. Perform insulation-resistance tests in accordance with IEEE 43.

    1. Machines larger than 200 horsepower (150 kilowatts):
    Test duration shall be for ten minutes. Calculate polarization index.

    2. Machines 200 horsepower (150 kilowatts) and less:
    Test duration shall be for one minute. Calculate the dielectric-absorption ratio.

3. Perform dielectric withstand voltage tests on machines rated at 2300 volts and greater in accordance with:

* Optional

7.15.1 Rotating Machinery, AC Induction Motors and Generators *(continued)*

1. IEEE Standard 95 for dc dielectric withstand voltage test.

2. NEMA MG1 for ac dielectric withstand voltage test.

4. Perform phase-to-phase stator resistance test on machines 2300 volts and greater

*5. Perform insulation power-factor or dissipation-factor tests.

*6. Perform power-factor tip-up tests.

*7. Perform surge comparison tests.

8. Perform insulation-resistance test on insulated bearings in accordance with manufacturer's published data.

9. Test surge protection devices in accordance with Section 7.19 and Section 7.20.

10. Test motor starter in accordance with Section 7.16.

11. Perform resistance tests on resistance temperature detector (RTD) circuits.

12. Verify operation of machine space heater.

*13. Perform vibration test while machine is running under load.

*14. Measure bearing temperatures while machine is running under load.

*15. Perform current signature analysis while machine is running under load.

*16. Perform partial discharge test.

C. Test Values – Visual and Mechanical

1. Inspection (7.15.1.A.3)

   1. Air baffles should be clean.

   2. Filter media should be clean.

   3. Stator winding should be clean and show no evidence of overheating, cracking, tracking, or surface partial discharge activity.

   4. The stator core should be clean with no ventilation duct blockage, lamination looseness, burning, or rubs from contact with the rotor.

* Optional

7.15.1 Rotating Machinery, AC Induction Motors and Generators (*continued*)

5. Rotor should be clean and show no evidence of overheating or cracked bars or end rings and should show no evidence of rubs from contact with the stator.

6. Cooling fans should operate, not be loose on the rotor shaft, and have no cracks in them.

7. Slip ring to brush alignment shall be within manufacturer's published tolerances.

8. Brush rigging clearance shall be in accordance with manufacturer's published data.

9. Bearings should be inspected for evidence of overheating, rubs, rolling element damage, contamination, electrical damage, and lack of lubrication.

10. Slip ring wear should be within manufacturer's tolerances for continued use.

11. Brushes should be within manufacturer's tolerances for continued use.

12. Brush rigging should be intact.

2. Compare bolted connection resistance values to values of similar connections. Investigate values which deviate from those of similar bolted connections by more than 50 percent of the lowest value. (7.15.1.A.4.1)

3. Bolt-torque levels should be in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.12. (7.15.1.A.4.2)

4. Results of the thermographic survey shall be in accordance with Section 9. (7.15.1.A.4.3)

D. Test Values – Electrical Tests

1. Compare bolted connection resistance values to values of similar connections. Investigate any values that deviate from similar bolted connections by more than 50 percent of the lowest value.

2. The dielectric absorption ratio or polarization shall be compared to previously obtained results and should not be less than 1.0. The recommended minimum insulation resistance ($IR_{1\ min}$) test results in megohms should be corrected to 40° C and read as follows:

   1. $IR_{1\ min}$ = kV + 1 for most windings made before 1970, all field windings, and others not described in 2.2 and 2.3.

      (kV is the rated machine terminal-to-terminal voltage in rms kV)

   2. $IR_{1\ min}$ = 100 megohms for ac windings built after 1970 (form-wound coils).

* Optional

7. INSPECTION AND TEST PROCEDURES

7.15.1 Rotating Machinery, AC Induction Motors and Generators (*continued*)

3. IR $_{1\ min}$ = 5 megohms for most machines and random-wound stator coils and form-wound coils rated below 1 kV and dc armatures.

   NOTE: Dielectric withstand voltage and surge comparison tests shall not be performed on machines having values lower than those indicated above.

3. If no evidence of distress or insulation failure is observed by the end of the total time of voltage application during the dielectric withstand voltage test, the test specimen is considered to have passed the test.

4. Investigate phase-to-phase stator resistance values that deviate by more than five percent.

5. Power-factor or dissipation-factor values shall be compared with previous values of similar machines.

6. Tip-up values should indicate no significant increase in power factor.

7. If no evidence of distress, insulation failure, or lack of waveform nesting is observed by the end of the total time of voltage application during the surge comparison test, the test specimen is considered to have passed the test.

8. Bearing insulation-resistance measurements should be within manufacturer's published tolerances. In the absence of manufacturer's published tolerances, the comparison shall be made to similar machines.

9. Test results of surge protection devices shall be in accordance with Section 7.19 and Section 7.20.

10. Test results of motor starter equipment shall be in accordance with Section 7.16.

11. RTD circuits should conform to design intent and/or machine protection device manufacturer's specifications.

12. Heaters should be operational.

13. Vibration amplitudes of the uncoupled and unloaded machine should not exceed values shown in Table 100.10. If values exceed those in Table 100.10, perform complete vibration analysis.

14. Verify stator winding RTD's indicate the correct temperature.

15. Investigate cause if current signature analysis tests show evidence of cracks or breaks in squirrel-cage windings, non-uniform air gaps, rolling element bearing defects, etc. Sideband magnitudes less than 40 dB from the power frequency are considered indicators of broken bars, whereas magnitudes less than 60 dB may be a warning of a developing problem. Non-

* Optional

uniform air gaps may be present when the average of the difference between the current magnitudes of the side frequencies from the highest magnitude frequency component are less than 15 dB, and a warning if less than 25 dB.

16. Perform visual inspections and off-line tests on stator winding if partial discharge levels are high based on comparison to previous tests, tests on similar machines, or values determined by the test instrument manufacturer.

* Optional

7.15.2 Rotating Machinery, Synchronous Motors and Generators

A. Visual and Mechanical Inspection

1. Inspect physical and mechanical condition.

2. Inspect anchorage, alignment, and grounding.

3. Inspect air baffles, filter media, cooling fans, stator winding, stator core, rotor, slip rings, brushes, and brush rigging or brushless exciter.

4. Inspect bearings.

5. Inspect bolted electrical connections for high resistance using one or more of the following:

    1. Use of low-resistance ohmmeter in accordance with Section 7.15.2.B.1.

    2. Verify tightness of accessible bolted electrical connections by calibrated torque-wrench method in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.12.

    3. Perform thermographic survey in accordance with Section 9.

*6. Perform special tests such as air-gap spacing and machine alignment.

7. Verify the application of appropriate lubrication and lubrication systems.

8. Verify the absence of unusual mechanical or electrical noise or signs of overheating.

9. Verify that resistance temperature detector (RTD) circuits conform to drawings.

B. Electrical Tests

1. Perform resistance measurements through bolted connections with a low-resistance ohmmeter in accordance with Section 7.15.2.A.5.1.

2. Perform insulation-resistance tests in accordance with IEEE 43.

    1. Machines larger than 200 horsepower (150 kilowatts):
    Test duration shall be for ten minutes. Calculate polarization index.

    2. Machines 200 horsepower (150 kilowatts) and less:
    Test duration shall be for one minute. Calculate the dielectric-absorption ratio.

3. Perform dc dielectric withstand voltage tests on machines rated at 2300 volts and greater in accordance with IEEE 95.

* Optional

7.15.2 Rotating Machinery, Synchronous Motors and Generators (*continued*)

4. Perform phase-to-phase stator resistance test on machines 2300 volts and greater.

*5. Perform insulation power-factor or dissipation-factor tests.

*6. Perform power-factor tip-up tests.

*7. Perform surge comparison tests.

8. Perform insulation-resistance test on insulated bearings in accordance with manufacturer's published data.

9. Test surge protection devices in accordance with Section 7.19 and Section 7.20.

10. Test motor starter in accordance with Section 7.16.

11. Perform resistance tests on resistance temperature detector (RTD) circuits.

12. Verify operation of machine space heater.

*13. Perform vibration test while machine is running under load.

14. Perform insulation-resistance tests on the main rotating field winding, the exciter-field winding, and the exciter-armature winding in accordance with IEEE 43.

*15. Perform an ac voltage-drop test on all rotating field poles.

*16. Perform a high-potential test on the excitation system in accordance with IEEE 421.3.

17. Measure resistance of machine-field winding, exciter-stator winding, exciter-rotor windings, and field discharge resistors.

*18. Perform front-to-back resistance tests on diodes and gating tests of silicon-controlled rectifiers for field application semiconductors.

19. Prior to re-energizing, apply voltage to the exciter supply and adjust exciter-field current to nameplate value.

*20. Verify that the field application timer and the enable timer for the power-factor relay have been tested and set to the motor drive manufacturer's recommended values.

*21. Record stator current, stator voltage, and field current for the complete acceleration period including stabilization time for a normally loaded starting condition. From the recording determine the following information:

    1. Bus voltage prior to start.

* Optional

2. Voltage drop at start.

3. Bus voltage at machine full-load.

4. Locked-rotor current.

5. Current after synchronization but before loading.

6. Current at maximum loading.

7. Acceleration time to near synchronous speed.

8. RPM just prior to synchronization.

9. Field application time.

10. Time to reach stable synchronous operation.

*22. Plot a V-curve of stator current versus excitation current at approximately 50 percent load to check correct exciter operation.

*23. If the range of exciter adjustment and machine loading permit, reduce excitation to cause power factor to fall below the trip value of the power-factor relay. Verify relay operation.

*24. Measure bearing temperatures while machine is running under load.

25. Perform current signature analysis test while machine is running under load.

*26. Perform partial discharge test.

C. Test Values – Visual and Mechanical

1. Inspection (7.15.2.A.3)

   1. Air baffles should be clean.

   2. Filter media should be clean.

   3. Stator winding should be clean showing no evidence of overheating, cracking, tracking, or surface partial discharge activity.

   4. The stator core should be clean with no ventilation duct blockage, lamination looseness, burning, or rubs from contact with the rotor.

   5. The rotor field winding should be clean, show no evidence of burning, movement of coils on poles, or cracks in insulation or pole washers.

* Optional

7.15.2 Rotating Machinery, Synchronous Motors and Generators (*continued*)

6. Cooling fans should operate.

7. Bearings should be inspected for evidence of overheating, rubs, rolling element damage contamination, electrical damage, and lack of lubrication.

8. Slip ring wear should be within manufacturer's tolerances for continued use.

9. Brushes should be within manufacturer's tolerances for continued use.

10. Brush rigging should be intact.

11. Brushless exciter, including rotating diode rectifier and field discharge resistor should be inspected for contamination, signs of overheating, or other defects.

2. Compare bolted connection resistance values to values of similar connections. Investigate any values that deviate from similar bolted connections by more than 50 percent of the lowest value. (7.15.2.A.5.1)

3. Bolt-torque levels should be in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.12. (7.15.2.A.5.2)

4. Results of thermographic survey shall be in accordance with Section 9. (7.15.2.A.5.3)

D. Test Values – Electrical

1. Compare bolted connection resistance values to values of similar connections. Investigate any values that deviate from similar bolted connections by more than 50 percent of the lowest value.

2. The dielectric absorption ratio or polarization index shall be compared to previously obtained results and should not be less than 1.0. The recommended minimum insulation resistance ($IR_{1\,min}$) test results in megohms should be corrected to 40° C and read as follows:

   1. $IR_{1\,min}$ = kV + 1 for most windings made before 1970, all field windings and others not described in 2.2 and 2.3.

      (kV is the rated machine terminal-to-terminal voltage in rms kV)

   2. $IR_{1\,min}$ = 100 megohms for ac windings built after 1970 (form-wound coils).

   3. $IR_{1\,min}$ = 5 megohms for most machines and random-wound stator coils and form-wound coils rated below 1 kV and dc armatures.

      NOTE: Dielectric withstand voltage, high-potential, and surge comparison tests shall not be performed on machines having values lower than those indicated above**.**

* Optional

7.15.2 Rotating Machinery, Synchronous Motors and Generators (*continued*)

3. If no evidence of distress or insulation failure is observed by the end of the total time of voltage application during the dielectric withstand voltage test, the test specimen is considered to have passed the test.

4. Investigate phase-to-phase stator resistance values that deviate by more than five percent.

5. Power-factor or dissipation-factor values shall be compared with previous values of similar machines.

6. Tip-up values should indicate no significant increase in power factor.

7. If no evidence of distress, insulation failure, or lack of waveform nesting is observed by the end of the total time of voltage application during the surge comparison test, the test specimen is considered to have passed the test.

8. Insulation resistance of bearings should be within manufacturer's published tolerances. In the absence of manufacturer's published tolerances, the comparison shall be made to similar machines.

9. Test results of surge protection devices shall be in accordance with Section 7.19 and Section 7.20.

10. Test results of motor starter equipment shall be in accordance with Section 7.16.

11. RTD circuits should conform to design intent and/or machine protection device manufacturer's specifications.

12. Heaters should be operational.

13. Vibration amplitudes of the uncoupled and unloaded machine should not exceed values shown in Table 100.10. If values exceed those in Table 100.10, perform complete vibration analysis.

14. The recommended minimum insulation resistance ($IR_{1\ min}$) test results in megohms should be corrected to 40° C and read as follows:

    1. $IR_{1\ min}$ = kV + 1 for most windings made before 1970, all field windings, and others not described in 2.2 and 2.3.

       (kV is the rated machine terminal-to-terminal voltage in rms kV)

    2. $IR_{1\ min}$ = 100 megohms for ac windings built after 1970 (form-wound coils).

* Optional

7.15.2 Rotating Machinery, Synchronous Motors and Generators (*continued*)

3. IR $_{1\ min}$ = 5 megohms for most machines and random-wound stator coils and form-wound coils rated below 1 kV and dc armatures.

   NOTE: Dielectric withstand voltage, high-potential, and surge comparison tests shall not be performed on machines having values lower than those indicated above.

15. The pole-to-pole ac voltage drop shall not exceed 10 percent variance between poles.

16. If no evidence of distress or insulation failure is observed by the end of the total time of voltage application during the dielectric withstand voltage, test, the winding is considered to have passed the test.

17. The measured resistance values of motor-field windings, exciter-stator windings, exciter-rotor windings, and field-discharge resistors shall be compared to manufacturer's recommended values.

18. Resistance test results of diodes and gating tests of silicon-controlled rectifiers should be in accordance with industry standards and system design requirements.

19. Exciter power supply should allow exciter-field current to be adjusted to nameplate value.

20. Application timer and enable timer for power factor relay test results should comply with manufacturer's recommended values.

21. Recorded values should be in accordance with system design requirements.

22. Plotted V-curve should indicate correct exciter operation.

23. When reduced excitation falls below trip value for the power-factor relay, the relay should operate

24. Verification that bearing temperature detectors are indicating the correct temperatures.

25. Investigate cause if current signature analysis tests show evidence of non-uniform air gaps, rolling element bearing defects, etc. Sideband magnitudes less than 40 dB from the power frequency are considered indicators of broken bars, whereas, magnitudes of less than 60 dB may be a warning of a developing problem. Non-uniform air gaps may be present when the average of the difference between the current magnitudes of the side frequencies from the highest magnitude frequency component are less than 15 dB, and a warning if less than 25 dB.

26. Perform visual inspections and off-line tests on stator windings if partial discharge levels are high based on comparison to previous tests, tests on similar machines, or values determined by the test instrument manufacturer.

* Optional

7.15.3 Rotating Machinery, DC Motors and Generators

A. Visual and Mechanical Inspection

1. Inspect physical and mechanical condition.

2. Inspect anchorage, alignment, and grounding.

3. Inspect air baffles, field poles and yokes, armature, cooling fans, commutator, brushes, and brush rigging, and bearings.

4. Inspect bolted electrical connections for high resistance using one or more of the following:

    1. Use of low-resistance ohmmeter in accordance with Section 7.15.3.B.1.

    2. Verify tightness of accessible bolted electrical connections by calibrated torque-wrench method in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.12.

    3. Perform thermographic survey in accordance with Section 9.

5. Inspect commutator and tachometer generator.

*6. Perform special tests such as air-gap spacing and machine alignment.

B. Electrical Tests

1. Perform resistance measurements through bolted connections with a low-resistance ohmmeter in accordance with Section 7.15.3.A.4.1.

2. Perform insulation-resistance tests on all windings in accordance with IEEE 43.

    1. Machines larger than 200 horsepower (150 kilowatts):
       Test duration shall be for ten minutes. Calculate polarization index.

    2. Machines 200 horsepower (150 kilowatts) and less:
       Test duration shall be for one minute. Calculate the dielectric absorption ratio.

3. Perform high-potential test in accordance with NEMA MG 1, paragraph 3.01.

*4. Perform an ac voltage-drop test on all field poles.

5. Measure armature running current and field current or voltage. Compare to nameplate.

*6. Perform vibration tests while machine is running under load.

* Optional

7.15.3 Rotating Machinery, DC Motors and Generators (*continued)*

7. Verify that all protective devices are in accordance with Section 7.16.

C. Test Values – Visual and Mechanical

1. Inspection (7.15.3.A.3)

    1. Air baffles should be clean

    2. Filter media should be clean.

    3. Field poles and yokes should show no evidence of overheating, contamination, or other defects.

    4. The armature and commutator should show no evidence of overheating, contamination, or other defects. Commutator wear and marking patterns should be evaluated to determine if they indicate defects.

    5. Cooling fans should operate.

    6. Brushes should be within manufacturer's tolerances for continued use.

    7. Brush rigging should be intact.

    8. Bearings should be inspected for evidence of overheating, rubs, rolling element damage, contamination, electrical damage, and lack of lubrication.

2. Compare bolted connection resistance values to values of similar connections. Investigate any values that deviate from similar bolted connections by more than 50 percent of the lowest value. (7.15.3.A.4.1)

3. Bolt-torque levels should be in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.12. (7.15.3.A.4.2)

4. Results of thermographic survey shall be in accordance with Section 9. (7.15.3.A.4.3)

5. Commutator and tachometer generator should be in accordance with manufacturer's published data and/or system design. (7.15.3.A.5)

6. Air-gap spacing and machine alignment should be within manufacturer's recommendations. (7.15.3.A.6)

* Optional

7.15.3 Rotating Machinery, DC Motors and Generators (*continued)*

D. Test Values – Electrical

1. Compare bolted connection resistance values to values of similar connections. Investigate any values that deviate from similar bolted connections by more than 50 percent of the lowest value.

2. The dielectric absorption ratio or polarization index shall be compared to previously obtained results and should not be less than 1.0. The recommended minimum insulation resistance
($IR_{1\ min}$) test results in megohms should be corrected to 40° C and read as follows:

   1. $IR_{1\ min}$ = kV + 1 for most windings made before 1970, all field windings, and others not described in 2.2 and 2.3.

      (kV is the rated machine terminal-to-terminal voltage, in rms kV)

   2. $IR_{1\ min}$ = 100 megohms for ac windings built after 1970 (form-wound coils).

   3. $IR_{1\ min}$ = 5 megohms for most machines and random-wound stator coils and form-wound coils rated below 1 kV and dc armatures.

      NOTE: Dielectric withstand voltage, high-potential, and surge comparison tests shall not be performed on machines having values lower than those indicated above.

3. If no evidence of distress or insulation failure is observed by the end of the total time of voltage application during the dielectric withstand test, the winding is considered to have passed the test.

4. The individual pole-to-pole AC voltage drop shall not exceed ten percent variance from the average value (average value = test voltage divided by number of coils) applicable to all of the poles.

5. Measured running current and field current or voltage should be comparable to nameplate data.

6. Vibration amplitudes of the uncoupled and unloaded machine should not exceed values shown in Table 100.10. If values exceed those in Table 100.10, perform complete vibration analysis.

7. Test results of motor starter equipment shall be in accordance with Section 7.16.

* Optional

7.16.1.1 Motor Control, Motor Starters, Low-Voltage

A. Visual and Mechanical Inspection

1. Inspect physical and mechanical condition.

2. Inspect anchorage, alignment, and grounding.

*3. Prior to cleaning the unit, perform as-found tests.

4. Clean the unit.

5. Inspect contactors.

    1. Verify mechanical operation.

    2. Inspect and adjust contact gap, wipe, alignment, and pressure in accordance with manufacturer's published data.

*6. Motor-Running Protection

    1. Verify overload element rating/motor protection settings are correct for application.

    2. If motor-running protection is provided by fuses, verify correct fuse rating considering motor characteristics and power-factor correction capacitors.

7. Inspect bolted electrical connections for high resistance using one or more of the following:

    1. Use of a low-resistance ohmmeter in accordance with Section 7.16.1.1.B.1.

    2. Verify tightness of accessible bolted electrical connections by calibrated torque-wrench method in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.12.

    3. Perform a thermographic survey in accordance with Section 9.

8. Use appropriate lubrication on moving current-carrying parts and on moving and sliding surfaces.

9. Perform as-left tests.

* Optional

7. INSPECTION AND TEST PROCEDURES

7.16.1.1 Motor Control, Motor Starters, Low-Voltage (*continued*)

B. Electrical Tests

1. Perform resistance measurements through bolted connections with a low-resistance ohmmeter in accordance with Section 7.16.1.1.A.7.1.

2. Perform insulation-resistance tests for one minute on each pole, phase-to-phase and phase-to-ground with starter closed, and across each open pole. Test voltage shall be in accordance with manufacturer's published data or Table 100.1.

*3. Perform insulation-resistance tests on all control wiring with respect to ground. The applied potential shall be 500 volts dc for 300-volt rated cable and 1000 volts dc for 600-volt rated cable. Test duration shall be one minute. For units with solid-state components, follow manufacturer's recommendation.

4. Test motor protection devices in accordance with manufacturer's published data. In the absence of manufacturer's data, use Section 7.9.

5. Test circuit breakers in accordance with Section 7.6.1.1.

6. Perform operational tests by initiating control devices.

C. Test Values – Visual and Mechanical

1. Compare bolted connection resistance values to values of similar connections. Investigate values which deviate from those of similar bolted connections by more than 50 percent of the lowest value. (7.16.1.1.A.7.1)

2. Bolt-torque levels should be in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.12. (7.16.1.1.A.7.2)

3. Results of the thermographic survey shall be in accordance with Section 9. (7.16.1.1.A.7.3)

D. Test Values – Electrical

1. Compare bolted connection resistance values to values of similar connections. Investigate values which deviate from those of similar bolted connections by more than 50 percent of the lowest value.

2. Insulation-resistance values should be in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.1. Values of insulation resistance less than this table or manufacturer's recommendations should be investigated.

3. Insulation-resistance values of control wiring should be comparable to previously obtained results but not less than two megohms.

* Optional

7.16.1.1 Motor Control, Motor Starters, Low-Voltage (*continued*)

4. Motor protection parameters shall be in accordance with manufacturer's published data or Section 7.9.

5. Circuit breaker test results shall be in accordance with Section 7.6.1.1.

6. Control devices should perform in accordance with system design and/or requirements.

* Optional

7.16.1.2 Motor Control, Motor Starters, Medium-Voltage

A. Visual and Mechanical Inspection

1. Inspect physical and mechanical condition.

2. Inspect anchorage, alignment, and grounding.

*3. Prior to cleaning the unit, perform as-found tests.

4. Clean the unit.

5. Inspect bolted electrical connections for high resistance using one or more of the following:

    1. Use of a low-resistance ohmmeter in accordance with Section 7.16.1.2.B.1.

    2. Verify tightness of accessible bolted electrical connections by calibrated torque-wrench method in accordance with manufacturer’s published data. In the absence of manufacturer’s published data, use Table 100.12.

    3. Perform a thermographic survey in accordance with Section 9.

6. Test electrical and mechanical interlock systems for correct operation and sequencing.

7. Verify correct barrier and shutter installation and operation.

8. Exercise active components and confirm correct operation of indicating devices.

9. Inspect contactors.

    1. Verify mechanical operation.

    2. Inspect and adjust contact gap, wipe, alignment, and pressure in accordance with manufacturer’s published data.

10. Compare overload protection rating with motor nameplate to verify correct size. Set adjustable or programmable devices according to the protective device coordination study.

11. Use appropriate lubrication on moving current-carrying parts and on moving and sliding surfaces.

12. Perform as-left tests.

* Optional

7.16.1.2 Motor Control, Motor Starters, Medium-Voltage (*continued*)

B. Electrical Tests

1. Perform resistance measurements through bolted connections with a low-resistance ohmmeter in accordance with Section 7.16.1.2.A.5.1.

2. Perform insulation-resistance tests on contactor(s) for one minute, phase-to-ground and phase-to-phase with the contactor closed, and across each open contact. Test voltage shall be in accordance with manufacturer's published data or Table 100.1.

*3. Perform insulation-resistance tests on all control wiring with respect to ground. The applied potential shall be 500 volts dc for 300-volt rated cable and 1000 volts dc for 600-volt rated cable. Test duration shall be one minute. For units with solid-state components, follow manufacturer's recommendation.

*4 Perform magnetron atmospheric condition (MAC) test on each vacuum interrupter.

*5. Perform a dielectric withstand voltage test in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.19.

6. Perform vacuum bottle integrity test (dielectric withstand voltage) across each vacuum bottle with the contacts in the open position in strict accordance with manufacturer's published data. Do not exceed maximum voltage stipulated for this test.

7. Perform contact resistance tests.

8. Measure blowout coil circuit resistance.

9. Measure resistance of power fuses.

10. Energize contactor using an auxiliary source. Adjust armature to minimize operating vibration where applicable.

11. Test control power transformers in accordance with Section 7.1.B.8.

12. Test starting transformers in accordance with Section 7.2.1.

13. Test starting reactors, in accordance with Section 7.20.3.

14. Test motor protection devices in accordance with manufacturer's published data. In the absence of manufacturer's data, test in accordance with Section 7.9.

*15. Perform system function test in accordance with Section 8.

16. Verify operation of cubicle space heater.

17. Test instrument transformers in accordance with Section 7.10.

* Optional

7.16.1.2 Motor Control, Motor Starters, Medium-Voltage (*continued*)

C. Test Values – Visual and Mechanical

1. Compare bolted connection resistance values to values of similar connections. Investigate values which deviate from those of similar bolted connections by more than 50 percent of the lowest value. (7.16.1.2.A.5.1)

2. Bolt-torque levels should be in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.12. (7.16.1.2.A.5.2)

3. Results of the thermographic survey shall be in accordance with Section 9. (7.16.1.2.A.5.3)

4. Electrical and mechanical interlocks should operate in accordance with system design. (7.16.1.2.A.6)

5. Barrier and shutter installation and operation should be in accordance with manufacturer's design. (7.16.1.2.A.7)

6. Indicating devices should operate in accordance with system design. (7.16.1.2.A.8)

D. Test Values – Electrical

1. Compare bolted connection resistance values to values of similar connections. Investigate values which deviate from those of similar bolted connections by more than 50 percent of the lowest value.

2. Insulation-resistance values should be in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.5. Values of insulation resistance less than this table or manufacturer's recommendations should be investigated.

3. Insulation-resistance values of control wiring should be comparable to previously obtained results but not less than two megohms.

4. Evaluate each vacuum interrupter in accordance with test equipment manufacturer's instructions.

5. If no evidence of distress or insulation failure is observed by the end of the total time of voltage application during the dielectric withstand voltage test, the test specimen is considered to have passed the test.

6. If no evidence of distress or insulation failure is observed by the end of the total time of voltage application during the vacuum bottle integrity test, the vacuum bottle is considered to have passed the test.

7. Microhm or dc millivolt drop values should not exceed the high levels of the normal range as indicated in the manufacturer's published data. If manufacturer's data is not available,

* Optional

7.16.1.2 Motor Control, Motor Starters, Medium-Voltage (*continued*)

investigate values which deviate from those of similar connections by more than 50 percent of the lowest value.

8. Resistance values of blowout coils should be in accordance with manufacturer's published data.

9. Resistance values should not deviate by more than 15 percent between identical fuses.

10. Contactor coil should operate with minimal vibration and noise.

11. Control power transformer test results shall be in accordance with Section 7.1.B.8.

12. Starting transformer test results shall be in accordance with Section 7.2.1.

13. Starting reactor test results shall be in accordance with Section 7.20.3.

14. Motor protection parameters shall be in accordance with manufacturer's published data.

15. System function test results should be in accordance with manufacturer's published data and/or system design.

16. Heaters should be operational.

17. The results of instrument transformer tests shall be in accordance with Section 7.10.

7. INSPECTION AND TEST PROCEDURES

7.16.2.1 Motor Control, Motor Control Centers, Low-Voltage

1. Refer to Section 7.1 for appropriate inspections and tests of the motor control center bus.

2. Refer to Section 7.5.1.1 for appropriate inspections and tests of the motor control center switches.

3. Refer to Section 7.6 for appropriate inspections and tests of the motor control center circuit breakers.

4. Refer to Section 7.16.1.1 for appropriate inspections and tests of the motor control center starters.

* Optional

7.16.2.2 Motor Control, Motor Control Centers, Medium-Voltage

1. Refer to Section 7.1 for appropriate inspections and tests of the motor control center bus.

2. Refer to Section 7.5.1.2 for appropriate inspections and tests of the motor control center switches.

3. Refer to Section 7.6 for appropriate inspections and tests of the motor control center circuit breakers.

4. Refer to Section 7.16.1.2 for appropriate inspections and tests of the motor control center starters.

* Optional

7. INSPECTION AND TEST PROCEDURES

7.17 Adjustable-Speed Drive Systems

A. Visual and Mechanical Inspection

1. Inspect physical and mechanical condition.

2. Inspect anchorage, alignment, and grounding.

*3. Prior to cleaning the unit, perform as-found tests.

4. Clean the unit.

5. Ensure vent path openings are free from debris and that heat transfer surfaces are not contaminated by oil, dust, or dirt.

6. Verify correct connections of circuit boards, wiring, disconnects, and ribbon cables.

7. Motor running protection

    1. Compare drive overcurrent set points with motor full-load current rating to verify correct settings.

    2. If drive is used to operate multiple motors, compare individual overload element ratings with motor full-load current ratings.

    3. Apply minimum and maximum speed set points. Confirm set points are within limitations of the load coupled to the motor.

8. Inspect bolted electrical connections for high resistance using one or more of the following:

    1. Use of a low-resistance ohmmeter in accordance with Section 7.17.B.1.

    2. Verify tightness of accessible bolted electrical connections by calibrated torque-wrench method in accordance with manufacturer's published data or Table 100.12.

    3. Perform a thermographic survey in accordance with Section 9.

9. Verify correct fuse sizing in accordance with manufacturer's published data.

10. Perform as-left tests.

* Optional

7.17 Adjustable-Speed Drive Systems (*continued*)

B. Electrical Tests

1. Perform resistance measurements through bolted connections with low-resistance ohmmeter in accordance with Section 7.17.A.8.1.

2. Test the motor overload relay elements by injecting primary current through the overload circuit and monitoring trip time of the overload element.

3. Test input circuit breaker by primary injection in accordance with Section 7.6.

*4. Perform insulation resistance tests on all control wiring with respect to ground. The applied potential shall be 500 volts dc for 300-volt rated cable and 1000 volts dc for 600-volt rated cable. Test duration shall be one minute. For units with solid-state components, follow manufacturer's recommendation.

5. Test for the following parameters in accordance with relay calibration procedures outlined in Section 7.9 or as recommended by the manufacturer:

    1. Input phase loss protection
    2. Input overvoltage protection
    3. Output phase rotation
    4. Overtemperature protection
    5. DC overvoltage protection
    6. Overfrequency protection
    7. Drive overload protection
    8. Fault alarm outputs

6. Perform continuity tests on bonding conductors in accordance with Section 7.13.

7. Perform operational tests by initiating control devices.

    1. Slowly vary drive speed between minimum and maximum. Observe motor and load for unusual noise or vibration.
    2. Verify operation of drive from remote start/stop and speed control signals.

* Optional

7.17 Adjustable-Speed Drive Systems (*continued*)

C. Test Values – Visual and Mechanical

1. Compare bolted connection resistance values to values of similar connections. Investigate values which deviate from those of similar bolted connections by more than 50 percent of the lowest value. (7.17.A.8.1)

2. Bolt-torque levels should be in accordance with manufacturer’s published data. In the absence of manufacturer’s published data, use Table 100.12. (7.17.A.8.2)

3. Results of the thermographic survey shall be in accordance with Section 9. (7.17.A.8.3)

D. Test Values – Electrical

1. Compare bolted connection resistance values to values of similar connections. Investigate values which deviate from those of similar bolted connections by more than 50 percent of the lowest value.

2. Overload test trip times at 300 percent of overload element rating should be in accordance with manufacturer’s published time-current curve.

3. Input circuit breaker test results shall be in accordance with Section 7.6.

4. Insulation-resistance values of control wiring should be comparable to previously obtained results but not less than two megohms.

5. Relay calibration test results shall be in accordance with Section 7.9.

6. Continuity of bonding conductors shall be in accordance with Section 7.13.

7. Control devices should perform in accordance with system requirements.

7. INSPECTION AND TEST PROCEDURES

7.18.1.1 Direct-Current Systems, Batteries, Flooded Lead-Acid

A. Visual and Mechanical Inspection

1. Verify the battery area ventilation system is operable.
2. Verify existence of suitable eyewash equipment.
3. Inspect physical and mechanical condition.
4. Inspect battery support racks, mounting, battery spill containment system, anchorage, clearances, alignment, and grounding.

*5. Prior to cleaning or adding electrolyte, perform as-found tests.

6. Verify electrolyte level. Measure electrolyte specific gravity and temperature levels.
7. Verify presence of flame arresters.
8. Neutralize acid on exterior surfaces and clean with water.
9. Clean corroded/oxidized terminals and apply an oxide inhibitor.
10. Inspect bolted electrical connections for high resistance using one or more of the following:
    1. Use of a low-resistance ohmmeter.
    2. Verify tightness of accessible connections by calibrated torque-wrench in accordance with manufacturer's published data.
    3. Perform a thermographic survey under load in accordance with Section 9.
11. Perform as-left tests.

B. Electrical Tests

1. Perform resistance measurements through all bolted connections with a low-resistance ohmmeter in accordance with Section 7.18.1.1.A.10.1.
2. Measure charger float and equalizing voltage levels. Adjust to battery manufacturer's recommended settings.
3. Measure each monoblock/cell voltage and total battery voltage with charger energized and in float mode of operation.
4. Measure intercell connection resistances.
5. Perform internal ohmic measurement tests.

* Optional

7.18.1.1 Direct-Current Systems, Batteries, Flooded Lead-Acid (*continued*)

*6. Perform a load test in accordance with manufacturer's specifications or IEEE 450.

7. Measure the battery system voltage from positive-to-ground and negative-to-ground.

C. Test Values – Visual and Mechanical

1. Electrolyte level and specific gravity should be within normal limits. (7.18.1.1.A.6)

2. Compare bolted connection resistance values to values of similar connections. Investigate values which deviate from those of similar bolted connections by more than 50 percent of the lowest value. (7.18.1.1.A.10.1)

3. Bolt-torque levels should be in accordance with manufacturer's published data. (7.18.1.1.A.10.2)

4. Results of the thermographic survey shall be in accordance with Section 9. (7.18.1.1.A.10.3)

D. Test Values – Electrical

1. Compare bolted connection resistance values to values of similar connections. Investigate values which deviate from those of similar bolted connections by more than 50 percent of the lowest value.

2. Charger float and equalize voltage levels should be in accordance with battery manufacturer's published data.

3. Cell voltages should be within 0.05 volt of each other or in accordance with manufacturer's published data.

4. Compare bolted connection resistance values to values of similar connections. Investigate values which deviate from those of similar bolted connections by more than 50 percent of the lowest value.

5. Cell internal ohmic values (resistance, impedance or conductance) should not vary by more than 25 percent between identical cells that are in a fully charged state.

6. Results of load tests should be in accordance with manufacturer's published data or IEEE 450.

7. Voltage measured from positive-to-ground shall be equal in magnitude to the voltage measured from negative-to-ground.

* Optional

7.18.1.2 Direct-Current Systems, Batteries, Vented Nickel-Cadmium

A. Visual and Mechanical Inspection

1. Verify that battery area ventilation system is operable.

2. Verify existence of suitable eyewash equipment.

3. Inspect physical and mechanical condition.

4. Verify adequacy of battery support racks or cabinets, mounting, battery spill containment system, anchorage, alignment, grounding, and clearances.

*5. Prior to cleaning or adding electrolyte, perform as-found tests.

6. Verify electrolyte level. Measure pilot-cell electrolyte temperature.

7. Verify the units are clean.

8. Clean corroded/oxidized terminal surfaces and apply oxide inhibitor.

9. Inspect bolted electrical connections for high resistance using one or more of the following:

    1. Use of a low-resistance ohmmeter.

    2. Verify tightness of accessible electrical connections by calibrated torque-wrench in accordance with manufacturer's published data.

    3. Perform thermographic survey in accordance with Section 9.

10. Perform as-left tests.

B. Electrical Tests

1. Perform resistance measurements through all bolted connections with a low-resistance ohmmeter in accordance with Section 7.18.1.2.A.9.1.

2. Measure charger float and equalizing voltage levels. Adjust to battery manufacturer's recommended settings.

3. Measure each monoblock/cell voltage and total battery voltage with charger energized and in float mode of operation.

4. Measure intercell connection resistances.

* Optional

7.18.1.2 Direct-Current Systems, Batteries, Vented Nickel-Cadmium (*continued*)

*5. Perform a load test in accordance with manufacturer's published data or IEEE 1106.

6. Measure the battery system voltage from positive-to-ground and negative-to-ground.

C. Test Values – Visual and Mechanical

1. Electrolyte level shall be within normal limits. (7.18.1.2.A.6)

2. Compare bolted connection resistance values to values of similar connections. Investigate values which deviate from those of similar bolted connections by more than 50 percent of the lowest value. (7.18.1.2.A.9.1)

3. Bolt-torque levels should be in accordance with manufacturer's published data. (7.18.1.2.A.9.2)

4. Results of the thermographic survey shall be in accordance with Section 9. (7.18.1.2.A.9.3)

D. Test Values – Electrical

1. Compare bolted connection resistance values to values of similar connections. Investigate values which deviate from those of similar bolted connections by more than 50 percent of the lowest value.

2. Charger float and equalize voltage levels should be in accordance with battery manufacturer's published data.

3. Cell voltages should be in accordance with manufacturer's published data.

4. Compare bolted connection resistance values to values of similar connections. Investigate values which deviate from those of similar bolted connections by more than 50 percent of the lowest value.

5. Results of load tests should be in accordance with manufacturer's published data or IEEE 1106.

6. Voltage measured from positive-to-ground should be equal in magnitude to the voltage measured from negative-to-ground.

* Optional

7.18.1.3 Direct-Current Systems, Batteries, Valve-Regulated Lead-Acid

A. Visual and Mechanical Inspection

1. Verify the battery area ventilation system is operable.

2. Verify the existence of suitable eyewash equipment.

3. Inspect physical and mechanical condition.

4. Inspect battery support racks or cabinets, mounting, anchorage, alignment, grounding, and clearances.

*5. Prior to cleaning, perform as-found tests.

6. Neutralize acid on exterior surfaces and clean with water.

7. Clean corroded/oxidized terminals and apply an oxide inhibitor.

8. Inspect bolted electrical connections for high resistance using one or more of the following:

   1. Use of a low-resistance ohmmeter.

   2. Verify tightness of accessible connections by calibrated torque-wrench in accordance with manufacturer's published data.

   3. Perform a thermographic survey under load in accordance with Section 9.

9. Perform as-left tests.

B. Electrical Tests

1. Perform resistance measurements through all bolted connections with a low-resistance ohmmeter.

2. Measure negative post temperature.

3. Measure each monoblock/cell voltage and total battery voltage with charger energized and in float mode of operation.

4. Measure intercell connection resistances.

5. Perform internal ohmic measurement tests.

*6. Perform an annual load test in accordance with manufacturer's specifications or IEEE 1188.

7. Measure the battery system voltage from positive-to-ground and negative-to-ground.

* Optional

7.18.1.3 Direct-Current Systems, Batteries, Valve-Regulated Lead-Acid (*continued*)

C. Test Values – Visual and Mechanical

1. Compare bolted connection resistance values to values of similar connections. Investigate values which deviate from those of similar bolted connections by more than 50 percent of the lowest value. (7.18.1.3.A.8.1)

2. Bolt-torque levels should be in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.12. (7.18.1.3.A.8.2)

3. Results of the thermographic survey shall be in accordance with Section 9. (7.18.1.3.A.8.3)

D. Test Values – Electrical

1. Compare bolted connection resistance values to values of similar connections. Investigate values that deviate from those of similar bolted connections by more than 50 percent of the lowest value.

2. Negative post temperature should be within manufacturer's published data or IEEE 1188.

3. Monoblock/cell voltages should be in accordance with manufacturer's published data.

4. Compare bolted connection resistance values to values of similar connections. Investigate values which deviate from those of similar bolted connections by more than 50 percent of the lowest value.

5. Monoblock/cell internal ohmic values (resistance, impedance, or conductance) should not vary by more than 25 percent between identical monoblocks/cells that are in a fully charged state.

6. Results of load tests should be in accordance with manufacturer's published data or IEEE 1188.

7. Voltage from positive-to-ground should be similar in magnitude to the voltage measured from negative-to-ground.

* Optional

7.18.2 Direct-Current Systems, Chargers

A. Visual and Mechanical Inspection

1. Inspect for physical and mechanical condition.

2. Inspect anchorage, alignment, and grounding.

*3. Prior to cleaning the unit, perform as-found tests.

4. Clean the unit.

5. Inspect all bolted electrical connections for high resistance using one or more of the following:

    1. Use of a low-resistance ohmmeter.

    2. Verify tightness of accessible bolted electrical connections by calibrated torque-wrench method in accordance with manufacturer's published data or Table 100.12.

    3. Perform a thermographic survey under load in accordance with Section 9.

6. Inspect filter and tank capacitors.

7. Verify operation of cooling fans. Clean filters if provided.

8. Perform as-left tests.

B. Electrical Tests

1. Perform resistance measurements through all bolted connections with a low-resistance ohmmeter.

2. Verify float voltage and equalize voltage.

*3. Verify high-voltage shutdown settings.

4. Verify correct load sharing (parallel chargers).

5. Verify calibration of meters in accordance with Section 7.11.

6. Verify operation of alarms.

7. Measure and record input and output voltage and current.

8. Measure and record ac ripple current and voltage imposed on battery.

*9. Perform full load testing and verify current limit of charger.

* Optional

7.18.2 Direct-Current Systems, Chargers (*continued*)

C. Test Values – Visual and Mechanical

1. Compare bolted connection resistance values to values of similar connections. Investigate values which deviate from those of similar bolted connections by more than 50 percent of the lowest value. (7.18.2.A.5.1)

2. Bolt-torque levels should be in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.12. (7.18.2.A.5.2)

3. Results of the thermographic survey shall be in accordance with Section 9. (7.18.2.A.5.3)

D. Test Values – Electrical

1. Compare bolted connection resistance values to values of similar connections. Investigate values that deviate from those of similar bolted connections by more than 50 percent of the lowest value.

2. Float and equalize settings should be in accordance with the battery manufacturer's published data.

3. High-voltage shutdown settings should be in accordance with manufacturer's published data.

4. Results of load sharing between parallel chargers should be in accordance with system design specifications.

5. Results of meter calibration should be in accordance with Section 7.11.

6. Results of alarm operation should be in accordance with manufacturer's published data and system design.

7. Input and output voltage should be in accordance with manufacturer's published data.

8. AC ripple current and voltage imposed on the battery should be within charger manufacturer specification and in accordance with the battery manufacturer's published data.

9. Charger should be capable of producing rated full-load output and current limit should be in accordance with the specified settings.

* Optional

7.18.3 Direct-Current Systems, Rectifiers

– RESERVED –

* Optional

7. INSPECTION AND TEST PROCEDURES

7.19.1 Surge Arresters, Low-Voltage Surge Protection Devices

A. Visual and Mechanical Inspection

1. Inspect physical and mechanical condition.

2. Inspect anchorage, alignment, and grounding.

*3. Prior to cleaning the unit, perform as-found tests.

4. Clean the unit.

5. Inspect bolted electrical connections for high resistance using one or more of the following:

    1. Use of a low-resistance ohmmeter in accordance with Section 7.19.1.B.1.

    2. Verify tightness of accessible bolted electrical connections by calibrated torque-wrench method in accordance with manufacturer's published data or Table 100.12.

6. Verify that the ground lead on each device is individually attached to a ground bus or ground electrode.

7. Perform as-left tests.

8. Verify healthy status in accordance with manufacturer's data.

B. Electrical Tests

1. Perform resistance measurements through bolted connections with a low-resistance ohmmeter in accordance with Section 7.19.1.A.5.1.

2. Perform insulation-resistance test on each arrester, from the phase terminal to ground. Apply voltage in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.1.

3. Test grounding connection in accordance with Section 7.13.

* Optional

7.19.1 Surge Arresters, Low-Voltage Surge Protection Devices (*continued*)

C. Test Values – Visual and Mechanical

1. Compare bolted connection resistance values to values of similar connections. Investigate values which deviate from those of similar bolted connections by more than 50 percent of the lowest value. (7.19.1.A.5.1)

2. Bolt-torque levels should be in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.12. (7.19.1.A.5.2)

D. Test Values – Electrical

1. Compare bolted connection resistance values to values of similar connections. Investigate values which deviate from those of similar bolted connections by more than 50 percent of the lowest value.

2. Insulation-resistance values should be in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.1. Values of insulation resistance less than this table or manufacturer's recommendations should be investigated.

3. Resistance between the arrester ground terminal and the ground system should be less than 0.5 ohm and in accordance with Section 7.13.

7.19.2 Surge Arresters, Medium- and High-Voltage Surge Protection Devices

A. Visual and Mechanical Inspection

1. Inspect physical and mechanical condition.

2. Inspect anchorage, alignment, and grounding.

*3. Prior to cleaning the unit, perform as-found tests.

4. Clean the unit.

5. Inspect bolted electrical connections for high resistance using one or more of the following:

    1. Use of a low-resistance ohmmeter in accordance with Section 7.19.2.B.1.

    2. Verify tightness of accessible bolted electrical connections by calibrated torque-wrench method in accordance with manufacturer's published data or Table 100.12.

6. Verify that the ground lead on each device is individually attached to a ground bus or ground electrode.

7. Verify that stroke counter, if present, is correctly mounted and electrically connected.

8. Perform as-left tests.

B. Electrical Tests

1. Perform resistance measurements through bolted connections with a low-resistance ohmmeter in accordance with Section 7.19.2.A.5.1.

2. Perform insulation-resistance tests from phase terminals(s) to case for one minute. Test voltage and minimum resistance shall be in accordance with manufacturer's published data. In the absence of manufacturer's published data, refer to Table 100.1.

3. Test the grounding connection in accordance with Section 7.13.

*4. Perform a watts-loss test in accordance with test equipment manufacturer's published data.

* Optional

7.19.2 Surge Arresters, Medium- and High-Voltage Surge Protection Devices (*continued*)

C. Test Values – Visual and Mechanical

1. Compare bolted connection resistance values to values of similar connections. Investigate values which deviate from those of similar bolted connections by more than 50 percent of the lowest value. (7.19.2.A.5.1)

2. Bolt-torque levels should be in accordance with manufacturer’s published data. In the absence of manufacturer’s published data, use Table 100.12. (7.19.2.A.5.2)

D. Test Values – Electrical

1. Compare bolted connection resistance values to values of similar connections. Investigate values which deviate from those of similar bolted connections by more than 50 percent of the lowest value.

2. Insulation-resistance values should be in accordance with manufacturer’s published data. In the absence of manufacturer’s published data, use Table 100.1. Values of insulation resistance less than this table or manufacturer’s recommendations should be investigated

3. Resistance between the arrester ground terminal and the ground system should be less than 0.5 ohm and in accordance with Section 7.13.

4. Watts loss values are evaluated on a comparison basis with similar units and test equipment manufacturer’s published data.

* Optional

7. INSPECTION AND TEST PROCEDURES

7.20.1 Capacitors and Reactors, Capacitors

A. Visual and Mechanical Inspection

1. Inspect physical and mechanical condition.

2. Inspect anchorage, alignment, grounding, and required clearances.

*3. Prior to cleaning the unit, perform as-found tests.

4. Clean the unit.

5. Verify that capacitors are electrically connected in their specified configuration.

6. Inspect bolted electrical connections for high resistance using one or more of the following:

    1. Use of a low-resistance ohmmeter in accordance with Section 7.20.1.B.1.

    2. Verify tightness of accessible bolted electrical connections by calibrated torque-wrench method in accordance with manufacturer's published data or Table 100.12.

    3. Perform a thermographic survey in accordance with Section 9.

7. Perform as-left tests.

B. Electrical Tests

1. Perform resistance measurements through bolted connections with a low-resistance ohmmeter in accordance with Section 7.20.1.A.6.1.

2. Perform insulation-resistance tests from phase terminal(s) to case for one minute. Test voltage and minimum resistance should be in accordance with manufacturer's published data. In the absence of manufacturer's published data, refer to Table 100.1.

3. Measure the capacitance of all terminal combinations.

4. Measure resistance of internal discharge resistors.

* Optional

7. INSPECTION AND TEST PROCEDURES

7.20.1 Capacitors and Reactors, Capacitors (*continued*)

C. Test Values – Visual and Mechanical

1. Compare bolted connection resistance values to values of similar connections. Investigate values which deviate from those of similar bolted connections by more than 50 percent of the lowest value. (7.20.1.A.6.1)

2. Bolt-torque levels should be in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.12. (7.20.1.A.6.2)

3. Results of the thermographic survey shall be in accordance with Section 9. (7.20.1.A.6.3)

D. Test Values – Electrical

1. Compare bolted connection resistance values to values of similar connections. Investigate values which deviate from those of similar bolted connections by more than 50 percent of the lowest value.

2. Insulation-resistance values should be in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.1. Values of insulation resistance less than this table or manufacturer's recommendations should be investigated

3. Investigate capacitance values differing from manufacturer's published data.

4. Investigate discharge resistor values differing from manufacturer's published data. In accordance with NFPA 70, Article 460, residual voltage of a capacitor shall be reduced to 50 volts in the following time intervals after being disconnected from the source of supply:

Rated Voltage	Discharge Time
≤ 600 volts	1 minute
> 600 volts	5 minutes

* Optional

7. INSPECTION AND TEST PROCEDURES

7.20.2 Capacitors and Reactors, Capacitor Control Devices

– RESERVED –

* Optional

7.20.3.1 Capacitors and Reactors, Reactors (Shunt and Current-Limiting), Dry Type

A. Visual and Mechanical Inspections

1. Inspect physical and mechanical condition.

2. Inspect anchorage, alignment, and grounding.

*3. Prior to cleaning the unit, perform as-found tests.

4. Clean the unit.

5. Inspect bolted electrical connections for high resistance using one or more of the following:

    1. Use of a low-resistance ohmmeter in accordance with Section 7.20.3.1.B.1.

    2. Verify tightness of accessible bolted electrical connections by calibrated torque-wrench method in accordance with manufacturer's published data or Table 100.12.

    3. Perform a thermographic survey in accordance with Section 9.

6. Verify that tap connections are as specified.

7. Perform as-left tests.

B. Electrical Tests

1. Perform resistance measurements through bolted connections with low-resistance ohmmeter in accordance with Section 7.20.3.1.A.5.1.

2. Perform winding-to-ground insulation-resistance tests. Apply voltage in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.1.

3. Measure winding resistance.

*4. Perform dielectric withstand voltage tests on each winding-to-ground.

*5. Perform online partial-discharge survey in accordance with Section 11.

* Optional

7.20.3.1 Capacitors and Reactors, Reactors (Shunt and Current-Limiting), Dry Type (*continued*)

C. Test Values – Visual and Mechanical

1. Compare bolted connection resistance values to values of similar connections. Investigate values which deviate from those of similar bolted connections by more than 50 percent of the lowest value. (7.20.3.1.A.5.1)

2. Bolt-torque levels should be in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.12. (7.20.3.1.A.5.2)

3. Results of the thermographic survey shall be in accordance with Section 9. (7.20.3.1.A.5.3)

D. Test Values – Electrical

1. Compare bolted connection resistance values to values of similar connections. Investigate values which deviate from those of similar bolted connections by more than 50 percent of the lowest value.

2. Insulation-resistance values should be in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.1. Values of insulation resistance less than this table or manufacturer's recommendations should be investigated.

3. If winding-resistance test results vary by more than one percent from factory acceptance tests, consult the manufacturer.

4. AC dielectric withstand test voltage shall not exceed 75 percent of factory acceptance test voltage for one-minute duration. DC dielectric withstand test voltage shall not exceed 100 percent of the factory acceptance rms test voltage for one-minute duration. If no evidence of distress or insulation failure is observed by the end of the total time of voltage application during the dielectric withstand voltage test, the test specimen is considered to have passed the test.

5. Results of online partial-discharge survey should be in accordance with manufacturer's published data. In the absence of manufacturer's published data, refer to Table 100.23.

* Optional

7.20.3.2 Capacitors and Reactors, Reactors (Shunt and Current-Limiting), Liquid-Filled

A. Visual and Mechanical Inspection

1. Inspect physical and mechanical condition.

2. Inspect anchorage, alignment, and grounding.

*3. Prior to cleaning the unit, perform as-found tests.

4. Clean the unit.

5. Verify settings and operation of all temperature devices.

6. Verify that cooling fans and pumps operate correctly and that fan and pump motors have correct overcurrent protection.

7. Verify operation of all alarm, control, and trip circuits from temperature and level indicators, pressure relief device, and fault pressure relay.

8. Inspect bolted electrical connections for high resistance using one or more of the following:

    1. Use of a low-resistance ohmmeter in accordance with Section 7.20.3.2.B.1.

    2. Verify tightness of accessible bolted electrical connections by calibrated torque-wrench method in accordance with manufacturer's published data or Table 100.12.

    3. Perform a thermographic survey in accordance with Section 9.

9. Perform as-left tests.

10. Verify correct liquid level in all tanks and bushings.

11. Verify that positive pressure is maintained on nitrogen-blanketed reactors.

12. Perform mechanical maintenance, inspections, and tests as recommended by the manufacturer.

13. Verify that tap connections are as specified.

* Optional

7.20.3.2 Capacitors and Reactors, Reactors (Shunt and Current-Limiting), Liquid-Filled

B. Electrical Tests

1. Perform resistance measurements through bolted connections with a low-resistance ohmmeter in accordance with Section 7.20.3.2.A8.1.

2. Perform winding-to-ground insulation-resistance tests. Apply voltage in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.1. Calculate polarization index.

3. Perform insulation power-factor or dissipation-factor tests on windings in accordance with the test equipment manufacturer's published data.

4. Perform power-factor or dissipation-factor tests on each bushing equipped with a power-factor/ capacitance tap. In the absence of a power-factor/ capacitance tap, perform hot-collar tests. These tests shall be in accordance with the test equipment manufacturer's published data.

5. Measure winding resistance.

6. Measure the percentage of oxygen in the nitrogen gas blanket.

7. Remove a sample of insulating liquid in accordance with ASTM D923. Sample shall be tested for the following:

    1. Dielectric breakdown voltage: ASTM D1816

    2. Acid neutralization number: ASTM D974

    3. Specific gravity: ASTM D1298

    4. Interfacial tension: ASTM D971

    5. Color: ASTM D1500

    6. Visual condition: ASTM D1524

    7. Water in insulating liquids: ASTM D1533

    *8. Measure power factor or dissipation factor in accordance with ASTM D924

8. Remove a sample of insulating liquid in accordance with ASTM D3613 and perform dissolved-gas-analysis (DGA) in accordance with IEEE C57.104 or ASTM D3612.

* Optional

7.20.3.2 Capacitors and Reactors, Reactors (Shunt and Current-Limiting), Liquid-Filled

C. Test Values – Visual and Mechanical

1. Operation of temperature devices should be in accordance with system requirements. (7.20.3.2.A.5)

2. Operation of cooling fans and pumps should be in accordance with manufacturer's recommendations and system design. (7.20.3.2.A.6)

3. Compare bolted connection resistance values to values of similar connections. Investigate values which deviate from those of similar bolted connections by more than 50 percent of the lowest value. (7.20.3.2.A.8.1)

4. Bolt-torque levels should be in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.12. (7.20.3.2.A.8.2)

5. Results of the thermographic survey shall be in accordance with Section 9. (7.20.3.2.A.8.3)

6. Liquid levels should be in accordance with manufacturer's indicated tolerances. (7.20.3.2.A.10)

7. Positive pressure should be indicated on the pressure gauge for gas-blanketed reactors. (7.20.3.2.A.11)

D. Test Values – Electrical

1. Compare bolted connection resistance values to values of similar connections. Investigate values which deviate from those of similar bolted connections by more than 50 percent of the lowest value.

2. Insulation-resistance values should be in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.1. Values of insulation resistance less than this table or manufacturer's recommendations should be investigated. The polarization index should be greater than 1.0 and should be compared to previous data.

3. Maximum power-factor or dissipation-factor values of liquid-filled reactors should be in accordance with manufacturer's published data. In the absence of manufacturer's data, compare to test equipment manufacturer's published data. Representative values are indicated in Table 100.3.

4. Power-factor or dissipation-factor and capacitance values should be within ten percent of nameplate rating for bushings. Hot collar tests are evaluated on a milliampere/milliwatt loss basis, and the results should be compared to values of similar bushings.

* Optional

7.20.3.2 Capacitors and Reactors, Reactors (Shunt and Current-Limiting), Liquid-Filled

5. Consult the manufacturer if winding-resistance values vary more than two percent from factory acceptance tests.

6. Investigate presence of oxygen in the nitrogen gas blanket.

7. Insulating liquid values should be in accordance with Table 100.4.

8. Evaluate results of dissolved-gas analysis in accordance with IEEE C57.104. Compare results to previous test data.

* Optional

7.20.4 RESISTORS

A. Visual and Mechanical Inspection

1. Inspect physical and mechanical condition.

2. Inspect anchorage, alignment, and grounding.

*3. Prior to cleaning the unit, perform as-found tests.

4. Clean the unit.

5. Inspect bolted electrical connections for high resistance using one or more of the following:

    1. Use of a low-resistance ohmmeter in accordance with Section 7.20.3.2.B.1.

    2. Verify tightness of accessible bolted electrical connections by calibrated torque-wrench method in accordance with manufacturer's published data or Table 100.12.

    3. Perform a thermographic survey in accordance with Section 9.

6. Perform as-left tests.

7. Verify correct liquid level in all tanks and bushings.

8. Perform mechanical maintenance, inspections, and tests as recommended by the manufacturer.

B. Electrical Tests

1. Perform resistance measurements through bolted connections with a low-resistance ohmmeter in accordance with Section 7.20.3.2.A8.1.

2. Perform resistor-to-ground insulation-resistance tests. Apply voltage in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.1. Calculate polarization index.

3. Perform power-factor or dissipation-factor tests on each bushing equipped with a power-factor/ capacitance tap. In the absence of a power-factor/ capacitance tap, perform hot-collar tests. These tests shall be in accordance with the test equipment manufacturer's published data.

5. Measure resistor resistance.

6. Perform electrical tests on current transformer in accordance with Section 7.10.1.

* Optional

7.20.4 RESISTORS

C. Test Values – Visual and Mechanical

1. Compare bolted connection resistance values to values of similar connections. Investigate values which deviate from those of similar bolted connections by more than 50 percent of the lowest value. (7.20.3.2.A.8.1)

2. Bolt-torque levels should be in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.12. (7.20.3.2.A.8.2)

3. Results of the thermographic survey shall be in accordance with Section 9. (7.20.3.2.A.8.3)

D. Test Values – Electrical

1. Compare bolted connection resistance values to values of similar connections. Investigate values which deviate from those of similar bolted connections by more than 50 percent of the lowest value.

2. Insulation-resistance values should be in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.1. Values of insulation resistance less than this table or manufacturer's recommendations should be investigated.

3. Power-factor or dissipation-factor and capacitance values should be within ten percent of nameplate rating for bushings. Hot collar tests are evaluated on a milliampere/milliwatt loss basis, and the results should be compared to values of similar bushings.

4. Consult the manufacturer if resistance values vary more than two percent from factory acceptance tests.

7. INSPECTION AND TEST PROCEDURES

7.21 Outdoor Bus Structures

A. Visual and Mechanical Inspection

1. Inspect physical and mechanical condition.

2. Inspect anchorage, alignment, and grounding.

*3. Prior to cleaning the unit, perform as-found tests.

4. Inspect bolted electrical connections for high resistance using one or more of the following:

    1. Use of a low-resistance ohmmeter in accordance with Section 7.21.B.1.

    2. Verify tightness of accessible bolted electrical connections by calibrated torque-wrench method in accordance with manufacturer's published data or Table 100.12.

    3. Perform a thermographic survey in accordance with Section 9.

5. Clean the insulators.

6. Perform as-left tests.

B. Electrical Tests

1. Perform resistance measurements through bolted connections with a low-resistance ohmmeter in accordance with Section 7.21.A.4.1.

2. Measure insulation resistance of each bus, phase-to-ground with other phases grounded for equipment rated less than 46 kV. Apply voltage in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.1.

*3. Perform dielectric withstand voltage test on each bus phase, phase-to-ground with other phases grounded in accordance with manufacturer's published data. In the absence of manufacturer's data use Table 100.19. Test duration shall be for one minute.

*4. Perform insulation power-factor or dissipation-factor tests on each bus phase, phase-to-ground with other phases grounded.

* Optional

7.21 Outdoor Bus Structures (*continued*)

C. Test Values – Visual and Mechanical

1. Compare bolted connection resistance values to values of similar connections. Investigate values which deviate from those of similar bolted connections by more than 50 percent of the lowest value. (7.21.A.4.1)

2. Bolt-torque levels should be in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.12. (7.21.A.4.2)

3. Results of the thermographic survey shall be in accordance with Section 9. (7.21.A.4.3)

D. Test Values – Electrical

1. Compare bolted connection resistance values to values of similar connections. Investigate values which deviate from those of similar bolted connections by more than 50 percent of the lowest value.

2. Insulation-resistance values should be in accordance with manufacturer's published data or Table 100.1.

3. If no evidence of distress or insulation failure is observed by the end of the total time of voltage application during the dielectric withstand voltage test, the test specimen is considered to have passed the test.

4. Power-factor or dissipation-factor results shall be compared to manufacturer's published data. In absence of manufacturer's published data, the comparison shall be made to previous test results or data from the test equipment manufacturers.

* Optional

7. INSPECTION AND TEST PROCEDURES

7.22.1 Emergency Systems, Engine Generator

NOTE: The prime mover is not addressed in these specifications.

A. Visual and Mechanical Inspection

1. Inspect physical and mechanical condition.

2. Inspect anchorage, alignment, and grounding.

*3. Prior to cleaning the unit, perform as-found tests.

4. Clean the unit.

5. Perform as-left tests.

B. Electrical and Mechanical Tests

1. Perform insulation-resistance test on the generator winding-to-ground in accordance with IEEE 43. Calculate the polarization index.

2. Test protective relay devices in accordance with Section 7.9.

3. Functionally test engine shutdown for low oil pressure, overtemperature, overspeed, and other protection features as applicable.

*4. Perform vibration test for each main bearing cap.

5. Conduct performance test in accordance with NFPA 110.

6. Verify correct functioning of governor and regulator.

C. Test Values – Visual and Mechanical

1. Anchorage, alignment, and grounding should be in accordance with manufacturer's published data and system design.

* Optional

7.22.1 Emergency Systems, Engine Generator (*continued*)

D. Test Values – Electrical

1. Insulation resistance values should be in accordance with IEEE 43.

2. The dielectric absorption ratio or polarization shall be compared to previously obtained results and should not be less than 1.0. The recommended minimum insulation resistance ($IR_{1\ min}$) test results in megohms should be corrected to 40° C and read as follows:

    1. $IR_{1\ min}$ = kV + 1 for most windings made before 1970, all field windings, and others not described in 2.2 and 2.3.

       (kV is the rated machine terminal-to-terminal voltage, in rms kV)

    2. $IR_{1\ min}$ = 100 megohms for most dc armature and ac windings built after 1970 (form-wound coils).

    3. $IR_{1\ min}$ = 5 megohms for most machines and random-wound stator coils and form-wound coils rated below 1 kV.

       NOTE: Dielectric withstand voltage high-potential, and surge comparison tests shall not be performed on machines having values lower than those indicated above.

3. Protective relay device test results shall be in accordance with Section 7.9.

4. Low oil pressure, overtemperature, overspeed, and other protection features should operate in accordance with manufacturer's and system design requirements.

5. Vibration levels should be in accordance with manufacturer's published data and shall be compared to baseline data.

6. Performance tests should conform to manufacturer's published data and NFPA 110.

7. Governor and regulator should operate in accordance with manufacturer's and system design requirements.

* Optional

7.22.2 Emergency Systems, Uninterruptible Power Systems

NOTE: There are many configurations of uninterruptible power supply installations. Some are as simple as a static switch selecting between two highly reliable sources, while others are complex systems using a combination of rectifiers, batteries, inverters, motor/generators, static switches and bypass switches. It is the intent of these specifications to list possible tests and maintenance of the major components of the system and more specifically the system as a whole. It is important that the manufacturer's recommended maintenance be performed**.**

A. Visual and Mechanical Inspection

1. Inspect physical and mechanical condition.

2. Check for anchorage, alignment, grounding, and required clearances.

3. Verify that fuse sizes and types correspond to drawings.

*4. Prior to cleaning the unit, perform as-found tests.

5. Clean the unit.

6. Test all electrical and mechanical interlock systems for correct operation and sequencing.

7. Inspect bolted electrical connections for high resistance using one or more of the following:

    1. Use of a low-resistance ohmmeter in accordance with Section 7.22.2.B.1.

    2. Verify tightness of accessible bolted electrical connections by calibrated torque-wrench method in accordance with manufacturer's published data or Table 100.12.

    3. Perform a thermographic survey in accordance with Section 9.

8. Perform as-left tests.

9. Check operation of forced ventilation.

10. Verify that filters are in place, filters are clean and free from debris, and/or vents are clear.

* Optional

7.22.2 Emergency Systems, Uninterruptible Power Systems (*continued*)

B. Electrical Tests

1. Perform resistance measurements through bolted connections with a low-resistance ohmmeter in accordance with Section 7.22.2.A.7.1.

2. Test static transfer from inverter to bypass and back. Use normal load, if possible.

3. Verify free running frequency of oscillator.

*4. Test dc undervoltage trip level on inverter input breaker. Verify setting is in accordance with manufacturer's published data

*5. Test alarm circuits.

6. Verify sync indicators for static switch and bypass switches.

7. Perform electrical tests for UPS system breakers in accordance with Section 7.6.

8. Perform electrical tests for UPS system automatic transfer switches in accordance with Section 7.22.3.

9. Perform electrical tests for UPS system batteries in accordance with Section 7.18.

10. Perform electrical tests for UPS rotating machinery in accordance with Section 7.15.

C. Test Values – Visual and Mechanical

1. Electrical and mechanical interlock systems should operate in accordance with system design requirements. (7.22.2.A.6)

2. Compare bolted connection resistance values to values of similar connections. Investigate values which deviate from those of similar bolted connections by more than 50 percent of the lowest value. (7.22.2.A.7.1)

3. Bolt-torque levels should be in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.12. (7.22.2.A.7.2)

4. Results of the thermographic survey shall be in accordance with Section 9. (7.22.2.A.7.3)

* Optional

7.22.2 Emergency Systems, Uninterruptible Power Systems (*continued*)

D. Test Values – Electrical

1. Compare bolted connection resistance values to values of similar connections. Investigate values which deviate from those of similar bolted connections by more than 50 percent of the lowest value.
2. Static transfer should function in accordance with manufacturer's published data.
3. Oscillator free running frequency should be within manufacturer's recommended tolerances.
4. DC undervoltage should trip inverter input breaker.
5. Alarm circuits should operate in accordance with design requirements.
6. Sync indicators should operate in accordance with design requirements.
7. Breaker performance shall be in accordance with Section 7.6.
8. Automatic transfer switch performance shall be in accordance with Section 7.22.3.
9. Battery test results shall be in accordance with Section 7.18.
10. Rotating machinery performance shall be in accordance with Section 7.15.

7.22.3 Emergency Systems, Automatic Transfer Switches

A. Visual and Mechanical Inspection

1. Inspect physical and mechanical condition.

2. Inspect anchorage, alignment, grounding, and required clearances.

*3. Prior to cleaning the unit, perform as-found tests.

4. Clean the unit.

5. Use appropriate lubrication on moving current-carrying parts and on moving and sliding surfaces.

6. Verify that warning labels are attached and visible.

7. Verify tightness of all control connections.

8. Inspect bolted electrical connections for high resistance using one or more of the following:

    1. Use of a low-resistance ohmmeter in accordance with Section 7.22.3.B.1.

    2. Verify tightness of accessible bolted electrical connections by calibrated torque-wrench method in accordance with manufacturer's published data or Table 100.12.

    3. Perform a thermographic survey in accordance with Section 9.

9. Perform manual transfer operation.

10. Verify positive mechanical interlocking between normal and alternate sources.

11. Perform as-left tests.

* Optional

7.22.3 Emergency Systems, Automatic Transfer Switches (*continued*)

B. Electrical Tests

1. Perform resistance measurements through bolted connections with a low-resistance ohmmeter in accordance with Section 7.22.3.A.8.1.

*2. Perform insulation resistance tests on all control wiring with respect to ground. The applied potential shall be 500 volts dc for 300-volt rated cable and 1000 volts dc for 600-volt rated cable. Test duration shall be one minute. For units with solid-state components or for control devices that cannot tolerate the applied voltage, follow manufacturer's recommendation.

3. Perform a contact/pole-resistance test.

4. Verify settings and operation of control devices.

5. Calibrate and set all relays and timers in accordance with Section 7.9.

6. Perform automatic transfer tests:

    1. Simulate loss of normal power.

    2. Return to normal power.

    3. Simulate loss of emergency power.

    4. Simulate all forms of single-phase conditions.

7. Verify correct operation and timing of the following functions:

    1. Normal source voltage-sensing and frequency sensing relays.

    2. Engine start sequence.

    3. Time delay upon transfer.

    4. Alternate source voltage-sensing and frequency sensing relays.

    5. Automatic transfer operation.

    6. Interlocks and limit switch function.

    7. Time delay and retransfer upon normal power restoration.

    8. Engine cool down and shutdown feature.

* Optional

7.22.3 Emergency Systems, Automatic Transfer Switches (*continued*)

C. Test Values – Visual and Mechanical

1. Compare bolted connection resistance values to values of similar connections. Investigate values which deviate from those of similar bolted connections by more than 50 percent of the lowest value. (7.22.3.A.8.1)

2. Bolt-torque levels should be in accordance with manufacturer’s published data. In the absence of manufacturer’s published data, use Table 100.12. (7.22.3.A.8.2)

3. Results of the thermographic survey shall be in accordance with Section 9. (7.22.3.A.8.3)

D. Test Values – Electrical

1. Compare bolted connection resistance values to values of similar connections. Investigate values which deviate from those of similar bolted connections by more than 50 percent of the lowest value.

2. Insulation-resistance values of control wiring should be comparable to previously obtained results but not less than two megohms.

3. Microhm or dc millivolt drop values should not exceed the high levels of the normal range as indicated in the manufacturer’s published data. If manufacturer’s data is not available, investigate values that deviate from adjacent poles or similar switches by more than 50 percent of the lowest value.

4. Control devices should operate in accordance with manufacturer’s published data.

5. Relay test results shall be in accordance with Section 7.9.

6. Automatic transfers should operate in accordance with manufacturer’s design.

7. Operation and timing should be in accordance with manufacturer’s and/or system design requirements.

7. INSPECTION AND TEST PROCEDURES

7.23 Communications

– RESERVED –

* Optional

7. INSPECTION AND TEST PROCEDURES

7.24.1 Automatic Circuit Reclosers and Line Sectionalizers, Automatic Circuit Reclosers, Oil/Vacuum

A. Visual and Mechanical Inspection

1. Record model numbers, style numbers, serial numbers, firmware revisions, software revisions, and rated control voltage.
2. Inspect the physical and mechanical condition of equipment and wiring.
3. Verify operation of light-emitting diodes, displays, and targets.
4. Verify equipment is clean.
5. Check tightness of all connections.
6. Verify appropriate grounding equipment and circuits.
7. Verify backup batteries are healthy and connected.
8. Check with setting engineer for applicable firmware updates and product recalls.
9. Verify settings and application configurations are in accordance with engineered settings.
10. Verify devices display the correct date and time.
11. Reset and clear events, maintenance data, statistical data, and alarms from the devices after any tests.
12. Download or document settings and logic from the devices.

B. Electrical Tests

1. Perform metering tests on all analog inputs and verify metering values on the human machine interface (HMI) and at remote terminals.
2. Verify operation of all enabled digital inputs.
3. Verify operation of all digital outputs by operating the controlled device.
4. Verify all communication links are operational and function test failovers of redundant communication links.
5. Verify operation of all internal logic functions including tagging, lockouts, and local/remote control.

* Optional

7.24.1 Automatic Circuit Reclosers and Line Sectionalizers, Automatic Circuit Reclosers, Oil/Vacuum *(continued)*

6. Verify alarm and status points.

7. Verify deadband and zero deadband settings and operation.

8. Reset and clear events, maintenance data, statistical data, and alarms from the devices after completion of tests.

C. Test Values – Visual and Mechanical

1. Light-emitting diodes, displays, and targets should illuminate.

2. Equipment should be clean and operational.

3. Settings and logic should agree with the most recent engineered setting files.

4. Verify equipment displays the correct date and time and events are accurately timestamp.

D. Test Values – Electrical

1. Metering readings shall be in accordance with manufacturer's published tolerances.

2. Inputs shall operate as per the design.

3. Outputs shall operate as per the design.

4. Communication links shall be operational and failover times shall be in accordance with the manufacturer's published data.

5. Internal logic functions shall operate as per the design.

6. Alarms and statuses shall be as per the design.

7. Deadbands shall operate in accordance with manufacturer's published data.

8. **Indications and statistical data should operate in accordance with manufacturer's published data.**

* Optional

7.24.1 Automatic Circuit Reclosers and Line Sectionalizers, Automatic Circuit Reclosers, Oil/Vacuum *(continued)*

A. Visual and Mechanical Inspection

1. Inspect physical and mechanical condition.

2. Inspect anchorage, alignment, and grounding.

*3. Prior to cleaning the unit, perform as-found tests.

4. Clean the unit.

5. Perform all mechanical operation and contact alignment maintenance and tests on both the recloser and its operating mechanism in accordance with manufacturer's published data.

6. Inspect bolted electrical connections for high resistance using one or more of the following:

   1. Use of a low-resistance ohmmeter in accordance with Section 7.24.1.B.1.

   2. Verify tightness of accessible bolted electrical connections by calibrated torque-wrench method in accordance with manufacturer's published data or Table 100.12.

   3. Perform a thermographic survey in accordance with Section 9.

7. Inspect for correct insulating liquid level.

8. Perform as-left tests.

B. Electrical Tests

1. Perform resistance measurements through bolted connections with a low-resistance ohmmeter in accordance with Section 7.24.1.A.6.1.

2. Perform insulation-resistance tests for one minute on each pole, phase-to-phase and phase-to-ground with recloser closed and across each pole with the recloser open. Apply voltage in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.1.

3. Perform a contact/pole-resistance test.

*4. Perform insulation-resistance tests on all control wiring with respect to ground. The applied potential shall be 500 volts dc for 300-volt rated cable and 1000 volts dc for 600-volt rated cable. Test duration shall be one minute. For units with solid-state components, follow manufacturer's recommendation.

* Optional

7.24.1 Automatic Circuit Reclosers and Line Sectionalizers, Automatic Circuit Reclosers, Oil/Vacuum (*continued*)

5. Remove a sample of insulating liquid in accordance with ASTM D923. Sample shall be tested in accordance with the referenced standard.

    1. Dielectric breakdown voltage: ASTM D1816

    2. Color: ASTM D1500

    3. Visual condition: ASTM D1524

*6. Perform minimum pickup voltage tests on trip and close coils in accordance with Table 100.20.

*7. Perform power-factor or dissipation-factor tests on each pole with the recloser open and each phase with the recloser closed.

*8. Perform power-factor or dissipation-factor tests on each bushing equipped with a power-factor/capacitance tap. In the absence of a power-factor/capacitance tap, perform hot-collar tests. These tests should be in accordance with the test equipment manufacturer's published data.

9. Perform vacuum bottle integrity test (dielectric withstand voltage) across each vacuum bottle with the recloser in the open position in strict accordance with manufacturer's published data.

10. Perform dielectric withstand voltage test on each phase with the recloser closed and the poles not under test grounded. Test voltage should be in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.19.

11. Verify operation of heaters.

12. Test all protective functions in accordance with Section 7.9.

13. Test all metering and instrumentation in accordance with Section 7.11.

14. Test instrument transformers in accordance with Section 7.10.

* Optional

7.24.1 Automatic Circuit Reclosers and Line Sectionalizers, Automatic Circuit Reclosers, Oil/Vacuum (*continued*)

C. Test Values – Visual and Mechanical

1. Mechanical operation and contact alignment should be in accordance with manufacturer's published data. (7.24.1.A.5)

2. Compare bolted connection resistance values to values of similar connections. Investigate values which deviate from those of similar bolted connections by more than 50 percent of the lowest value. (7.24.1.A.6.1)

3. Bolt-torque levels should be in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.12. (7.24.1.A.6.2)

4. Results of the thermographic survey shall be in accordance with Section 9. (7.24.1.A.6.3)

5. Insulating liquid level should be in accordance with manufacturer's recommended tolerances. (7.24.1.A.7)

D. Test Values – Electrical

1. Compare bolted connection resistance values to values of similar connections. Investigate values which deviate from those of similar bolted connections by more than 50 percent of the lowest value.

2. Insulation-resistance values should be in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.1. Values of insulation resistance less than this table or manufacturer's recommendations should be investigated.

3. Microhm or dc millivolt drop values should not exceed the high levels of the normal range as indicated in the manufacturer's published data. If manufacturer's data is not available, investigate values that deviate from adjacent poles or similar reclosers by more than 50 percent of the lowest value.

4. Insulation-resistance values of control wiring should be comparable to previously obtained results but not less than two megohms.

5. Insulating liquid test results should be in accordance with Table 100.4.

6. Minimum pickup for trip and close coils should conform with manufacturer's published data. In the absence of manufacturer's published data refer to Table 100.20.

* Optional

7.24.1 Automatic Circuit Reclosers and Line Sectionalizers, Automatic Circuit Reclosers, Oil/Vacuum (*continued*)

7. Power-factor or dissipation-factor values and tank loss index shall be compared to manufacturer's published data. In the absence of manufacturer's published data, the comparison shall be made to test data from similar circuit reclosers or sectionalizers or data from test equipment manufacturers.

8. Power-factor or dissipation-factor and capacitance values should be within ten percent of nameplate rating for bushings. Hot collar tests are evaluated on a milliampere/milliwatt loss basis, and the results should be compared to values of similar bushings.

9. If no evidence of distress or insulation failure is observed by the end of the total time of voltage application during the vacuum bottle integrity test, the test specimen is considered to have passed the test

10. If no evidence of distress or insulation failure is observed by the end of the total time of voltage application during the dielectric withstand voltage test, the test specimen is considered to have passed the test.

11. Heaters should be operational.

12. Protective device function test results shall be in accordance with Section 7.9.

13. Metering and instrumentation test results shall be in accordance with Section 7.11.

14. Instrument transformer test results shall be in accordance with Section 7.10.

* Optional

7.24.2 Automatic Circuit Reclosers and Line Sectionalizers, Automatic Line Sectionalizers, Oil

A. Visual and Mechanical Inspection

1. Inspect physical and mechanical condition.

2. Inspect anchorage, alignment, and grounding.

*3. Prior to cleaning the unit, perform as-found tests.

4. Clean the unit.

5. Perform all mechanical operation and contact alignment maintenance and tests on both the sectionalizer and its operating mechanism in accordance with manufacturer's published data.

6. Inspect bolted electrical connections for high resistance using one or more of the following:

    1. Use of a low-resistance ohmmeter in accordance with Section 7.24.2.B.1.

    2. Verify tightness of accessible bolted electrical connections by calibrated torque-wrench method in accordance with manufacturer's published data or Table 100.12.

    3. Perform a thermographic survey in accordance with Section 9.

7. Inspect for correct insulating liquid level.

8. Perform as-left tests.

B. Electrical Tests

1. Perform resistance measurements through bolted connections with a low-resistance ohmmeter in accordance with Section 7.24.2.A.6.1.

2. Perform insulation-resistance tests for one minute on each pole, phase-to-phase and phase-to-ground with sectionalizer closed and across each open pole. Apply voltage in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.1.

3. Perform a contact/pole-resistance test.

4. Remove a sample insulating liquid in accordance with ASTM D923. Sample shall be tested in accordance with the referenced standard.

* Optional

7.24.2 Automatic Circuit Reclosers and Line Sectionalizers, Automatic Line Sectionalizers, Oil (*continued*)

1. Dielectric breakdown voltage: ASTM D877 and/or ASTM D1816

2. Color: ASTM D15

3. Visual condition: ASTM D1524

5. Perform dielectric withstand voltage tests on each pole-to-ground and pole-to-pole.

6. Test sectionalizer counting function by application of simulated fault current (greater than 160 percent of continuous current rating).

7. Test sectionalizer lockout function for all counting positions.

8. Test for reset timing on trip actuator.

*9. Perform power-factor or dissipation-factor tests on each pole with the recloser open and each phase with the recloser closed.

* 10. Perform power-factor or dissipation-factor tests on each bushing equipped with a power-factor/capacitance tap. In the absence of a power-factor/capacitance tap, perform hot-collar tests. These tests should be in accordance with the test equipment manufacturer's published data.

C. Test Values – Visual and Mechanical

1. Mechanical operation and contact alignment should be in accordance with manufacturer's published data. (7.24.2.A.5)

2. Compare bolted connection resistance values to values of similar connections. Investigate values which deviate from those of similar bolted connections by more than 50 percent of the lowest value. (7.24.2.A.6.1)

3. Bolt-torque levels should be in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.12. (7.24.2.A.6.2)

4. Results of the thermographic survey shall be in accordance with Section 9. (7.24.2.A.6.3)

5. Insulating liquid level should be in accordance with manufacturer's recommended tolerances. (7.24.2.A.7)

D. Test Values – Electrical

* Optional

7.24.2 Automatic Circuit Reclosers and Line Sectionalizers, Automatic Line Sectionalizers, Oil (*continued*)

1. Compare bolted connection resistance values to values of similar connections. Investigate values which deviate from those of similar bolted connections by more than 50 percent of the lowest value.

2. Insulation-resistance values should be in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.1. Values of insulation resistance less than this table or manufacturer's recommendations should be investigated.

3. Microhm or dc millivolt drop values should not exceed the high levels of the normal range as indicated in the manufacturer's published data. If manufacturer's data is not available, investigate values that deviate from adjacent poles or similar switches by more than 50 percent of the lowest value.

4. Insulating liquid test results should be in accordance with Table 100.4.

5. If no evidence of distress or insulation failure is observed by the end of the total time of voltage application during the dielectric withstand voltage test, the test specimen is considered to have passed the test.

6. Operations counter should advance one digit per close-open cycle.

7. Lockout function should operate in accordance with manufacturer's published data.

8. Reset timing of trip actuator should operate in accordance with manufacturer's published data.

9. Power-factor or dissipation-factor values and tank loss index shall be compared to manufacturer's published data. In the absence of manufacturer's published data, the comparison shall be made to test data from similar circuit reclosers or sectionalizers or data from test equipment manufacturers.

10. Power-factor or dissipation-factor and capacitance values should be within ten percent of nameplate rating for bushings. Hot collar tests are evaluated on a milliampere/milliwatt loss basis, and the results should be compared to values of similar bushing.

* Optional

7. INSPECTION AND TEST PROCEDURES

7.25 FIBER-OPTIC CABLES

A. Visual and Mechanical Inspection

1. Inspect cable and connections for physical and mechanical damage.

2. Verify that all connectors and splices are correctly installed.

3. Verify that visible cable bends are not less than the manufacturer's minimum allowable bending radius.

B. Optical Tests

*1. Perform cable length measurement, fiber fracture inspection, and construction defect inspection using an optical time domain reflectometer.

*2. Perform connector and splice integrity test using an optical time domain reflectometer.

*3. Perform cable attenuation loss measurement with an optical power loss test set.

*4. Perform connector and splice attenuation loss measurement from both ends of the optical cable with an optical power loss test set.

5. Perform transmit and receive power readings from local device and remote device.

C. Test Values – Visual and Mechanical

1. Cable and connections should not have been subjected to physical or mechanical damage. (7.25.A.1)

2. Connectors and splices should be installed in accordance with industry standards. (7.25.A.2)

3. Cable bends should not be less than the manufacturer's minimum allowable bending radius.

D. Test Values – Optical

1. The optical time domain reflectometer signal should be analyzed for cable backscatter by viewing the reflected power/distance graph.

2. The optical time domain reflectometer signal should be analyzed for excessive connection or splice attenuation by viewing the reflected power/distance graph.

* Optional

7.25 FIBER-OPTIC CABLES (*continued*)

3. Cable attenuation loss measurement shall be expressed in dB/km. Losses should be within the manufacturer’s recommendations when no local site specifications are available.

4. Connector attenuation loss measurement shall be expressed in dB and splice attenuation loss measurement shall be expressed dB/km. Losses should be within the manufacturer’s recommendations when no local site specifications are available.

5. Transmit and receive results shall compare to previous test results and manufacturer’s published data.

* Optional

7.26 Electric Vehicle Charging Systems

A. Visual and Mechanical Inspections

1. Compare equipment nameplate data with drawings and specifications.

2. Inspect physical and mechanical condition of cords and connectors.

3. Inspect anchorage, alignment, grounding, and required area clearances.

4. Verify the unit is clean and all shipping bracing, loose parts, and documentation shipped inside cubicles have been removed.

5. Verify that wiring connections are tight and that wiring is secure to prevent damage during routine operation of moving parts.

6. Inspect bolted electrical connections for high resistance using one or more of the following:

    1. Use of a low-resistance ohmmeter in accordance with Section 7.26.B.1.

    2. Verify tightness of accessible bolted electrical connections by calibrated torque wrench method in accordance with manufacturer's published data or Table 100.12.

    3. Perform thermographic survey in accordance with Section 9.

7. Verify operation and sequencing of interlocking systems.

* Optional

7.26 Electric Vehicle Charging Systems (*continued*)

B. Electrical Tests

1. Perform resistance measurements through bolted electrical connections with a low-resistance ohmmeter in accordance with Section 7.26.A.6.1.

2. Perform system function tests in accordance with manufacturer's published data.

C. Test Values – Visual and Mechanical

1. Compare bolted connection resistance values to values of similar connections. Investigate values which deviate from those of similar bolted connections by more than 50 percent of the lowest value. (7.26.A.6.1)

2. Bolt-torque levels shall be in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.12. (7.26.A.6.2)

3. Results of the thermographic survey shall be in accordance with Section 9. (7.26.A.6.3)

D. Test Values – Electrical

1. Compare bolted connection resistance values to values of similar connections. Investigate values which deviate from those of similar bolted connections by more than 50 percent of the lowest value.

2. Results of system function tests should be in accordance with manufacturer's published data.

* Optional

8. SYSTEM FUNCTION TESTS

It is the purpose of system function tests to prove the correct interaction of all sensing, processing, and action devices. Refer to commissioning documentation for checklists and equipment system function tests to be performed.

Perform system function tests upon completion of the maintenance tests defined, as system conditions allow.

1. Develop test parameters and perform tests for the purpose of evaluating performance of all integral components and their functioning as a complete unit within design requirements and manufacturer's published data.
2. Verify the correct operation of all interlock safety devices for fail-safe functions in addition to design function.
3. Verify the correct operation of all sensing devices, alarms, and indicating devices.
4. Perform function tests to verify relay self-test, power supply failure, and trip coil monitor alarms are communicated correctly to SCADA systems.
5. Measure latency of communication lines for pilot wire, line differential and transfer trip protection schemes.
6. Function test lock-out relay, block close circuits, and block reclose (hotline tag) circuits.
7. Function test bus restoration and/or transfer switches.
8. Verify correct metering on protective relays and meters including values communicated to SCADA systems.
9. Verify protection, control circuits, and instrument transformer circuits are restored to normal operation.
10. Verify communication lines are operational for local and remote devices.
11. Verify control annunciation systems are left with no alarms and any alarms present shall be investigated.
12. Verify systems are left in normal operating mode or position, transfer and restoration schemes are enabled, and monitoring and protection devices are operational.

9. THERMOGRAPHIC SURVEY

1. Visual and Mechanical Inspection

1. Inspect physical and mechanical condition.
2. Remove panel covers or view the equipment through viewing ports designed to transmit applicable signals being measured.

2. Thermographic Survey Report

Provide a report which includes the following:

1. Description of equipment to be tested.
2. Discrepancies.
3. Temperature difference between the area of concern and the reference area.
4. Probable cause of temperature difference.
5. Areas inspected. Identify inaccessible and/or unobservable areas and/or equipment.
6. Identify load conditions at time of inspection.
7. Provide photographs and/or thermograms of the deficient area.
8. Provide recommended action for repair.

3. Test Parameters

1. Inspect distribution systems with imaging equipment capable of detecting a minimum temperature difference of 1° C at 30° C.
2. Equipment shall detect emitted radiation and convert detected radiation to visual signal.
3. Thermographic surveys should be performed during periods of maximum possible loading. Refer to NFPA 70B.

4. Test Results

Suggested actions based on temperature rise can be found in Table 100.18.

10. ELECTROMAGNETIC FIELD SURVEY

1. Procedure

1. Take detailed measurements of the magnetic flux density, vector direction, and temporal variations over the area, as necessary.

    1. Perform spot measurements of the magnetic fields (40 to 800 Hertz) at grid intervals one meter above the floor throughout the measured area. Record x, y, z, and resultant magnetic flux density values for each measurement point.

    2. Take additional detailed spot measurements directly at floor level, at two meters above the floor, at grid point locations, and directly on the wall surface separating the measured area from suspected magnetic field source.

    3. If measured magnetic flux densities at any perimeter wall appear to be above 3.0 mG, take additional spot measurements of the adjoining space utilizing the same measurement grid spacing at a height of one meter above floor.

    4. Take a baseline magnetic flux density reading at a specific point in the immediate area of the suspected magnetic field source.

    5. Determine magnetic field temporal variations as required by positioning the Gaussmeter at or near the location of highest magnetic flux density for 24 to 48 hours.

2. Obtain and record electrical system information including load measurements.

3. The magnetic field evaluation shall be performed in accordance with the recommended practices and procedures in accordance with IEEE 644.

2. Survey Report

Provide a report which includes the following:

1. Basis, description, purpose, and scope of the survey.

2. Tabulations and or attached graphical representations of the magnetic flux density measurements corresponding to the time and area or space where the measurements were taken.

3. Descriptions of each of the operating conditions evaluated and identification of the condition that resulted in the highest magnetic flux density.

4. Descriptions of equipment performance issues that could be related to measured magnetic flux density.

5. Description of magnetic field test equipment.

6. Conclusions and recommendations.

11. ONLINE PARTIAL DISCHARGE SURVEY FOR SWITCHGEAR

1. Inspection

1. Inspect physical and mechanical condition.
    1. Inspect for audible indications of partial discharge.
    2. Inspect for indications of ozone.
2. Select sensor technology based upon the type of equipment under survey.
    1. Transient-earth voltage (TEV).
    2. High-frequency current transformer (HFCT).
    3. Ultra-High Frequency (UHF).
    4. Coupler technology on voltage indicator system (VIS) ports.
    5. Contact acoustic.
    6. Airborne acoustic.
3. Connect sensor to an instrument that displays the signal intensity and phase relationship of the signal.
4. Record results.

2. Survey Report

Provide a report that includes the following:

1. List of equipment surveyed.
2. Discrepancies.
3. Background signal level for baseline reference.
4. Probable cause of signal difference and nature of activity.
5. Inaccessible or unobservable equipment.
6. Ambient temperature and humidity
7. System voltage and current at time of inspection.
8. Graphical or color-coded representation of the deficiencies.
9. Recommended action.

3. Test Results

Suggested actions can be found in Table 100.23.1, 100.23.2, 100.23.3, 100.23.4, or 100.23.5, according to applicable sensor technology.

TABLES

NETA®

STANDARD FOR
MAINTENANCE TESTING SPECIFICATIONS
FOR
ELECTRICAL POWER
EQUIPMENT AND SYSTEMS

TABLE 100.1

Insulation Resistance Test Values
Electrical Apparatus and Systems Other Than Rotating Machinery

Nominal Rating of Equipment (Volts)	Minimum Test Voltage (DC)	Recommended Minimum Insulation Resistance (Megohms)
250	500	25
600	1,000	100
1,000	1,000	100
2,500	1,000	500
5,000	2,500	1,500
8,000	2,500	2,500
15,000	2,500	5,000
25,000	5,000	10,000
34,000	5,000	100,000
46,000 and above	5,000	100,000

Notes:

1. In the absence of consensus standards dealing with insulation-resistance tests, the NETA Standards Review Council suggests the above representative values.
2. Test results are dependent on the temperature of the insulating material and the humidity of the surrounding environment at the time of the test. See Table 100.14 for temperature correction factors.
3. Insulation-resistance test data may be used to establish a trending pattern. Deviations from the baseline information permit evaluation of the insulation.
4. For rotating machinery insulation resistance test values, refer to Table 100.11.
5. For transformer insulation resistance test values, refer to Table 100.5.
6. For cables, the acceptability for continued energization should not be determined by the recommended minimum insulation resistance values shown above. These values may be used as guidance of typical magnitudes, however, conductor and insulation geometry, temperature, and overall cable length need to be taken into account per the manufacturer's published data for definitive minimum insulation resistance criteria.
7. The minimum insulation resistance values do not apply to low-voltage metal-enclosed busways. For minimum insulation resistance values of low-voltage metal-enclosed busways refer to Section 7.4.

TABLE 100.2
Switchgear Withstand Test Voltages

<table>
<tr><th rowspan="2">Type of Switchgear</th><th rowspan="2" colspan="2">Rated Maximum Voltage (kV) (rms)</th><th colspan="3">Maximum Test Voltage (kV)</th></tr>
<tr><th colspan="2">AC</th><th>DC</th></tr>
<tr><td rowspan="2">Low-Voltage Power Circuit Breaker Switchgear</td><td colspan="2">.254/.508/.635</td><td colspan="2">1.6</td><td>2.3</td></tr>
<tr><td colspan="2">.730/1.058</td><td colspan="2">2.2</td><td>3.1</td></tr>
<tr><td rowspan="5">Metal-Clad Switchgear</td><td colspan="2">4.76</td><td colspan="2">14</td><td>20</td></tr>
<tr><td colspan="2">8.25</td><td colspan="2">27</td><td>37.5</td></tr>
<tr><td colspan="2">15.0</td><td colspan="2">27</td><td>37.5</td></tr>
<tr><td colspan="2">27.0</td><td colspan="2">45</td><td>Note 3</td></tr>
<tr><td colspan="2">38.0</td><td colspan="2">60</td><td>Note 3</td></tr>
<tr><td rowspan="3">Station-Type Cubicle Switchgear</td><td colspan="2">15.5</td><td colspan="2">37</td><td>Note 3</td></tr>
<tr><td colspan="2">38.0</td><td colspan="2">60</td><td>Note 3</td></tr>
<tr><td colspan="2">72.5</td><td colspan="2">120</td><td>Note 3</td></tr>
<tr><td rowspan="6">Metal-Enclosed Interrupter Switchgear</td><td>With stress cone type terminations</td><td>With IEEE 386 type terminations</td><td>With stress cone type terminations</td><td>With IEEE 386 type terminations</td><td></td></tr>
<tr><td>4.76</td><td>4.76</td><td>14</td><td>14</td><td>Note 3</td></tr>
<tr><td>8.25</td><td>8.25</td><td>27</td><td>25</td><td>Note 3</td></tr>
<tr><td>15.0</td><td>14.4</td><td>27</td><td>25</td><td>Note 3</td></tr>
<tr><td>27</td><td>26.3</td><td>45</td><td>30</td><td>Note 3</td></tr>
<tr><td>38.0</td><td>26.6</td><td>60</td><td>37</td><td>Note 3</td></tr>
</table>

Notes:

1. Derived from IEEE C37.20.1-2015 S*tandard for Metal-Enclosed Low-Voltage Power Circuit-Breaker Switchgear*, Paragraph 5; IEEE C37.20.2-2015, *Standard for Metal-Clad Switchgear*, Paragraph 6.5; and IEEE C37.20.3-2013 *Standard for Metal-Enclosed Interrupter Switchgear*, Paragraph 6.19. This table includes 0.75 multiplier with fraction rounded down.
2. The column headed "DC" is given as a reference only for those using dc tests to verify the integrity of connected cable installations without disconnecting the cables from the switchgear. It represents values believed to be appropriate and approximately equivalent to the corresponding power frequency withstand test values specified for voltage rating of switchgear. The presence of this column in no way implies any requirement for a dc withstand test on ac equipment or that a dc withstand test represents an acceptable alternative to the power-frequency withstand tests specified in these specifications, either for design tests, production tests, conformance tests, or field tests. When making dc tests, the voltage should be raised to the test value in discrete steps and held for a period of one minute.
3. Because of the variable voltage distribution encountered when making dc withstand tests, the manufacturer should be contacted for recommendations before applying dc withstand tests to the switchgear. Voltage transformers above 34.5 kV should be disconnected when testing with dc. Refer to IEEE C57.13-2015 (*IEEE Standard Requirements for Instrument Transformers*) paragraph 8.5.

TABLE 100.3

Maintenance Test Values
Recommended Dissipation Factor/Power at 20° C
Liquid-Filled Transformers, Regulators, and Reactors

	Oil Maximum	Silicone Maximum	Tetrachloroethylene Maximum	High Fire Point Hydrocarbon Maximum	Ester Fluids
Power Transformers	1.0%	0.5%	3.0%	2.0%	3.0%
Distribution Transformers and Apparatus	2.0%	0.5%	3.0%	3.0%	3.0%

Notes:

In the absence of consensus standards dealing with transformer dissipation/power factor values, the NETA Standards Review Council suggests the above representative values. Maximum values for oil are based on data from Doble Engineering Company.

TABLE 100.4.1

Suggested Limits for Service-Aged Class I Mineral Oil

Mineral Oil [1]				
Test	ASTM Method	Acceptable Values		
		69 kV and Below	Above 69 kV – Below 230 kV	230 kV and Above
Dielectric breakdown, kV minimum [2]	D877	26	26	26
Dielectric breakdown, kV minimum @ 1 mm (0.04 inch) gap	D1816	23	28	30
Dielectric breakdown, kV minimum @ 2 mm (0.08 inch) gap	D1816	40	47	50
Interfacial tension mN/m minimum	D971	25	30	32
Neutralization number, mg KOH/g maximum	D974	0.20	0.15	0.10
Water content, ppm maximum @ 60º C [3]	D1533	35	25	20
Power factor at 25º C, %	D924	0.5	0.5	0.5
Power factor at 100º C, %	D924	5.0	5.0	5
Color [3]	D1500	1.5	1.5	1.5
Visual Condition	D1524	Bright, clear and free of particles	Bright, clear and free of particles	Bright, clear and free of particles
Specific Gravity (Relative Density) @ 15º C Maximum [4]	D1298	0.91	0.91	0.91

Notes:

1. IEEE C57.106-2015 Guide for Acceptance and Maintenance of Mineral Oil in Equipment, Table 3.
2. IEEE 637-1985 Guide for Reclamation of Insulating Oil and Criteria for Its Use, Table 1.
3. In the absence of consensus standards, NETA's Standard Review Council suggests these values.
4. IEEE C57.106 Guide for Acceptance and Maintenance of Insulating Mineral Oil in Electrical Equipment, Table 1.

TABLE 100.4.2
Suggested Limits
For Service-Aged Less-Flammable Hydrocarbon Insulating Liquid

Test	ASTM Method	Acceptable Values
Dielectric breakdown voltage, kV minimum	D877	24
Dielectric breakdown voltage for 1 mm (0.04 inch) gap, kV minimum	D1816	23
Dielectric breakdown voltage for 2 mm (0.08 inch) gap, kV minimum	D1816	34
Water content, ppm maximum	D1533 B	35
Dissipation/power factor, 60 Hertz, % max. @ 25º C	D924	1.0
Fire point, º C, minimum[3]	D92	300
Interfacial tension, mN/m, 25º C	D971	24
Neutralization number, mg KOH/g	D664	0.20

Notes:

1. IEEE C57.121-1998 *Guide for Acceptance and Maintenance of Less-Flammable Hydrocarbon Fluid in Transformers*, Table 4.
2. The values in this table are considered typical for acceptable service-aged LFH fluids as a general class. If actual test analysis approaches the values shown, consult the fluid manufacturer for specific recommendations.
3. If the purpose of the HMWH installation is to comply with the NFPA 70 *National Electrical Code*, this value is the minimum for compliance with NEC Article 450.23.

TABLE 100.4.3
Suggested Limits
for Service-Aged Silicone Insulating Liquid

Test	ASTM Method	Acceptable Values
Dielectric breakdown, kV minimum	D877	25
Visual	D2129	Colorless, clear, free of particles
Water content, ppm maximum	D1533	100
Dissipation/power factor, 60 Hertz, maximum @ 25º C	D924	0.2
Viscosity, cSt @ 25º C	D445	47.5 – 52.5
Fire Point, º C, minimum	D92	340
Neutralization number, Mg KOH/g max	D974	0.2

Notes:

1. IEEE C57.111-1989 (R1995) *Guide for Acceptance of Silicone Insulating Fluid and Its Maintenance in Transformers*, Table 3.

TABLE 100.4.4
Suggested Limits for Service-Aged Tetrachloroethylene Insulating Fluid

Test	ASTM Method	Acceptable Values
Dielectric breakdown, kV minimum	D877	26
Visual	D2129	Clear with purple iridescence
Water content, ppm maximum	D1533	35
Dissipation/power factor, % maximum @ 25º C	D924	12.0
Viscosity, cSt @ 25º C	D445	0
Fire point, º C, minimum	D92	-
Neutralization number, mg KOH/g maximum	D974	0.25
Neutralization number, mg KOH/g maximum	D664	-
Interfacial tension, mN/m minimum @ 25º C	D971	-

Notes:

1. Reference*: Instruction Book PC-2000 for Wecosol™ Fluid-Filled Primary and Secondary Unit Substation Transformers,* ABB Power T&D.

TABLE 100.4.5

Suggested Limits for Continued use of In-Service Natural Ester liquids Grouped by Voltage Class (See also IEEE C57.147-2018) [1]

Test and ASTM Method	IEEE Value For Voltage Class		
	≤ 69 kV	> 69 kV < 230 kV	≥ 230 kV [2]
Dielectric strength [3], ASTM D1816, kV, 1 mm gap 2 mm gap	23 40	28 47	30 50
Dissipation factor (power factor), ASTM D924, %, maximum 25º C	3%	3%	3%
Water content: ASTM D1533, mg/kg, maximum	450	350	200
Optional [3] Fire Point, ASTM D92, º C, minimum	300	300	300
Kinematic viscosity increase from initial value	≥ 10%	≥ 10%	≥ 10%

Notes:

1. The test limits shown in this table apply to natural ester insulating liquids as a class. Due to differences in their chemistry, certain values are significantly different than the limits for mineral oil. Refer to *IEEE C57.147-2018* for details. Specific typical values for each brand of insulating liquid should be obtained from each insulating liquid manufacturer. If test results, while in compliance with this table, are significantly different from published typical values, it is recommended that the insulating liquid manufacturer be contacted.
2. Provisional for ≥ 230 kV.
3. Most outdoor transformer installations do not require NEC "Less-Flammable" designation. After retrofilling with unused natural esters a reduction in the fire point of the ester liquid due to residual mineral oil. For transformers requiring a less-flammable rating, or otherwise known to require its insulating liquid to meet a minimum 300 °C fire point, any sample testing with a fire point below 300 °C indicates excessive residual mineral oil and should be investigated. For more information, please refer to *IEEE C57.147-2018 Fig.* B.4 and Annex C.1.

This table was derived from Reference *IEEE C57.147-2018*, Section 8.3, Table 4 and Annex B, Table B.2.

TABLE 100.5
Transformer Insulation Resistance Maintenance Testing

Transformer Coil Rating Type (Volts)	Minimum DC Test Voltage	Recommended Minimum Insulation Resistance (Megohms)	
		Liquid Filled	**Dry**
0 – 600	1,000	100	500
601 – 5,000	2,500	1,000	5,000
Greater than 5,000	5,000	5,000	25,000

Notes:

1. In the absence of consensus standards, the NETA Standards Review Council suggests the above representative values.
2. See Table 100.14 for temperature correction factors.
3. Since insulation resistance depends on insulation rating (kV) and winding capacity (kVA), values obtained should be compared to manufacturer's published data.

TABLE 100.6.1
Extruded Dieletric Shielded Power Cables
DC Test Voltages (kV)

Rated Voltage Phase-to-Phase	100% Insulation Level	133% Insulation Level	173% Insulation Level
2001 - 5000	9	11	11
5001 - 8000	11	14	14
8001 - 15000	18	20	20
15001 - 25000	25	30	31
25001 - 28000	26	31	33
28001 - 35000	31	39	41
35001 - 46000	41	54	56

Tables and notes from ANSI/NEMA WC74/ICEA S93-639-2017 *American National Standard for 5-46 kV Shielded Power Cable for Use in the Transmission and Distribution of Electric Energy*, Appendix F Table F-1.

Notes:

1. Apply voltage for 5 consecutive minutes.
2. After 5 years of service, DC Testing is not recommended.
3. Table voltages are intended for cables designed to meet this standard. Consult manufacturers of cables and accessories before applying this test.

TABLE 100.6.2

Laminated Dielectric Shielded Power Cable DC Test Voltages

System Voltage, kV rms, Phase-to-Phase	System BIL, kV Crest	Maintenance Test, kV dc, Phase-to-Ground
5	75	23
8	95	29
15	110	46
25	150	61
28	170	68
35	200	75
46	250	95
69	350	130
115	450	170
115	550	205
138	650	245
161	750	280
230	1,050	395
345	1,175	440
345	1,300	488
500	1,425	535
500	1,550	580
500	1,675	629

Notes:

1. Maximum test voltage should be maintained for 15 continuous minutes.
2. Voltages higher than those listed, up to 80% of system BIL, may be considered, but the age and operating environment of the system should be taken into account. The user is urged to consult the suppliers of the cable and any/all accessories before applying the high voltage.
3. When older cables or other types/classes of cables or other equipment such as transformers, switchgear, motors, etc. are connected to the cable to be tested, voltages lower than those shown in the table may be necessary to comply with the limitations imposed by such interconnected cables and equipment. See IEEE 95 [B5] and Table 1 of IEEE C37.20.2.
4. If the test voltage exceeds 50% of system BIL, surge protection against excessive overvoltages induced by flashovers at the termination should be provided.
5. It is strongly recommended that the user consult with the manufacturer(s) of all components that will be subjected to such testing before performing any tests on cables and cable accessories rated 115kV and higher.
6. It should be noted that this table and the test procedures suggested in this guide do not necessarily agree with the recommendations of other organizations, such as those of the Association of Edison Illuminating Companies AEIC CS1-12, CS2-14, CS3-16, CS4-18. Where there is concern, a user should consult the supplier of the cable and accessories to ascertain that the components will withstand the test.

Reproduction of table 1 from IEEE 400.1-2018 *IEEE Guide for Field Testing of Laminated Dielectric, Shielded Power Cable Systems Rated 5 kV and Above with High Direct Current Voltage.*

TABLE 100.6.3
Shielded Power Cable
VLF (0.1 Hz) Test Voltages

Very Low Frequency Testing Levels 0.1 Hz Test Voltage (rms)	
System Voltage Phase-to-Phase (kV RMS)	**Maintenance Test Voltage Phase-to-Ground (kV RMS)**
5	7
10	10
15	16
25	24
35	33
46	43
69	63
115	103
138	123

Notes:

1. Adaptation of Table 3 from IEEE 400.2-2013 Guide for field testing of shielded power cable systems using very low frequency (VLF).

2. Recommended duration of test is a minimum of 30 minutes without failure. Should the cable be considered of critical importance, a 60 minute withstand test is recommended. Should a failure occur, the test should be repeated in full, after the repair has been made.

TABLE 100.6.4
Shielded Power Cables
DAC Test Voltages for Offline PD Tests

Power cable rated voltage U [kV] phase-to-phase	**U_e [kV]**	**DAC test voltage level V_t [kV_{peak}] phase-to-ground**
3	2	5
5	3	6
6	4	10
8	5	11
10	6	14
15	9	21
20	12	28
25	15	35
30	18	41
35	21	48
45-47	26	60
60-69	35	80
110-115	64	145
132-138	77	150
150-161	87	170
220-230	127	204
275-287[1]	159	237
330-345[1]	191	282
380-400[1]	220	293

Notes:

1. Test voltage levels given above 230 kV are provisional and given for study purposes only.

Extracted from IEEE 400.4-2015, Guild for Field Testing of Shielded Power Cable Systems Rated 5 kV and above with Damped Alternating Current (DAC) Voltage, table A.2.

TABLE 100.6.5
Shielded Power Cable
AC (20 Hz-300 Hz) Test Voltages

System Voltage Phase-to-Phase (kV RMS)	Test Duration (Minutes)	Maintenance Test Voltage Phase-to-Ground (kV RMS)
3	15	3
5	15	5
6	15	6
8	15	8
10	15	10
15	15	15
20	15	20
25	15	25
30	15	30
35	15	35
45-47	60	52
60-69	60	72
110-115	60	128
132-138	60	132
150-161	60	150

Notes:

1. This table has been adapted from the recommended voltage levels and durations in IEC-60502-2-2014 and IEC-60840-2020. Those standards reference these voltages and durations as AC voltage test of the insulation after installation. The standard further acknowledges installations which have been in use, lower voltages than given in the table above, and / or shorter durations may be used.
2. It is strongly recommended that the user consult with the manufacturer(s) and system owners of all components that will be subjected to such testing before performing any tests on cables and cable accessories on time duration and voltage levels before testing. Duration and voltage levels should be negotiated, taking into account the age, environment, history of breakdowns, and the purpose of carrying out the test.

Table 100.6.6
VLF Tan Delta (TD)
MV/HV Cable Test Voltages

Step	Voltage	Min# of TD Test Data Points
1	0.5 Uo	6
2	1 Uo	6
3	1.5 Uo	6
*4	Ref table 100.6.3[3]	Duration of withstand

Notes:

1. Uo is defined as the Nominal RMS Rated Voltage, Phase to Ground
2. For new installations use nameplate voltage rating (ensure all components of circuit have same rating). For existing installations use operating voltage/system voltage. Example to calculate Uo and therefore 0.5 and 1.5 x Uo.
   - 2.1. For a 15 kV rated system the Uo for the new cable would be $15/\sqrt{3}=8.7$ kV.
   - 2.2. For a 15 kV rated system the Uo for an existing cable operating at 13.8 kV use $13.8/\sqrt{3}=8.0$ kV.
3. If performing the optional Monitored Withstand Test with TD at voltage and duration as specified in Table 100.6.3.
4. All voltage steps may be minimized to under Uo for degraded cables.

Table 100.6.7.1
Evaluation of Tan Delta Test Results
XLPE Cable

Criteria (10^{-3})	Acceptable (10^{-3})	Further Study (10^{-3})	Action Required (10^{-3})
Standard Deviation at any voltage step	< 0.1	0.1 – 0.5	> 0.5
Differential / Tip Up (1.5 Uo minus 0.5 Uo)	< 5	5 – 80	> 80
Mean Value at any voltage step	< 4	4 – 50	> 50

Notes:

1. Evaluation criteria are based on IEEE 400.2-2013 Table 4.
2. Evaluation is acceptable if all criteria are good. If any single criteria is in further study or action required, then the cable's evaluation falls into that more severe category.
3. All evaluation criteria is based on using VLF sinusoidal at 0.1 Hz.

Table 100.6.7.2
Evaluation of Tan Delta Test Results
EPR Cable

Criteria (10^{-3})	Acceptable (10^{-3})	Further Study (10^{-3})	Action Required (10^{-3})
Standard Deviation at any voltage step	< 0.1	0.1 – 1.3	> 1.3
Differential / Tip Up (1.5 Uo minus 0.5 Uo)	< 5	5 – 100	> 100
Mean Value at any voltage step	< 35	35 – 120	> 120

Notes:

1. Evaluation criteria are based on IEEE 400.2-2013 Table 5.
2. Evaluation is acceptable if all criteria are good. If any single criteria is in further study or action required, then the cable's evaluation falls into that more severe category.
3. All evaluation criteria is based on using VLF sinusoidal at 0.1 Hz.
4. Evaluation criteria for EPR cable are largely dependent on type of fill EPR insulation. Values shown are for unknown filled EPR insulation. For more detailed analysis, review Table 5 of IEEE 400.2.

TABLE 100.7
Molded-Case Circuit Breakers
Inverse Time Trip Test
(At 300% of Rated Continuous Current of Circuit Breaker)

Range of Rated Continuous Current (Amperes)	Maximum Trip Time in Seconds for Each Maximum Frame Rating [1]	
	≤ 250 V	250 – 600 V
0 – 30	50	70
31 – 50	80	100
51 – 100	140	160
101 – 150	200	250
151 – 225	230	275
226 – 400	300	350
401 – 600	- - - -	450
601 – 800	- - - -	500
801 – 1,000	- - - -	600
1,001 – 1,200	- - - -	700
1,201 – 1,600	- - - -	775
1,601 – 2,000	- - - -	800
2,001 – 2,500	- - - -	850
2,501 – 5,000	- - - -	900
6,000	- - - -	1,000

Notes:

Derived from Table 3, NEMA AB 4-2017, *Guidelines for Inspection and Preventative Maintenance of Molded-Case Circuit Breaker Used in Commercial and Industrial Applications.*

1. Trip times may be substantially longer for integrally-fused circuit breakers if tested with the fuses replaced by solid links (shorting bars).

TABLE 100.8
Molded-Case Circuit Breakers
Instantaneous Trip Tolerances

Breaker Type	Tolerance of Settings	Tolerances of Manufacturer's Published Trip Range	
		High Side	Low Side
Electronic Trip Units (1)	+30% -30%	- - - -	- - - -
Adjustable (1)	+40% -30%	- - - -	- - - -
Nonadjustable (2)	- - - -	+25%	-25%

Notes:

NEMA AB4-2017 *Guidelines for Inspection and Preventative Maintenance of Molded-Case Circuit Breaker Used in Commercial and Industrial Applications, Table 4.*

1. Tolerances are based on variations from the nominal settings.
2. Tolerances are based on variations from the manufacturer's published trip band (i.e., -25% below the low side of the band; +25% above the high side of the band.)

TABLE 100.9
Instrument Transformer Dielectric Tests
Field Maintenance

Nominal System (kV)	**BIL (kV)**	**Periodic Dielectric Withstand Test Field Test Voltage (kV)**	
		AC	**DC** [2]
0.6	10	2.6	4
1.1	30	6.5	10
2.4	45	9.7	15
4.8	60	12.3	19
8.32	75	16.9	26
13.8	95	22.1	34
13.8	110	26.0	34
25	125	32.5	40
25	150	32.5	50
34.5	150	45.5	50
34.5	200	61.7	70
46	250	91.0	Note 1
69	350	120.0	Note 1
115	450	149.0	Note 1
115	550	149.0	Note 1
138	550	178.0	Note 1
138	650	178.0	Note 1
161	650	211.0	Note 1
161	750	256.0	Note 1
230	900	256.0	Note 1
230	1,050	299.0	Note 1

Notes:

Table is derived from Paragraph 8.5.2 and Table 2 of IEEE C57.13-2016, *Standard Requirements for Instrument Transformers*.

1. Periodic dc potential tests are not recommended for transformers rated higher than 34.5 kV.
2. Under some conditions transformers may be subjected to periodic insulation test using direct voltage. In such cases the test direct voltage should not exceed the original factory acceptance test rms alternating voltage. Periodic direct-voltage tests should not be applied to (instrument) transformers of higher than 34.5 kV voltage rating.

TABLE 100.10 *Maximum Allowable Vibration Amplitude*					
RPM (at 60 Hz)	**Velocity (in/s peak)**	**Velocity (mm/s)**	**RPM (at 50 Hz)**	**Velocity (in/s peak)**	**Velocity (mm/s)**
3,600	0.15	3.8	3,000	0.15	3.8
1,800	0.15	3.8	1,500	0.15	3.8
1,200	0.15	3.8	1,000	0.13	3.3
900	0.12	3.0	750	0.10	2.5
720	0.09	2.3	600	0.08	2.0
600	0.08	2.0	500	0.07	1.7

Notes:

Derived from NEMA MG 1-7.08, Table 7-1.

1. Table is unfiltered vibration limits for resiliently mounted machines. For machines with rigid mounting multiply the limiting values by 0.8.

TABLE 100.11
Insulation Resistance Test Values for Rotating Machinery for One Minute at 40° C

Winding Rating Voltage [1] (V)	Test Voltage, DC	Recommended Minimum Insulation Resistance (Megohms): Windings Before 1970, Field Windings, Others Not Listed in Table 100.11[2]	Recommended Minimum Insulation Resistance (Megohms): DC Armature, AC Windings, (form-wound coils)	Recommended Minimum Insulation Resistance (Megohms): Random-Wound Stator Coils, Form-Wound Coils below 1 kV
< 1,000	500	kV + 1	100	5
1,000 – 2,500	500 – 1,000	kV + 1	100	N/A
2,501 – 5,000	1,000 – 2,500	kV + 1	100	N/A
5,001 – 12,000	2,500 – 5,000	kV + 1	100	N/A
> 12,000	5,000 – 10,000	kV + 1	100	N/A

Notes:

Refer to table 100.14 for temperature correction factors.

1. Rated line-to-line voltage for three-phase ac machines, line-to-ground voltage for single-phase machines, and rated direct voltage for dc machines or field windings.
2. kV is the rated machine terminal to terminal voltage.
3. Values based on IEEE 43-2013.

TABLE 100.12.1

Bolt-Torque Values for Electrical Connections
US Standard Fasteners [1]
Heat-Treated Steel – Cadmium or Zinc Plated [2]

Grade	**SAE 1&2**	**SAE 5**	**SAE 7**	**SAE 8**
Head Marking				
Minimum Tensile (Strength) (lbf/in^2)	64K	105K	133K	150K
Bolt Diameter (Inches)	**Torque (Pound-Feet)**			
1/4	4	6	8	8
5/16	7	11	15	18
3/8	12	20	27	30
7/16	19	32	44	48
1/2	30	48	68	74
9/16	42	70	96	105
5/8	59	96	135	145
3/4	96	160	225	235
7/8	150	240	350	380
1.0	225	370	530	570

Notes:

1. Consult manufacturer for equipment supplied with metric fasteners.
2. Table is based on national coarse thread pitch.

TABLE 100.12.2 *US Standard Fasteners* [1] *Silicon Bronze Fasteners* [2, 3] *Torque (Pound-Feet)*		
Bolt Diameter (Inches)	**Nonlubricated**	**Lubricated**
5/16	15	10
3/8	20	15
1/2	40	25
5/8	55	40
3/4	70	60

Notes:

1. Consult manufacturer for equipment supplied with metric fasteners.
2. Table is based on national coarse thread pitch.
3. This table is based on bronze alloy bolts having a minimum tensile strength of 70,000 pounds per square inch.

TABLE 100.12.3 ***US Standard Fasteners*** [1] ***Aluminum Alloy Fasteners*** [2, 3] ***Torque (Pound-Feet)***	
Bolt Diameter (Inches)	**Lubricated**
5/16	10
3/8	14
1/2	25
5/8	40
3/4	60

Notes:

1. Consult manufacturer for equipment supplied with metric fasteners.
2. Table is based on national coarse thread pitch.
3. This table is based on aluminum alloy bolts having a minimum tensile strength of 55,000 pounds per square inch.

TABLE 100.12.4 *US Standard Fasteners* [1] *Stainless Steel Fasteners* [2, 3] *Torque (Pound-Feet)*	
Bolt Diameter (Inches)	**Uncoated**
5/16	15
3/8	20
1/2	40
5/8	55
3/4	70

Notes:

1. Consult manufacturer for equipment supplied with metric fasteners.
2. Table is based on national coarse thread pitch.
3. This table is to be used for the following hardware types:
   Bolts, cap screws, nuts, flat washers, locknuts (18-8 alloy)
   Belleville washers (302 alloy).

Tables in 100.12 are compiled from Penn-Union Catalogue and Square D Company, Anderson Products Division, *General Catalog*: Class 3910 *Distribution Technical Data, Class 3930 Reference Data Substation Connector Products*.

TABLE 100.13
SF_6 Gas Test Limits

Test	< 30 PSIG Max Limits			≥ 30 PSIG Max Limits		
Water Content	-23° C	720 ppmv	90 ppmw	-36° C	200 ppmv	25 ppmw
SO_2 Decomposition Products	12 ppm			12ppm		
Total Decomposition Products	50 ul/l (50 ppm)			50 ul/l (50 ppm)		
Mineral Oil [1]	10 ppm			10 ppm		
Purity of SF_6	≥ 97.0% volume			≥ 97.0% volume		

Notes:

This table is derived from IEC 60480 Guidelines for the checking and treatment of sulfur hexafluoride taken from electrical equipment and specification for re-use, Table 2. The values have been adjusted to reflect common result parameters for 3 in 1 field gas test equipment.

1. If all gas handling equipment is oil-free (compressor, pump), then oil content measurement is not necessary.
2. This table is a guide for field testing and actual results should be compared to manufacturer's literature.
3. < 30 PSIG is for a vessel with no switching (GIS); > 30 PSIG is for MV, HV, and EHV applications where there is switching.
4. The values in the table are for applications using pure SF6 gas. For mixed gases refer to manufacturer's data.

TABLE 100.14.1
Insulation Resistance Conversion Factors (20° C)

Temperature		Multiplier	
° C	° F	Apparatus Containing Oil Immersed Insulation	Apparatus Containing Solid Insulation
-10	14	0.125	0.25
-5	23	0.180	0.32
0	32	0.25	0.40
5	41	0.36	0.50
10	50	0.50	0.63
15	59	0.75	0.81
20	68	1.00	1.00
25	77	1.40	1.25
30	86	1.98	1.58
35	95	2.80	2.00
40	104	3.95	2.50
45	113	5.60	3.15
50	122	7.85	3.98
55	131	11.20	5.00
60	140	15.85	6.30
65	149	22.40	7.90
70	158	31.75	10.00
75	167	44.70	12.60
80	176	63.50	15.80
85	185	89.789	20.00
90	194	127.00	25.20
95	203	180.00	31.60
100	212	254.00	40.00
105	221	359.15	50.40
110	230	509.00	63.20

Notes:

Derived from Megger, *Stitch in Time…The Complete Guide to Electrical Insulation Testing*.

Formula:
$R_c = R_a \times K$
Where: R_c is resistance corrected to 20° C
R_a is measured resistance at test temperature
K is applicable multiplier

Example: Resistance test on oil-immersion insulation at 104° F
R_a = 2 megohms @ 104° F
$K = 3.95$
$R_c = R_a \times K$
$R_c = 2.0 \times 3.95$
R_c = 7.90 megohms @ 20° C

TABLE 100.14.2 *Insulation Resistance Conversion Factors (40° C)*			
Temperature		**Multiplier**	
° C	**° F**	**Apparatus Containing oil Immersed Insulation**	**Apparatus Containing Solid Insulation**
-10	14	0.03	0.10
-5	23	0.04	0.13
0	32	0.06	0.16
5	41	0.09	0.20
10	50	0.13	0.25
15	59	0.18	0.31
20	68	0.25	0.40
25	77	0.35	0.50
30	86	0.50	0.63
35	95	0.71	0.79
40	104	1.00	1.00
45	113	1.41	1.26
50	122	2.00	1.59
55	131	2.83	2.00
60	140	4.00	2.52
65	149	5.66	3.17
70	158	8.00	4.00
75	167	11.31	5.04
80	176	16.00	6.35
85	185	22.63	8.00
90	194	32.00	10.08
95	203	45.25	12.70
100	212	64.00	16.00
105	221	90.51	20.16
110	230	128.00	25.40

Notes:

Derived from Megger, *Stitch in Time…The Complete Guide to Electrical Insulation Testing* and IEEE 43-2000 *IEEE Recommended Practice for Testing Insulation Resistance of Rotating Machinery*.

Notes: The insulation resistance coefficient is based on the halving of the insulation resistance to the change in temperature.

Apparatus Containing Immersed Oil Insulation Table uses 10° C change with temperature halving.
Apparatus Containing Solid Insulation Table uses 15° C change with temperature halving.

Formula:
$R_c = R_a \times K$
Where: R_c is resistance corrected to 40° C
R_a is measured resistance at test temperature
K is applicable multiplier

Example: Resistance test on oil-immersion insulation at 68° F/20° C
R_a = 2 megohms @ 68° F/20° C
K = 0.40
$R_c = R_a \times K$
R_c = 2.0 x 0.40 = 0.8 megohms @ 40° C

TABLE 100.15
High-Potential Test Voltage for Automatic Circuit Reclosers Maintenance Testing

Nominal Voltage Class (kV)	Maximum Voltage (kV)	Rated Impulse Withstand Voltage (kV)	Maximum AC [1] Field Test Voltage (kV)
14.4 (1 ø and 3 ø)	15.0	95	26.2
14.4 (1 ø and 3 ø)	15.5	110	37.5
24.9 (1 ø and 3 ø)	27.0	125	45.0
34.5 (1 ø and 3 ø)	38.0	150	52.5

Notes:

Derived from IEEE C37.61-1973(R1992) *Standard Guide for the Application, Operation, and Maintenance of Automatic Circuit Reclosers* and from IEEE C37.60-2012 High-voltage switchgear and controlgear – Part 111: Automatic circuit reclosers and fault interrupters for alternating current systems up to 38 kV.

1. Derived from IEEE C37.60-2012, Table 2. In accordance with IEEE C37.61, Section 6.2.2 (Servicing), a .75 multiplier has been applied to the values.

TABLE 100.16
High-Potential Test Voltage for Periodic Test of Line Sectionalizers

Nominal Voltage Class (kV)	Maximum Voltage (kV)	Rated Impulse Withstand Voltage (kV)	Maximum AC Field Test Voltage (kV)	15 Minute DC Withstand (kV)
14.4 (1 ø, 3 ø)	15.5	95	26.2	39
24.9 (1 ø)	27.0	125	30.0	58.5
34.5 (3 ø)	38.0	150	37.5	77.2

Notes:

1. Derived from IEEE C37.63-2013 *Standard Requirements for Overhead, Pad-Mounted, Dry-Vault, and Submersible Automatic Line Sectionalizers for Alternating Current Systems Up To 38 kV*, Table 2.
2. The table includes a 0.75 multiplier with fractions rounded down.
3. In the absence of consensus standards, the NETA Standards Review Council suggests the above representative values.
4. Values of ac voltage given are dry test, one-minute, factory acceptance test values.

TABLE 100.17

Metal-Enclosed Bus Dielectric Withstand Test Voltages

Type of Bus	Rated kV	Maximum Test Voltage (kV)	
		AC	DC
Isolated Phase for Generator Leads	24.5	37.0	52.0
	29.5	45.0	--
	34.5	60.0	--
Isolated Phase for Other than Generator Leads	15.5	37.0	--
	27.0	45.0	--
	38.0	60.0	--
Nonsegregated Phase	1.058	2.25	--
	4.76	14.2	--
	8.25	27.0	--
	15.0	27.0	--
	15.5	37.5	--
	27.0	45.0	--
	38.0	60.0	--
Segregated Phase	15.5	37.0	--
	27.0	45.0	--
	38.0	60.0	--
DC Bus Duct [1]	0.3/.325	1.6	2.3
	0.8	2.7	3.9
	1.2	3.6	5.4
	1.6	4.0	5.7
	3.2	6.6	9.3

Notes:

Derived from IEEE C37.23-2015 *Standard for Metal-Enclosed Bus and Calculating Losses in Isolated-Phase Bus*, Tables 3A, 3B, 3C, 3D and paragraph 6.4.3. The table includes a 0.75 multiplier with fractions rounded down.

1. The presence of the column headed "DC" does not imply any requirement for a dc withstand test on ac equipment. This column is given as a reference only for those using dc tests and represents values believed to be appropriate and approximately equivalent to the corresponding power frequency withstand test values specified for each class of bus.

   Direct current withstand tests are recommended for flexible bus to avoid the loss of insulation life that may result from the dielectric heating that occurs with rated frequency withstand testing.

   Because of the variable voltage distribution encountered when making dc withstand tests and variances in leakage currents associated with various insulation systems, the manufacturer should be consulted for recommendations before applying dc withstand tests to this equipment.

TABLE 100.18

Thermographic Survey
Suggested Actions Based on Temperature Rise

Temperature Difference (ΔT) Based on Comparisons Between Similar Components Under Similar Loading	Temperature Difference (ΔT) Based Upon Comparisons Between Component and Ambient Air Temperatures	Recommended Action
1° C - 3° C	1° C - 10° C	Possible deficiency; warrants investigation
4° C - 15° C	11° - 20° C	Indicates probable deficiency; repair as time permits
- - - - -	21° - 40° C	Monitor until corrective measures can be accomplished
> 15° C	> 40° C	Major discrepancy; repair immediately

Notes:

1. Temperature specifications vary depending on the exact type of equipment. Even in the same class of equipment (i.e., cables) there are various temperature ratings. Heating is generally related to the square of the current; therefore, the load current will have a major impact on ΔT. In the absence of consensus standards for ΔT, the values in this table will provide reasonable guidelines.
2. An alternative method of evaluation is the standards-based temperature rating system as discussed in Section 8.9.2, Conducting an IR Thermographic Inspection, *Electrical Power Systems Maintenance and Testing* by Paul Gill, PE, 2008.
3. It is a necessary and valid requirement that the person performing the electrical inspection be thoroughly trained and experienced concerning the apparatus and systems being evaluated as well as knowledgeable of thermographic methodology.

TABLE 100.19

Dielectric Withstand Test Voltages for Electrical Apparatus Other than Inductive Equipment

Nominal System (Line) Voltage [1] (kV)	Insulation Class (kV)	AC Factory Test (kV)	Maximum Field Applied AC Test (kV)	Maximum Field Applied DC Test (kV)
1.2	1.2	10	6.0	8.5
2.4	2.5	15	9.0	12.7
4.8	5.0	19	11.4	16.1
8.3	8.7	26	15.6	22.1
14.4	15.0	34	20.4	28.8
10.8	18.0	40	24.0	33.9
25.0	25.0	50	30.0	42.4
34.5	35.0	70	42.0	59.4
46.0	46.0	95	57.0	80.6
69.0	69.0	140	84.0	118.8

Notes:

In the absence of consensus standards, the NETA Standards Review Council suggests the above representative values.

1. Intermediate voltage ratings are placed in the next higher insulation class.

TABLE 100.20.1

Rated Control Voltages and Their Ranges for Circuit Breakers

Operating mechanisms are designed for rated control voltages listed with operational capability throughout the indicated voltage ranges to accommodate variations in source regulation, couple with low charge levels, as well as high charge levels maintained with floating charges. The maximum voltage is measured at the point of user connection to the circuit breaker [see notes (12) and (13)] with no operating current flowing, and the minimum voltage is measured with maximum operating current flowing.

	Direct Current Voltage Ranges (1)(2)(3)(4)(5) (8)(9)			**Rated Control Voltage (60 Hz)**	**Alternating Current Voltage Ranges (1)(2)(3)(4)(8)**
	Closing and Auxiliary Functions				**Closing, Tripping, and Auxiliary Functions**
(11) Rated Control Voltage	**Indoor Circuit Breakers**	**Outdoor Circuit Breakers**	**Opening Functions All Types**	**Single Phase**	**Single Phase**
24 (6)	--	--	14 - 28	120	104 – 127 (7)
48 (6)	38 – 56	36 – 56	28 – 56	240	208 – 254 (7)
125	100 – 140	90 – 140	70 – 140	Polyphase	Polyphase
250	200 – 280	180 – 280	140 – 280	208Y/120	180Y/104 – 220Y/127
--	--	--	--	240	208 – 254
--	--	--	--		

Notes:

Derived from Table 8, IEEE C37.06-2000, *AC High-Voltage Circuit Breakers Rated on a Symmetrical Current Basis – Preferred Ratings and Related Required Capabilities.*

TABLE 100.20.1 (continued)

Rated Control Voltages and Their Ranges for Circuit Breakers

Notes:

(1) Electrically operated motors, contactors, solenoids, valves, and the like, need not carry a nameplate voltage rating that corresponds to the control voltage rating shown in the table as long as these components perform the intended duty cycle (usually intermittent) in the voltage range specified.

(2) Relays, motors, or other auxiliary equipment that function as a part of the control for a device shall be subject to the voltage limits imposed by this standard, whether mounted at the device or at a remote location.

(3) Circuit breaker devices, in some applications, may be exposed to control voltages exceeding those specified here due to abnormal conditions such as abrupt changes in line loading. Such applications require specific study, and the manufacturer should be consulted. Also, application of switchgear devices containing solid-state control, exposed continuously to control voltages approaching the upper limits of ranges specified herein, require specific attention and the manufacturer should be consulted before application is made.

(4) Includes supply for pump or compressor motors. Note that rated voltages for motors and their operating ranges are covered by NEMA MG-1-1978.

(5) It is recommended that the coils of closing, auxiliary, and tripping devices that are connected continually to one dc potential should be connected to the negative control bus so as to minimize electrolytic deterioration.

(6) 24-volt or 48-volt tripping, closing, and auxiliary functions are recommended only when the device is located near the battery or where special effort is made to ensure the adequacy of conductors between battery and control terminals. 24-volt closing is not recommended.

(7) Includes heater circuits.

(8) Voltage ranges apply to all closing and auxiliary devices when cold. Breakers utilizing standard auxiliary relays for control functions may not comply at lower extremes of voltage ranges when relay coils are hot, as after repeated or continuous operation.

(9) Direct current control voltage sources, such as those derived from rectified alternating current, may contain sufficient inherent ripple to modify the operation of control devices to the extent that they may not function over the entire specified voltage ranges.

(10) This table also applies to circuit breakers in gas-insulated substation installations.

(11) In cases where other operational ratings are a function of the specific control voltage applied, tests in IEEE C37.09 may refer to the “Rated Control Voltage.” In these cases, tests shall be performed at the levels in this column.

(12) For an outdoor circuit breaker, the point of user connection to the circuit breaker is the secondary terminal block point at which the wires from the circuit breaker operating mechanism components are connected to the user’s control circuit wiring.

(13) For an indoor circuit breaker, the point of user connection to the circuit breaker is either the secondary disconnecting contact (where the control power is connected from the stationary housing to the removable circuit breaker) or the terminal block point in the housing nearest to the secondary disconnecting contact.

TABLE 100.20.2

Rated Control Voltages and Their Ranges for Circuit Breakers Solenoid-Operated Devices

Rated Voltage (DC)	Closing Voltage Ranges for Power Supply
125	90 – 115 or 105 – 130
250	180 – 230 or 210 – 260
230	190 – 230 or 210 – 260

Notes:

1. Some solenoid operating mechanisms are not capable of satisfactory performance over the range of voltage specified in the standard; moreover, two ranges of voltage may be required for such mechanisms to achieve an acceptable standard of performance.
2. The preferred method of obtaining the double range of closing voltage is by use of tapped coils. Otherwise it will be necessary to designate one of the two closing voltage ranges listed above as representing the condition existing at the device location due to battery or lead voltage drop or control power transformer regulation. Also, caution should be exercised to ensure that the maximum voltage of the range used is not exceeded.

TABLE 100.21

Accuracy of IEC Class TP Current Transformers Error Limit

WITHDRAWN

TABLE 100.22
Minimum Bending Radius for Medium Voltage Power Cables Single and Multiple Conductor

Cable Type	Overall Diameter of Cable					
	inches 0.75 and less	**mm** 190 and less	**inches** 0.76 to 1.50	**mm** 191 to 381	**inches** 1.51 and larger	**mm** 382 and Larger
	Minimum Bending Radius as a Multiple of Cable Diameter					
Smooth Aluminum Shealth Single Conductor Nonshielded, Multiple Conductor or Multiplexed, with Individually	10		12		15	
Single Conductor Shielded	12		12		15	
Multiple Conductor or Muliplexed, with Overall Shield	12		12		15	
Interlocked Armor or Corrugated Aluminum Sheath Nonshielded	7		7		7	
Multiple Conductor with Individually Shielded Conductor	12/7 [a]		12/7 [a]		12/7 [a]	
Multiple Conductor with Overall Shield	12		12		12	
Lead Sheath	12		12		12	

12x[a] individual shielded conductor diameter, or 7x overall cable diameter, whichever is greater.

Table 100.22 and notes from ANSI/NEMA WC 74/ICEA S93-639-2017 *American National Standard for 5-46 kV Shielded Power Cable for Use in the Transmission and Distribution of Electric Energy*, Appendix I.

NFPA 70 300.34 Conductor Bending Radius – over 1,000V – single conductor non-shielded 8X overall diameter, shielded 12X, multi-conductor 12X individual conductor or 7X entire table.

TABLE 100.22 (continued)

Minimum Bending Radius for Medium Voltage Power Cables Single and Multiple Conductor

Notes:

1. Interlocked-Armor and Metallic-Sheathed Cables use Table 100.22.
2. Flat-Tape Armored or Wire-Armored Cables minimum bending radius is twelve times the overall diameter of cable.

Shielded Cables without armor:

1. Tape-shielded with helically applied flat or corrugated tape or longitudinally applied corrugated tape-shielded cables.
2. The minimum bending radius for a single-conductor cable is twelve times the overall diameter.
3. For multiple-conductor or multiplexed single-conductor cables having individually taped shielded conductors, the minimum bending radius is twelve times the diameter of the individual conductors or seven times the overall diameter, whichever is greater.
4. For multiple-conductor cables having an overall tape shield over the assembly, the minimum bending radius is twelve times the overall diameter of the cable.

Wire-Shielded Cables.

1. The minimum bending radius for a single-conductor cable is eight times the overall diameter.
2. For multiple-conductor or multiplexed single-conductor cables having wire-shielded individual conductors, the minimum bending radius is eight times the diameter of the individual conductors or five times the overall diameter, whichever is greater.
3. For multiple-conductor cables having a wire shield over the assembly, the minimum bending radius is eight times the overall diameter of the cable.

TABLE 100.23
Online Partial Discharge Survey for Switchgear

Notes:

1. Any signal level present without a phase-angle related component is not indicative of partial-discharge activity.
2. Perform partial-discharge surveys every 12 months, or in accordance with Appendix B, Frequency of Maintenance Tests.
3. Compare measured levels to established background readings.
4. Background readings higher than values listed in these tables should be investigated and mitigated.

TABLE 100.23.1
Online Partial Discharge Survey for Switchgear
Suggested Actions Based on Nature and Strength of Signal
(TEV)

dB	Phase-Angle Related Component	Recommended Action
Any[1]	No	Survey in 12 months for trending [2]
< 20	Yes	Survey in 6 months for trending
20 – 30	Yes	Locate and repair as soon as practical
> 30	Yes	Locate and repair as soon as possible

TABLE 100.23.2
Online Partial Discharge Survey for Switchgear
Suggested Actions Based on Nature and Strength of Signal
(HFCT)

pC	dB	Phase Related Component	Recommended Action
Any	Any	No	Survey in 12 months for trending
< 250	< 10	Yes	Survey in 6 months for trending
200 – 500	10 – 20	Yes	Locate and repair as soon as practical
> 500	> 20	Yes	Locate and repair as soon as possible

Notes:

1. The use of HFCTs for cable testing is outside the scope of this standard.

TABLE 100.23.3 ***Online Partial Discharge Survey for Switchgear*** ***Suggested Actions Based on Nature and Strength of Signal*** ***(UHF)***		
dBmV	**Phase Related Component**	**Recommended Action**
Any	No	Survey in 12 months for trending
< 10	Yes	Survey in 6 months for trending
10 – 20	Yes	Locate and repair as soon as practical
> 20	Yes	Locate and repair as soon as possible

TABLE 100.23.4 ***Online Partial Discharge Survey for Switchgear*** ***Suggested Actions Based on Nature and Strength of Signal*** ***(VIS)***		
dB	**Phase Related Component**	**Recommended Action**
Any	No	Survey in 12 months for trending
20 – 29	Yes	Survey in 6 months for trending
30 – 39	Yes	Locate and repair as soon as practical
> 40	Yes	Locate and repair as soon as possible

TABLE 100.23.5 ***Online Partial Discharge Survey for Switchgear*** ***Suggested Actions Based on Nature and Strength of Signal*** ***(Contact/Airborne Acoustic)***		
dB	**Phase Related Component**	**Recommended Action**
Any	No	Survey in 12 months for trending
< 3	Yes	Survey in 6 months for trending
3 – 6	Yes	Locate and repair as soon as practical
> 6	Yes	Locate and repair as soon as possible

APPENDIX A

Definitions

NETA defines equipment voltage ratings in accordance with IEEE C37.84.1 *American National Standard for Electrical Power Systems and Equipment – Voltage Ratings (60 Hertz).*

As-found
Condition of the equipment when taken out of service, prior to maintenance.

As-left
Condition of equipment at the completion of maintenance. As-left values refer to test values obtained after all maintenance has been performed on the device under test.

Cable, Laminated Dielectric Insulation
Insulation formed in layers, typically from fluid-impregnated tapes of either cellulose paper, or polypropylene, or a combination of the two. Examples include: paper insulated lead covered (PILC) cable designs, Mass Impregnated Non Draining (MIND) cable designs, high pressure pipe type cable designs, and self-contained cable systems.

Cable, Extruded Dielectric Insulation
Insulation such as polyethylene (PE), crosslinked polyethylene (XLPE), tree retardant crosslinked polyethylene (TRXLPE), ethylene propylene rubber (EPR), etc. applied using an extrusion process.

Comment
Suggested revision, addition, or deletion in an existing section of the NETA specifications.

Commissioning, electrical
The systematic process of documenting and placing into service newly installed or retrofitted electrical power equipment and systems.

Electrical tests
Electrical tests involve application of electrical signals and observation of the response. It may be, for instance, applying a potential across an insulation system and measuring the resultant leakage current magnitude or power factor/dissipation factor. It may also involve application of voltage and/or current to metering and relaying equipment to check for correct response.

Equipment condition
Suitability of the equipment for continued operation in the intended environment as determined by evaluation of the results of inspections and tests.

Errata
A correction issued to a NETA Standard, published in NETA World, available on the NETA website and included in any further distribution of the Standard.

Exercise
To operate equipment in such a manner that it performs all its intended functions to allow observation, testing, measurement, and diagnosis of its operational condition.

Inspection
Examination or measurement to verify whether an item or activity conforms to specified requirements

Interim amendment
An interim amendment is made by NETA's Standards Review Council when there is a potential hazard prior to review by the Section Panel or the public

Maintenance, Condition of
The state of the electrical equipment considering the manufacturer's instructions, manufacturer's recommendations, and applicable industry codes, standards, and recommended practices.

Manufacturer's published data
Data provided by the manufacturer concerning a specific piece of equipment.

Mechanical inspection
Observation of the mechanical operation of equipment not requiring electrical stimulation, such as manual operation of circuit breaker trip and close functions. It may also include tightening of hardware, cleaning, and lubricating.

Proposal
Draft of a section that is currently "reserved" in one of the NETA specifications.

Ready-to-test condition
Preparing equipment which is to be tested in an isolated state, source and load disconnected, necessary protective grounds applied, and control and operating voltage sources identified.

Shall
Indicates mandatory requirements to be strictly followed in order to conform to the standard and from which no deviation is permitted (*shall* equals *is required to*).

Should
Indicates that among several possibilities one is recommended as particularly suitable, without mentioning or excluding others; or that a certain course of action is preferred but not necessarily required (*should* equals *is recommended that*).

System voltage
The root-mean-square (rms) phase-to-phase voltage of a portion of an alternating-current electric system. Each system voltage pertains to a portion of the system that is bounded by transformers or utilization equipment.

APPENDIX A

Definitions *(continued)*

Time-travel analysis
The measurement of the opening and closing strokes of circuit contacts from the time of an initiating signal to the complete at rest position of the contact. The analysis shall provide the information required to determine distance traveled, velocity of travel, contact insertion, contact bounce, and contact parting time.

Verify
To investigate by observation or by test to determine that a particular condition exists.

Visual inspection
Qualitative observation of physical characteristics, for example: cleanliness, physical integrity, evidence of overheating, lubrication, etc.

Voltage Class:

Low voltage[1]
A class of nominal system voltages 1000 volts or less.

Medium voltage[1]
A class of nominal system voltages greater than 1000 volts and less than 100,000 volts.

High voltage[1]
A class of nominal system voltages equal to or greater than 100,000 volts and equal to or less than 230,000 volts.

Extra-high voltage[1]
A class of nominal system voltages greater than 230,000 volts.

[1] NETA defines equipment voltage ratings in accordance with NEMA C37.84.1 *American National Standard for Electrical Power Systems and Equipment – Voltage Ratings (60 Hertz)*
.

APPENDIX B

Frequency of Maintenance Tests

NETA recognizes that the ideal maintenance program is reliability-based, unique to each plant and to each piece of equipment. NETA's Standards Review Council presents the following time-based maintenance schedule and matrix.

The following matrix is to be used in conjunction with Appendix B, Inspections and Tests. Application of the matrix is recognized as a guide only.

Specific condition, criticality, and desired reliability must be determined to correctly apply the matrix. Application of the matrix, along with the culmination of historical testing data and trending, should provide a quality electrical preventive maintenance program**.**

MAINTENANCE FREQUENCY MATRIX				
		Equipment Condition		
		Poor	**Average**	**Good**
Equipment Reliability Requirement	**Low**	1.0	2.0	2.5
	Medium	0.50	1.0	1.5
	High	0.25	0.50	0.75

APPENDIX B
Frequency of Maintenance Tests *(continued)*

Inspections and Tests
Frequency in Months
(Multiply These Values by the Factor in the Maintenance Frequency Matrix)

Section	Description	Visual	Visual & Mechanical	Visual & Mechanical & Electrical
7.1	Switchgear & Switchboard Assemblies	12	12	24
7.2	Transformers			
7.2.1.1	Small Dry-Type Transformers	2	12	36
7.2.1.2	Large Dry-Type Transformers	1	12	24
7.2.2	Liquid-Filled Transformers	1	12	24
	Sampling	–	–	12
7.3	Cables			
7.3.1	Low-Voltage, Low-Energy	–	–	–
7.3.2	Low-Voltage, 600-Volt Maximum	2	12	36
7.3.3	Medium- and High-Voltage	2	12	36
7.4	Metal-Enclosed Busways	2	12	24
	Infrared Only	–	–	12
7.5	Switches			
7.5.1.1	Air, Low-Voltage	2	12	36
7.5.1.2	Air, Medium-Voltage, Metal-Enclosed	–	12	24
7.5.1.3	Air, Medium- and High-Voltage Open	1	12	24
7.5.2	Oil, Medium-Voltage	1	12	24
7.5.3	Vacuum, Medium-Voltage	1	12	24
7.5.4	Medium-Voltage, SF_6	1	12	24
7.5.5	Cutouts	12	24	24
7.6	Circuit Breakers			
7.6.1.1	Air, Insulated-Case/Molded-Case	1	12	36
7.6.1.2	Air, Low-Voltage Power	1	12	36
7.6.1.3	Air, Medium-Voltage	1	12	36
7.6.2	Oil, Medium- and High-Voltage	1	12	36
	Sampling	–	–	12
7.6.3	Vacuum, Medium-Voltage	1	12	36
7.6.4	SF_6	1	12	36
7.7	Circuit Switchers	1	12	36
7.8	Network Protectors	12	12	24

APPENDIX B
Frequency of Maintenance Tests (*continued*)

Inspections and Tests
Frequency in Months
(Multiply These Values by the Factor in the Maintenance Frequency Matrix)

Section	Description	Visual	Visual & Mechanical	Visual & Mechanical & Electrical
7.9	Protective Relays			
7.9.1	Electromechanical and Solid State	1	12	24
7.9.2	Microprocessor-Based	1	12	36
7.10	Instrument Transformers			
7.10.1	Current Transformers	12	12	36
7.10.2	Voltage Transformers	12	12	36
7.10.3	Coupling-Capacitor Transformers	12	12	36
7.11	Metering Devices			
7.11.1	Electromechanical and Solid-State	12	12	36
7.11.2	Microprocessor-Based	12	12	36
7.12	Regulating Apparatus			
7.12.1.1	Step-Voltage Regulators	1	12	24
	Sample Liquid	–	–	12
7.12.1.2	Induction Regulators	12	12	24
7.12.2	Current Regulators	1	12	24
7.12.3	Load Tap-changers	1	12	24
	Sample Liquid	–	–	12
7.13	Grounding Systems	2	12	24
7.14	Ground-Fault Protection Systems	2	12	24
7.15	Rotating Machinery			
7.15.1	AC Induction Motors and Generators	1	12	24
7.15.2	Synchronous Motors and Generators	1	12	24
7.15.3	DC Motors and Generators	1	12	24
7.16	Motor Control			
7.16.1.1	Motor Starters, Low-Voltage	2	12	24
7.16.1.2	Motor Starters, Medium-Voltage	2	12	24
7.16.2.1	Motor Control Centers, Low-Voltage	2	12	24
7.16.2.2	Motor Control Centers, Medium-Voltage	2	12	24
7.17	Adjustable-Speed Drive Systems	1	12	24
7.18	Direct-Current Systems			
7.18.1	Batteries			
7.18.1.1	Flooded Lead-Acid	1	12	12
7.18.1.2	Vented Nickel-Cadmium	1	12	12
7.18.1.3	Valve-Regulated Lead-Acid	1	12	12
7.18.2	Battery Chargers	1	12	12

APPENDIX B
Frequency of Maintenance Tests (*continued*)

Inspections and Tests
Frequency in Months
(Multiply These Values by the Factor in the Maintenance Frequency Matrix)

Section	Description	Visual	Visual & Mechanical	Visual & Mechanical & Electrical
7.18.3	Rectifiers	1	12	24
7.19	Surge Arresters			
7.19.1	Low-Voltage Surge Protection Devices	2	12	24
7.19.2	Medium- and High-Voltage Surge Protection Devices	2	12	24
7.20	Capacitors and Reactors			
7.20.1	Capacitors	1	12	36
7.20.2	Capacitor Control Devices	1	12	12
7.20.3.1	Reactors, (Shunt and Current-Limiting) Dry-Type	2	12	36
7.20.3.2	Reactors, (Shunt and Current-Limiting) Liquid-Filled	1	12	36
	Sampling	–	–	12
7.21	Outdoor Bus Structures	1	12	36
7.22	Emergency Systems			
7.22.1	Engine Generator	1	2	12
	Functional Testing	–	–	2
7.22.2	Uninterruptible Power Systems	1	12	12
	Functional Testing	–	–	2
7.22.3	Automatic Transfer Switches	1	12	12
	Functional Testing	–	–	2
7.23	Communications - Reserved			
7.24	Automatic Circuit Reclosers and Line Sectionalizers			
7.24.1	Automatic Circuit Reclosers, Oil/Vacuum	1	12	24
	Sample	–	–	12
7.24.2	Automatic Line Sectionalizers, Oil	1	12	24
	Sample	–	–	12
9.	Thermographic Survey	–	12	–
10.	Electromagnetic Field Survey – As Needed	–	–	–
11.	Online Partial Discharge Survey for Switchgear	–	–	12

APPENDIX C
Frequency of Power System Studies

Frequency Matrix					
Circumstance	**One-Line Diagram Update**	**Short-circuit Study**	**Coordination Study**	**Incident Energy Analysis**	**Load Flow Study**
New installation or system modification	X	X	X	X	X
Annual data verification	X	X	X	X	X
Change in utility or source	X	X	X	X	X
Change in system impedance, configuration, or loading[1]	X	X		X	
Change in protection devices or settings	X		X	X	X
5 years since last update	X	X		X	

Notes:

1. Examples of changes: switch between delta and wye, solidly-grounded to resistance-grounded, kVA change, impedance, conductor length or size, adding motors, replacing motors with high-efficiency motors, change in horsepower, adding or removing power-factor correction capacitors.

APPENDIX D

Guidance for Circuit Reliability Considerations for Medium and High Voltage Cable Methods of Test

When considering the most effective field testing method for medium-voltage power cables, consensus standards from standards-making organizations such as ICEA, IEC, IEEE, manufacturer's published data, and other entities often have differing opinions as to the most effective method for performing cable assessment tests. These differing opinions create difficulties for the owners of the cables to make informed maintenance testing decisions.

Only after careful review of all standards, test methods, availability of replacement cable and accessories, and review of the site-specific cable system, can an end-user, engineer, or testing company make an informed decision as to the best value for testing of their medium-voltage cable system.

A key factor to determining the best course of action when performing cable testing is the system reliability requirements, which are different and unique for each facility, and unique to the type and construction of the cable system under test.

In the absence of guidance from existing consensus standards, and to aid in the selection of the appropriate test[s], NETA's Standard Review Council presents the following matrix based on reliability requirements of the cable system and the type of cables to be tested.

These recommendations are primarily a compilation of the current IEEE standards and recommendations from NETA accredited companies' field experience on cost benefit analysis. In the absence of a full evaluation, the following table may be used to determine the most appropriate tests and the time duration.

Circuit Reliability Requirement			
Cable Type	**Low**	**Medium**	**High**
MV Tape Shield	• 30 min VLF withstand OR • Diagnostic TD with Tip-Up	• 30 min VLF Monitored withstand and • Diagnostic TD with Tip-Up	• 60 min VLF Monitored withstand and • Diagnostic TD with Tip-Up
MV Concentric Neutral	• 30 min VLF withstand OR • Diagnostic TD with Tip-Up	• 30 min VLF Monitored withstand and • Diagnostic TD with Tip-Up	• 60 min VLF Monitored withstand and • Diagnostic TD with Tip-Up and • Offline PD

APPENDIX D

Guidance for Circuit Reliability Considerations for Medium and High Voltage Cable Methods of Test *(continued)*

HV Concentric Neutral	• 60 min VLF withstand OR • Diagnostic TD with Tip-Up	• 30 min VLF Monitored withstand and • Diagnostic TD with Tip-Up and • Offline PD	• 60 min VLF Monitored withstand and • Diagnostic TD with Tip-Up and • Offline PD
MV / HV PILC	• 15 min DC withstand	• 15 min Monitored DC withstand with Voltage Recovery	• 30 min VLF Monitored withstand and • Diagnostic TD with Tip-Up and

Notes:

1. Tape shielding attenuation properties of medium and high-voltage cables can affect the results of offline partial discharge tests, therefore is not a preferred test method for this type of cable.
2. Monitored withstand test is defined as a test in which a test with a predermined magnitude is applied for predetermined time. During the test, Tan Delta or particial discharge values are monitored and used together with the withstand results to determine the cable condition.
3. Frequencies outside of 0.1 Hz are not recommended for tan-delta analysis.
4. Very Low Frequency (VLF) testing is the most widely accepted and specified cable test due to the lower cost and availability of the test equipment. Power frequency cable testing (20-300 Hz) may be utilized in place of a recommended VLF testing mentioned above.
5. Cable type is identified as medium-voltage or high-voltage and the type of insulation shield present. Depending on the type, different service-aged insulation shields have an impact on the ability to perform tan-delta and partial discharge testing. The IEEE standards reference the above shielding types and have indicated the value of tests for each type of shielding.
6. Low criticality circuits are those that will have little impact to the overall facility, system, personnel, or life safety equipment if they fail.
7. Medium criticality circuits are those that when a failure occurs it can impact the overall facility, but likely will not completely shut it down if they fail. Medium criticality circuits will financially impact the organization if failure occurs, but these circuits will typically be able to be repaired, replaced, or repowered from a different source within a reasonable amount of time.
8. High criticality circuits are those that can affect life safety and/or completely shut down a significant portion of the facility or system if they fail. These circuits will likely have a major impact, both financially and operationally, to the organization. The mitigation of these failures can have a severe impact from a downtime, cost, or environmental perspective.

APPENDIX E

About the InterNational Electrical Testing Association

The InterNational Electrical Testing Association (NETA) is an accredited standards developer for the American National Standards Institute (ANSI) and defines the standards by which electrical equipment is deemed safe and reliable. NETA Certified Technicians conduct the tests that ensure this equipment meets the Association's stringent specifications. NETA is the leading source of specifications, procedures, testing, and requirements, not only for commissioning new equipment but for testing the reliability and performance of existing equipment.

CERTIFICATION

Certification of competency is particularly important in the electrical testing industry. Inherent in the determination of the equipment's serviceability is the prerequisite that individuals performing the tests be capable of conducting the tests in a safe manner and with complete knowledge of the hazards involved. They must also evaluate the test data and make an informed judgment on the continued serviceability, deterioration, or nonserviceability of the specific equipment. NETA, a nationally-recognized certification agency, provides recognition of four levels of competency within the electrical testing industry in accordance with *ANSI/NETA ETT Standard for Certification of Electrical Testing Technicians.*

QUALIFICATIONS OF THE TESTING ORGANIZATION

An independent overview is the only method of determining the long-term usage of electrical apparatus and its suitability for the intended purpose. NETA Accredited Companies best support the interest of the owner, as the objectivity and competency of the testing firm is as important as the competency of the individual technician. NETA Accredited Companies are part of an independent, third-party electrical testing association dedicated to setting world standards in electrical maintenance and acceptance testing. Hiring a NETA Accredited Company assures the customer that:

- The NETA Technician has broad-based knowledge -- this person is trained to inspect, test, maintain, and calibrate all types of electrical equipment in all types of industries.
- NETA Technicians meet stringent educational and experience requirements in accordance with *ANSI/NETA ETT Standard for Certification of Electrical Testing Technicians*.
- A Registered Professional Engineer will review all engineering reports.
- All tests will be performed objectively, according to NETA specifications, using calibrated instruments traceable to the National Institute of Science and Technology (NIST).
- The firm is a well-established, full-service electrical testing business.

APPENDIX E

About the InterNational Electrical Testing Association
(continued)

SPECIFICATIONS AND PUBLICATIONS

As a part of its service to the industry, the InterNational Electrical Testing Association provides nationally-recognized publications:

ANSI/NETA ETT	*Standard for Certification of Electrical Testing Technicians*
ANSI/NETA MTS	*Standard for Maintenance Testing Specifications for Electrical Power Equipment and Systems*
ANSI/NETA ATS	*Standard for Acceptance Testing Specifications for Electrical Power Equipment and Systems*
ANSI/NETA ECS	*Standard for Electrical Commissioning Specifications for Electrical Power Equipment and Systems*

The Association also produces a quarterly technical journal, *NETA World Journal*, which features articles of interest to electrical testing and maintenance companies, consultants, engineers, architects, and plant personnel directly involved in electrical testing and maintenance.

Additional information can be found at www.netaworld.org.

EDUCATIONAL PROGRAMS

PowerTest, NETA's annual technical conference, draws hundreds of qualified industry professionals from around the globe. This conference provides a forum for current industry advances, critical informational updates, networking, and more. Regular attendees include technicians from electrical testing and maintenance companies, consultants, engineers, architects, and plant personnel directly involved in electrical testing and maintenance. Paper presentations from field-experienced industry experts share practical knowledge and experience while in-depth seminars offer interactive training. At the Trade Show attendees enjoy the highest-quality gathering of industry-specific suppliers displaying state-of-the-art products and services directly related to the electrical testing industry. Attendance of PowerTest is the best opportunity for interaction and input in a professional technical environment.

Additional information can be found at www.powertest.org.

APPENDIX F

Form for Comments

(This appendix is not part of American National Standard ANSI/NETA MTS-2023)

Anyone may comment on this document using this form:

Type of Comment (Check one) ❑ Technical ❑ Editorial
Paragraph Number ______________________________
Recommend (Check One) ❑ *New Text ❑ *Revised Text ❑ *Deleted Text
❑This Comment is original material (Note: Original material is considered to be the submitter's own idea based on or as a result of his/her own experience, thought, or research and to the best of his/her knowledge is not copied from another source.)
❑This Comment is not original material; its source (if known) is ______________________________
Please Check One: ❑User ❑Producer ❑General Interest
Date ______________________________
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Organization represented, if any ______________________________

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Signature (required) ______________________________

*1. All comments must be relevant to the proposed standard.
*2. Suggested changes must include (1) proposed text, including the wording to be added, revised (and how revised), or deleted, (2) a statement of the problem and substantiation for a technical change, and (3) signature of submitter. (Note: State the problem that will be resolved by your recommendation; give the specific reason for your comment, including copies of texts, research papers, testing experience, etc. If more than 200 words, it may be abstracted for publication.)
*3. Editorial comments are welcome, but they cannot serve as the sole basis for a suggested change.

A comment that does not include all required information may be rejected by the Standards Review Council for that reason. Must use separate form for each comment. All comments must be typed or printed neatly. Illegible comments will be interpreted to the best of the staff's ability.

This form is available electronically on NETA's website at www.netaworld.org under Standards Activities.

Send to: Standards Review Council
3050 Old Centre Road, Suite 101, Portage, MI 49024
Phone: 888.300.6382 FAX: 269.488.6383 Email: neta@netaworld.org

APPENDIX G

Form for Proposals

(This appendix is not part of American National Standard ANSI/NETA MTS-2023)

Anyone may propose a new section for this document using the following form:

When drafting a proposed section:
Use the most recent edition of the specifications as a guideline for format and wording.
Remember that NETA specifications are "what to do" documents and do not include "how to do" information.
Include references.
When applicable, use the standard base format:

A. Visual and Mechanical Inspection
B. Electrical Tests
C. Test Values – Visual and Mechanical
D. Test Values – Electrical

Date ____

Name ____ Tel No. ________________

Company _____________________________________ Fax No. ________________

Address _____________________________________ E-Mail ________________

Please indicate organization represented (if any) ____________________________

NETA document title __________________________ Year __________________

Section/Number ____

Note 1: Type or print legibly in black ink.
Note 2: If supplementary material (photographs, diagrams, reports, etc.) is included, you may be required to submit sufficient copies for all Members and Alternates of the Section Panel.

I hereby grant NETA the nonexclusive, royalty-free rights, including nonexclusive, royalty-free rights in copyright, in this proposal. I understand that I acquire no rights in any publication of NETA in which this proposal in this or another similar analogous form is used.

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